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Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation

Master Thesis at Ulm University

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1 Introduction

1.1 Context

The General Data Protection Regulation (GDPR) requires organizations to maintain a Record of Processing Activities (ROPA), a formal document under Article 30 GDPR that records how personal data is collected, processed, stored, and shared. The content of a ROPA is rigidly specified (controller details, purposes, lawful bases, categories of data and recipients, retention periods, and technical and organizational measures), yet its substance is project-specific prose that has to be written from a blank page and kept current as processing activities change. Maintaining a ROPA is therefore a sustained cost rather than a one-off task; it requires legal and data-protection expertise that small and medium-sized organizations rarely have, and falls disproportionately on those least equipped to bear it [29].

ROPAgen, the tool evaluated in this thesis, is a web-based application developed at the Institute of Databases and Information Systems (DBIS) at the University of Ulm to address exactly this gap. It uses Large Language Models (LLMs) to assist novice users in drafting GDPR-compliant ROPA documentation, offering three interaction modes, each with a different level of LLM support:

Form mode, in which the user fills structured fields directly, with LLM suggestions available on demand.

Ask mode, in which the user fills a free-form text field per section while a chatbot explains the section and answers questions.

Chat mode, in which the user describes their requirements conversationally, and the LLM autonomously drafts and populates the ROPA. The three modes are described in detail in Section 3.1; by comparing them, the thesis evaluates how different levels of LLM support affect user performance and trust.

Although the application provides a functional framework for assisted ROPA generation, its effectiveness, usability, and impact on documentation quality have not yet been systematically evaluated. To ensure that such a system can be confidently integrated into practical compliance workflows, both qualitative and quantitative assessments are needed. These evaluations determine whether users can efficiently and correctly create ROPA documents, how the varying levels of LLM support influence performance and trust, and whether the system produces outputs aligned with GDPR requirements.

1.2 Research Problem and Research Questions

The goal of this thesis is to conduct a comprehensive evaluation of the existing LLM-supported ROPA generation application. The evaluation asks how the three interaction modes shape the experience of novice users, how they shape the documents those users produce, and whether the two are aligned. RQ1 is broken into three sub-questions tied to the user-study instruments; RQ2 is broken into two sub-questions tied to the document-quality streams. RQ3 is a single cross-stream question that reads the two evaluation streams against each other; it is reserved for the Discussion chapter, where the cross-signal interpretation belongs.

RQ1 (User experience). *How do varying degrees of LLM support influence novice users during ROPA document creation?*

RQ1.1 How do novice users perceive the *usability* of the three interaction modes (Form, Ask, Chat)?

RQ1.2 How does *cognitive load* differ across the three modes for users with no prior GDPR knowledge?

RQ1.3 How do novice users subjectively evaluate each mode: in their perception of the LLM's support, in their confidence about the documents they produce, and in the mode they ultimately prefer?

RQ2 (Document quality). *How do varying degrees of LLM support influence the quality of ROPA documents produced by novice users?*

RQ2.1 How does document quality differ across the three modes when measured by reference-based NLP similarity to an expert-authored ROPA (BLEU, ROUGE, METEOR, BERTScore, SBERT)?

RQ2.2 As a supplementary, reference-free signal, what additional GDPR-specific quality properties (completeness, scenario faithfulness, legal correctness, hallucination) does an LLM-as-a-judge surface across the three modes?

RQ3 (Confidence–quality alignment). *Is there a discrepancy between user confidence (RQ1.3) and the document quality actually achieved (RQ2.1, RQ2.2)?* RQ3 is not answered by any single instrument: it requires reading the user-study and document-quality findings against each other, and is therefore taken up only in the Discussion chapter. Going into the study, the implicit prediction was that more LLM support would translate into a better user experience and a more preferred mode: the ordering Chat $\succ$ Form $\succ$ Ask was anticipated, with Chat (the free-form conversational mode where the LLM does the most work) expected to be the overall favorite among novice users. The evaluation that follows tests this prediction against the experience and outputs of 73 novice participants.

1.3 Thesis Structure

The remainder of this thesis is organized into five further chapters and three appendices.

Chapter 2 (Background) defines the GDPR and the Article 30 ROPA obligation and locates this thesis in the related work on LLM-supported legal-document generation. **Chapter 3** (Methodology) introduces ROPAgen and its three interaction modes, then describes the user study: design, instruments, sample sizes, and the analytical approach through which the tool is evaluated. **Chapter 4** (Evaluation and Results) reports the empirical findings from the four user-study instruments and from the document-quality analyses, descriptively first and then inferentially. **Chapter 5** (Discussion) interprets the confidence–quality gap that emerges across the user-study and document-quality streams, draws the design implications for novice-targeting LLM-assisted tooling, and names the study's limitations. **Chap-**

ter 6 (Conclusion) answers RQ1, RQ2, and RQ3 and draws the design implications for LLM-assisted compliance tooling that follow. **Appendices A–C** contain the ROPAgen backend prompts and judge configuration, the verbatim study materials, and the full inferential statistics referenced from the Evaluation chapter.

2 Background

This chapter establishes the legal and methodological context for the thesis. The first two sections introduce the GDPR and the Article 30 obligation to maintain a Record of Processing Activities (ROPA), explaining why a ROPA is both legally required and practically demanding to produce. The third section surveys related work along four lenses — GDPR-specific compliance automation, structured legal-document generation, evaluation methodologies, and user-facing legal systems — and the chapter closes by identifying the gap that this thesis addresses.

2.1 General Data Protection Regulation (GDPR)

The General Data Protection Regulation (GDPR), in force since 25 May 2018, is the European Union's framework for personal-data protection [9]. Its scope is extraterritorial: any organization that processes data on individuals within the EU is subject to it, regardless of its own location. The regulation rests on privacy as a fundamental right under the 1950 European Convention on Human Rights and modernizes the 1995 Data Protection Directive in response to the growth of cloud services, social media, and large-scale data breaches. [9, 8]

Under the GDPR, personal data is any information relating to an identifiable individual, ranging from names and email addresses to biometric data, location information, and web cookies. The regulation distinguishes between the data subject (the individual), the data controller (the entity determining the purposes and means of processing), and the data processor (a third party handling data on behalf of a controller). All processing must adhere to seven principles: lawfulness, fairness and transparency; purpose limitation; data minimization; accuracy; storage limitation; integrity and confidentiality; and accountability. Accountability shifts the burden of proof to the organization, which must produce documented evidence of compliance. [9]

The regulation mandates that processing is only legal if justified by one of six specific bases, most notably through consent, which must be freely given, specific, and unambiguous. Other justifications include contractual necessity, legal obligations, or a legitimate interest that does not override the subject's rights. Furthermore, the GDPR introduces the concept of data protection by design and by default, requiring that privacy considerations be integrated into the initial development phases of any product or service. Large-scale processors or public authorities are often required to appoint a Data Protection Officer (DPO) to oversee these internal processes and serve as a liaison with regulatory bodies [9].

The GDPR grants individuals a defined set of privacy rights, including the rights to be informed, of access, of rectification, and of erasure (often termed the "right to be forgotten"). Data subjects also hold the right to data portability and to protections against automated decision-making and profiling. Compliance is enforced through substantial administrative fines under Article 83 GDPR, which has placed data protection among the core legal and operational concerns of organizations that handle personal data. [9]

2.2 Records of Processing Activities under Article 30 GDPR

Article 30 of the General Data Protection Regulation (GDPR) establishes the obligation for controllers and, where applicable, processors to maintain a Record of Processing Activities (ROPA). The ROPA functions as a core accountability instrument within the GDPR framework, operationalizing the principle that organizations must not only comply with data protection requirements but also be able to demonstrate such compliance to supervisory authorities. Its primary purpose is to provide a structured, transparent, and up-to-date overview of how personal data are processed within an organization, thereby enabling effective oversight, risk assessment, and regulatory control [8, 29].

From a regulatory perspective, the ROPA serves several interrelated objectives.

First, it enhances organizational awareness of data processing operations by requiring a systematic mapping of purposes, data flows, and safeguards. Second, it supports data protection by design and by default, as organizations must explicitly consider data categories, retention periods, and security measures. Third, it enables supervisory authorities to efficiently assess compliance, particularly in the context of audits, investigations, or following personal data breaches. As such, the ROPA is not merely a formal record, but a living compliance document that reflects actual processing practices [8].

Article 30 distinguishes between the obligations of data controllers and those of data processors. Controllers are required to document processing activities carried out under their responsibility, while processors must maintain records of processing performed on behalf of controllers. These records must be maintained in writing, including in electronic form, and must be made available to the competent supervisory authority upon request. Although the GDPR provides a limited exemption for organizations employing fewer than 250 persons, this exemption does not apply where processing is non-occasional, involves special categories of personal data, concerns criminal convictions, or is likely to result in risks to the rights and freedoms of data subjects. Consequently, in practice, many small and medium-sized organizations remain subject to the ROPA requirement [8].

The content of a ROPA is explicitly defined by Article 30 GDPR and is intended to be concise, structured, and easily comprehensible. Excessive or irrelevant information should be avoided, as clarity and usability are essential for both internal governance and external supervision. At a minimum, a ROPA should include the following elements.

- Name and contact details of the data controller (and, where applicable, joint controllers, representatives, and the data protection officer).
- Purposes of the processing activities.
- Categories of data subjects.
- Categories of personal data processed.
- Categories of recipients, including recipients in third countries or international organizations.

- Details of transfers of personal data to third countries or international organizations, including safeguards where applicable.
- Time limits for the erasure of different categories of personal data.
- A general description of the technical and organizational security measures implemented pursuant to Article 32 GDPR.

Where an organization acts as a data processor, the ROPA must additionally reflect the processing carried out on behalf of each controller, including the identity of those controllers, the categories of processing involved, relevant international data transfers, and applicable security measures. In both cases, the ROPA should be sufficiently detailed to provide a faithful representation of processing activities, while remaining accessible to non-technical stakeholders [8].

The GDPR does not prescribe a fixed update cycle for ROPAs; instead, it requires that records remain accurate and up to date. Consequently, any substantive change in processing purposes, data categories, recipients, or safeguards should trigger a review and update of the ROPA. Many organizations delegate the responsibility for maintaining the ROPA to a data protection officer, where one is appointed, to ensure consistency, avoid duplication of effort, and reduce the risk of omissions [8, 24].

Finally, Article 30 emphasizes the external relevance of ROPA. Organizations must be prepared to present their records to supervisory authorities upon request, and such requests may be accompanied by demands for supplementary documentation, including privacy notices, consent records, or contractual arrangements. In this sense, ROPA operates as a foundational compliance artifact, anchoring broader data protection governance and demonstrating an organization's commitment to accountability under the GDPR [8, 24].

2.3 Related Work

Prior work relevant to this thesis can be grouped along four lenses that mirror the design choices of the present study. The first concerns *domain*: research that targets GDPR compliance specifically and the automated production of regulatory artifacts such as ROPA. The second concerns *task*: architectures for generating structured, template-bound legal documents in adjacent jurisdictions. The third concerns *evaluation*: how the legal-AI literature measures the quality of generated documents, increasingly through a combination of surface-overlap metrics, semantic-similarity metrics, and either expert or LLM-based judgment. The fourth concerns the *user*: how legal AI is presented to non-expert audiences and whether it is accessible to people without prior legal training. Reviewing the literature through these four lenses makes clear what has been studied, what has not, and where the present thesis sits.

2.3.1 GDPR Compliance Automation

The closest prior work to this thesis targets the automation of GDPR compliance documentation directly. von Schwerin et al. [29] address the generation of data categories (e.g., contact information, payment data) for Records of Processing Activities, observing that existing legal benchmarks such as LawBench or LegalBench primarily evaluate static knowledge recall rather than the ability to produce documents that adhere to the precise formatting and terminology required by the GDPR. Their study compares four open-source models (Qwen2-7B, Meta-Llama-3-8B, zephyr-7b-beta, and orca_mini_v7_7b) against GPT-4 across four experiments—basic performance, model-specific prompting, few-shot prompting, and Retrieval-Augmented Generation (RAG)—and assesses outputs using a weighted Model Utility Score covering latency, structure and style, grammar, validity, and contextual understanding. The finding that open-source models, particularly Qwen2-7B, can match or exceed GPT-4 on structured ROPA outputs validates the use of locally deployable models for privacy-sensitive compliance work, and is the empirical foundation on which ROPAgen builds. The study also documents persistent risks—ambiguous legal language, jurisdictional variation, hallucination, and the resulting need for a "human-in-the-loop"—that motivate examining how real users behave when asked to produce ROPA documents with such tools.

Abualhaija et al. [1] approach the GDPR from a complementary angle, extracting technical software requirements for specific data subject rights (right of access, right to portability) from the regulation and supplementary legal sources. Their XTRAREG pipeline uses GPT-3.5 and GPT-4o together with RAG and prompting strategies (Zero-Shot Learning, Chain-of-Thought) to generate requirements, validated against a manual baseline of 108 expert-extracted reference requirements. The work confirms two patterns that recur throughout this thesis: RAG meaningfully improves both lexical and semantic alignment with expert references, and high BERTScores can coexist with low BLEU when the model captures the meaning of a requirement without matching its wording. Their reported failure modes—incomplete coverage, perspective drift (writing as the data subject rather than as the system), and citation of non-binding recitals—further reinforce the case for evaluating LLM-generated GDPR artifacts against a fixed reference rather than trusting model output at face value.

Together, these two works establish GDPR documentation as a viable LLM target while leaving the user-side question unanswered: how do non-experts fare when asked to produce such documents themselves? ROPAgen, and this thesis, extends this line of work by testing the user-facing realization of an open-source GDPR pipeline rather than the model performance in isolation.

2.3.2 Structured Legal-Document Generation

A second body of work targets the broader problem of producing structurally coherent legal documents from limited user input, exploring the architectural strategies—retrieval, wrappers, fine-tuning—that constrain LLMs to follow rigid legal templates. Nigam et al. [23] address private legal-document drafting in the Indian system through their Model-Agnostic Wrapper (MAW), a two-phase framework in which the model first generates a structured list of section titles (which the user can manually refine) and then iteratively fills each section, using a vector database (ChromaDB) to summarize prior sections and maintain long-form coherence. The system is exposed through a Human-in-the-Loop interface and evaluated on the Vidhik-Dastaavej dataset of 489 anonymized documents using ROUGE, BLEU, METEOR, BERTScore, an automatic G-Eval LLM judge, and a Likert-scale expert assessment. The headline finding—that direct fine-tuning on small datasets degrades quality, while a structured wrapper around a smaller open-source model can match GPT-4o—argues for orchestration over scale.

Lin and Cheng [20] take the opposite architectural route, fine-tuning an open-source BLOOM-560M model locally on 74,823 unlabeled Traditional Chinese fraud verdicts (Taiwan Judicial Yuan, 2011–2021) without Chinese word segmentation. Their structural-correctness metric defines a fixed order of legal constitutive elements—Subject of Crime, Subjective Legal Element, Act, Victim, Causation, Result of Hazard—and checks whether the generated draft follows that order, supplemented by perplexity and training/validation loss. The work demonstrates that domain-specific structured drafting is feasible without cloud infrastructure on consumer-grade hardware, addressing the same privacy concerns that motivate locally deployed GDPR tooling. Li et al. [18] push the structural framing further with CaseGen, a benchmark that decomposes Chinese case-document drafting into four sequential stages—defense statements, trial facts, legal reasoning, and judgment results—over 500 expert-annotated cases. Their finding that general-domain LLMs (GPT-4o, Qwen2.5) outperform legal-specific models like ChatLaw or LexiLaw, while still scoring below 6 out of 10 overall, suggests that careful staging of the generation task matters at least as much as legal-domain pretraining. Together, these three works frame structured legal-document generation as primarily an architectural problem and explore wrappers, fine-tuning, and multi-stage decomposition as alternative solutions. ROPAgen takes a related but distinct stance: rather than comparing architectures, it fixes the architecture and varies the *interaction modality* (Form, Ask, Chat), asking what the user-facing surface contributes once the underlying model is held constant.

2.3.3 Evaluation Methodologies for Legal AI

A third strand of the literature concerns how generated legal documents are assessed, and a clear methodological pattern is emerging: dual evaluation that combines automated NLP metrics with either expert annotation or LLM-as-a-Judge scoring. Su et al. [30] formalize this for Chinese judgment-document generation through JuDGE, a benchmark of 2,505 fact-judgment pairs supplemented by over 55,000 statutory articles and 103,000 past judgments. The framework evaluates outputs along four dimensions—Penalty Accuracy (normalized absolute difference on prison terms and fines), Convicting Accuracy (Recall, Precision, F1 over charges), Referencing Accuracy (correctness of statutory citations), and Documenting Similarity (METEOR and BERTScore against ground truth)—explicitly arguing that lexical metrics alone cannot capture the legal reasoning that distinguishes a sound judgment from a plausible-sounding one. Their results across direct generation, in-context learning, supervised fine-tuning, and Multi-source RAG show that even the strongest pipelines remain visibly short of professional ground truth, and that semantic-similarity metrics behave as proxies rather than ceilings.

Vayadande et al. [32], viewed from the methodological angle, illustrate the second template: comparative quantitative benchmarking against existing market tools (Kira Systems, Evisort) using accuracy percentages, time-efficiency in hours, and document-processing speed, supplemented by a debugging phase in which legal experts contribute balanced comments that refine the system. The reported 93–97 % accuracy and reduction of document time from 20 to 5 hours sit alongside an acknowledgment of remaining failures on judicial nuance and on input-quality dependencies, foreshadowing the same tension between aggregate metrics and qualitative caveats that emerges in this thesis. Together, JuDGE and Vayadande et al. converge on a shared template: surface-overlap metrics (BLEU, ROUGE, METEOR), semantic-similarity metrics (BERTScore, sentence embeddings), and a complementary qualitative judgment layer—human expert or LLM judge—that catches what the lexical and semantic metrics miss. This is the methodological template adopted in the present study for evaluating ROPAgen's three modes against a single expert reference document.

2.3.4 User-Facing Legal AI Systems

The fourth lens concerns how legal AI is presented to non-expert users, where prior work is markedly thinner. Surya et al. [31] target the accessibility gap directly with an AI-powered interactive legal chatbot for the Indian Department of Justice, aiming to democratize legal information for citizens facing complex jargon and the cost of professional counsel. Their three-tier architecture—a JavaScript single-page frontend, a Python Flask orchestrator handling translation, and an LLM service running LLaMA 3.1 via Ollama—supports real-time legal queries, document summarization (e.g., First Information Reports), and multilingual interaction by translating Hindi queries to English for the LLM and back. The system also surfaces citation links to government portals, addressing the trust problem that arises whenever non-experts consume LLM output on a high-stakes topic. Notably, however, the paper evaluates these features through illustrative use cases rather than a controlled user study, leaving open how non-expert users actually interact with such a system in practice.

Vayadande et al. [32], considered from the user-facing angle rather than the methodological one, fit the same pattern. Their assistant integrates ChatGPT 3.5 with OpenAI embeddings, LangChain, and a vector retrieval stack to support legal-document generation, comprehension, and abnormality detection, framing the tool as a means of reducing the practical effort of legal work. The system is again validated through accuracy and efficiency comparisons rather than through evidence of how non-expert users navigate the interface, interpret the output, or judge their own confidence in it. Both works establish that accessibility, multilingual support, and citation transparency are necessary properties of user-facing legal systems, but stop short of empirically measuring whether the resulting tools serve the non-experts they target.

2.4 Summary

Under the GDPR, accountability is a documented obligation rather than a stated principle: organizations must demonstrate compliance to supervisory authorities, and the Record of Processing Activities under Article 30 is one of the principal artifacts through which that demonstration is performed. The content of a ROPA is rigidly specified (controller details, purposes, lawful bases, categories of data and recipients, retention periods, and technical and organizational measures), yet its substance is project-specific prose that has to be written from a blank page and kept current as processing activities change. Maintaining a ROPA is therefore a sustained cost rather than a one-off task, and one that falls disproportionately on small and medium organizations that lack dedicated legal or data-protection staff. This is the practical gap into which an LLM-assisted ROPA tool such as ROPAgen is built.

Across the four lenses surveyed in the related-work section — *domain* (GDPR-specific compliance automation), *task* (structured legal-document generation more broadly), *evaluation methodology* (the emerging dual-evaluation template of NLP metrics paired with expert or LLM-as-a-Judge scoring), and *user* (legal AI presented to non-expert audiences) — the literature converges on three established points. LLM-based legal-document generation is technically feasible. Architectural choices — retrieval-augmented generation, model wrappers, multi-stage decomposition, local fine-tuning — measurably improve output. And dual evaluation combining surface-form and semantic NLP metrics with either expert annotation or LLM-as-a-Judge scoring has become the de facto template for assessing the resulting documents.

What remains absent is the intersection of three concerns. No prior work simultaneously (a) targets GDPR-specific structured output, (b) exposes the system to *novice users* without legal training, and (c) measures the gap between user confidence and actual document quality under varying levels of LLM support. This thesis fills that gap by holding the underlying ROPAgen pipeline constant, varying the interaction modality across Form, Ask, and Chat, and applying the dual-evaluation template from the legal-AI literature to a within-subjects user study against a single shared scenario. The instruments, sample, and analysis plan through which that gap is closed are described next in Chapter 3.

3 Methodology

This chapter is divided into two complementary parts. The first introduces ROPAgen, the tool under investigation, covering each of its three interaction modes and the corresponding prompts submitted to the LLM. The second details the design of the user study, including participants, instruments, sample sizes, and the analytical approach through which the tool is evaluated. Together, these sections establish the scope of what is measured and the means by which it is assessed, prior to any presentation of results.

3.1 ROPAgen

The tool evaluated in this study is ROPAgen [14], a web-based application that assists users in drafting GDPR Article 30 Records of Processing Activities through three different levels of LLM support. ROPAgen was developed by the author of this thesis at the Institute of Databases and Information Systems (DBIS), University of Ulm, prior to the present evaluation; this section describes the tool as it stood at the time of the study.

ROPAgen exists to ease the burden of ROPA drafting: the document has rigid structural requirements (Art. 30 GDPR) but most of its substance is project-specific prose that has to be written from a blank page. The tool offers three interaction modes — Form, Ask, and Chat — which offer different levels of LLM support;

Form mode places the user in full control: they select pre-defined options and fill in structured fields, with LLM support available only on demand via a suggestion button. **Ask** mode is a hybrid: users work through the same structured form section by section, but an LLM chatbot is available to answer free-form questions and explain each section as they go. **Chat** mode is fully autonomous: the user describes their requirements in free-form conversation, and the LLM independently determines which structured fields to populate and drafts the ROPA content on its own. This thesis evaluates the three modes: how the level of LLM support shapes user experience and how it shapes the documents that are actually produced. The remainder of this section describes the internal representation of a ROPA document in the tool, walks through the three modes from the user's perspective, and details the LLM backend that all three modes share.

Table 3.1: The three ROPAgen interaction modes at a glance, described in terms of who populates which part of the shared `DocumentData` object that the final-generation step (`finalropa`) receives.

Mode	LLM's role in the section	Predefined booleans set by	Free-text fields written by
Form	On-demand suggestion only (the *AI Suggest* button)	User (clicks), or LLM if *AI Suggest* is used	User, or LLM if *AI Suggest* is used
Ask	Explainer only; emits no structured update	Nobody — left at the default `false`	User only
Chat	Conversational extractor; emits `DATA_UPDATE` blocks merged into `DocumentData`	LLM (via `DATA_UPDATE`)	LLM (via `DATA_UPDATE`)

3.1.1 ROPA Documents in ROPAgen

Internally, ROPAgen represents a ROPA as a single structured `DocumentData` object organized around eight thematic groups that together cover the disclosures required under Art. 30(1) GDPR:

1. **Organization and controller** — name, postal address, e-mail, phone number, and the contact details of the data protection officer and any external representative.
2. **Legal basis** — Boolean flags for each lawful-basis option in Art. 6(1) GDPR (consent, contract, legal obligation, legitimate interest), the German national-

law bases relevant to the study scenario (§ 26 BDSG, § 28b IfSG, § 257/239 HGB, § 147 AO), and a free-text fallback for other grounds.

3. **Data categories** — nine grouped categories (personal master data, employment, finance, health, legal, behavioral, documentation, vehicle, and other) each with predefined Boolean sub-fields such as surname, date of birth, working hours, or holiday entitlement.

4. **Data sources** — Boolean flags for common upstream systems (Office 365, Azure, Google Workspace, AWS, e-mail, etc.) plus a free-text field for sources not covered by the predefined list.

5. **Person categories** — affected persons, internal recipients, external recipients within the EU, and authorized personnel, each populated through Boolean role tags with an optional free-text override.

6. **Retention periods** — Boolean flags for the standard retention triggers (after termination of employment, after termination of contract, upon revocation of consent), a free-text field for the concrete deletion period, and flags for legal-retention exceptions.

7. **Free-text fields** — three short prose fields covering the purpose of processing, the technical and organizational measures in place, and any additional information that does not fit elsewhere.

8. **Metadata** — the document identifier, title, and natural language in which the final ROPA should be generated.

This structured form is the tool's working representation while the user is drafting. The final ROPA itself is rendered as a Markdown document — with optional PDF export through a server-side rendering route — that the LLM produces in a single generation step from the assembled `DocumentData` object. A complete example of the resulting document, the reference baseline against which all participant outputs are scored by the NLP metric pipeline, is reproduced in Appendix A.3.

3.1.2 The Three Interaction Modes

ROPAgen offers three generation modes, each with a different level of LLM support: Form has the least LLM support, Chat is fully autonomous, and Ask is a hybrid mode that sits in the middle. The subsections that follow describe each mode in detail; Table 3.1 summarizes them at a glance. All three act on the same internal `DocumentData` schema and, when the user clicks the final *Generate* button, hand that schema to the same prose-generation prompt (`finalropa`); the output ROPA is therefore produced by an identical generation step in all three conditions, and only the path the user takes to populate `DocumentData` differs between modes.

Form Mode

Form mode offers the lightest LLM support: the user sees and does everything. Every field of the `DocumentData` schema is visible and edited by hand, laid out as a linear questionnaire of checkboxes and short text fields across the eight thematic groups (see the Form mode screenshot in Appendix A.2.1). The LLM is invoked only when the user explicitly asks for it: each section offers an optional *AI Suggest* button that asks the LLM, given the document's current state, to propose values for that section's fields; the suggestions populate the form but remain freely editable, and the user is under no obligation to use them. Once the user is satisfied with the form, clicking *Generate* sends the assembled `DocumentData` object to the `finalropa` prompt, which synthesizes a complete ROPA document in formal prose. Form mode therefore exposes the entire structure of a ROPA upfront and uses the LLM only at the margins. In short, the user populates the predefined boolean scaffolding and the free-text fields directly, and the LLM is involved only as an optional per-section suggestion engine and at the final prose-synthesis step.

Ask Mode

Ask mode pairs each section with an LLM chat panel and a free-form text field; there are no checkboxes or pre-defined options. The user can ask the LLM, section by section, to explain what the section requires or to suggest how to fill it, but the user is responsible for the final wording entered into the section's text field. Each chat panel (see Appendix A.2.2) opens with the LLM explaining the section's purpose and asking the user for the relevant information; subsequent turns maintain the conversation history for that section and ask follow-up questions. The model is given the current `DocumentData` state alongside the user's message so its explanations can be context-aware, but its reply is conversational only: it does not return a structured update block and no field is auto-populated. The predefined boolean flags of the `DocumentData` schema are not touched by either the user or the LLM in Ask mode. The final *Generate* step is identical to Form mode: the same `finalropa` prompt receives the same `DocumentData` object. The `DocumentData` object that reaches the final-generation step in Ask mode therefore carries only the user's per-section free-text writing; the predefined boolean scaffolding is left at its default. Any section the user does not type into reaches the generator with no scaffolded content to work from. In short, the user only chats with the LLM and populates the free-text `other` field of each section themselves; the predefined boolean scaffolding is never touched, so the final-generation step has no scaffolded content to work from beyond the user's own typing.

Chat Mode

Chat mode hands control to the LLM. The user does not see or fill the structured fields at all; instead, after entering the organization's metadata (name, contact information, and title of the processing activity), the user describes their processing in their own words, and the LLM decides which structured fields to populate from those descriptions (see Appendix A.2.3). The user navigates section by section through a chat interface only; the structured form is hidden from view. The LLM is explicitly instructed to extract structured data from the conversation and to emit it as a `DATA_UPDATE` block alongside its natural-language reply; the front-end parses that block and merges it into the shared `DocumentData` object. A mentioned vendor name therefore flips the corresponding data-source boolean, a mentioned data category populates the corresponding boolean, and any specific item that does not match a predefined option lands in the section's `other` free-text field. The user therefore never directly interacts with the underlying `DocumentData` object, although the same object is being built up under the hood. The final *Generate* step is once again identical to the other two modes: the same `finalropa` prompt receives the same `DocumentData` object. Chat mode is therefore a conversational-first workflow in which both the scaffolded booleans and the free-text fields are populated by the LLM, and only the final generation step is shared with the other modes. In short, the user describes the processing in their own words and the LLM populates both the predefined boolean scaffolding and the free-text fields on the user's behalf; the user never directly interacts with the structured form.

3.1.3 LLM Backend

All three modes are backed by the same model: Mistral Large 3 (model identifier `mistralai/mistral-large-2512`), accessed through the OpenRouter API gateway via the Vercel AI SDK. The choice of Mistral rests on three considerations relevant to a GDPR-compliance tool. First, Mistral AI is a European model provider [22], which keeps the generation workload aligned, in spirit, with the data-protection regime ROPAgen is helping users to document. Second, Mistral AI has a public commitment to open-weight releases such as Mistral 7B [12] and Mixtral 8x7B [13]; although the Mistral Large 3 variant used here is the company's commercial flagship rather than an open release, this lineage offers a credible path to a fully self-hosted deployment. Third, the Mistral family has shown competitive performance on the long-form reasoning and structured-output tasks relevant to legal-document generation [12]. Two temperature regimes are used (Table 3.2). Conversational and suggestion calls — per-section LLM suggestions in Form mode, dialogue turns in Ask and Chat modes, and the silent data-extraction calls in Chat mode — use temperature 0.2 and top-p 0.5, which permits variation in phrasing while keeping the responses on-task. The final ROPA generation step, in contrast, uses temperature 0.05 and top-p 0.05, which makes the output close to deterministic given the same `DocumentData` input.

Table 3.2: ROPAgen LLM call parameters. The same model (`mistralai/mistral-large-2512`) is used for both regimes.

Call type	Temperature	top-p
Conversational / suggestion	0.2	0.5
Final ROPA generation	0.05	0.05

Crucially, the prompt used in the final generation step is fixed across all three modes: the only thing that differs between a Form-mode output and a Chat-mode output is the contents of the `DocumentData` object that has been built up along the way. The full text of the backend prompts is reproduced in Appendix A.1, as the counterpart to the LLM-as-a-judge configuration recorded in Appendix A.4.

Whether the differing paths to `DocumentData` produce documents of differing quality, and whether those quality differences are visible to the user filling out the form, are the questions taken up by the user study described next.

3.2 User Study

The user study is a structured survey: after each interaction mode, participants completed a fixed battery of questionnaire items, and after all three modes they completed a final ranking task and a demographics block. For consistency, the remainder of this thesis refers to the data collection as the *user study* rather than the survey.

3.2.1 Study Design

The study followed a within-subjects experimental design with counterbalanced ordering. The recruited sample consisted of $N = 73$ university students, all novice users with no prior GDPR experience, and each session lasted approximately 90 minutes per participant. Counterbalancing was implemented through a Balanced Latin Square: the three modes (Form, Ask, Chat) were arranged into six ordered sequences, and participants were randomly assigned to one of the six groups, with each participant completing all three modes in their group's predetermined order. This design controls for order effects, learning effects, and fatigue. Demographic information was collected at the end of the session, after the three modes and the final ranking. Figure 3.1 situates the per-session flow within the broader project arc, from the development of ROPAgen through data collection to the analysis pipeline reported in the chapters that follow.

3.2.2 Task Scenario

A single standardized scenario was used across all participants and modes to ensure consistency. The scenario described a relatable workplace data processing

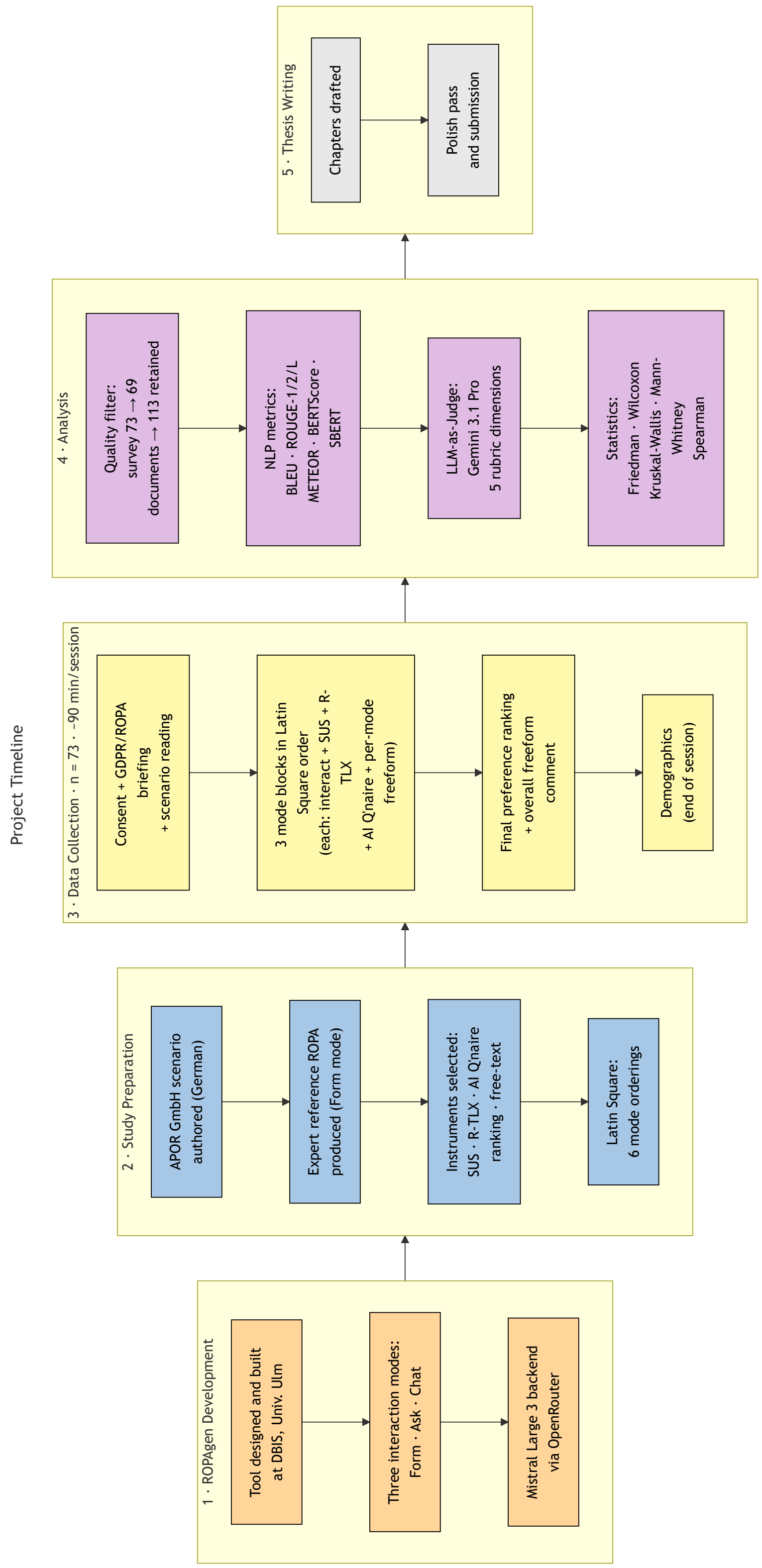


Figure 3.1: Project timeline. Five phases run left-to-right: ROPAgen development at DBIS; study preparation (scenario authoring, reference ROPA, instrument selection, Latin-Square construction); per-participant data collection in 90-minute sessions ($n = 73$); analysis (quality filter, NLP metric pipeline, LLM-as-Judge, statistical layer); and thesis writing. Demographics were administered at the end of each session rather than at the start.

activity at *APOR GmbH*, a fictional small software company in Ulm with approximately 50 employees: the two processing activities under consideration were the administration of employee working time and leave, and the management of parking authorizations for the company car park. The scenario covers all GDPR Article 30 fields and is accessible to participants without legal expertise; it is reproduced verbatim in both German (the version actually shown to participants) and English in Appendix B.1. The LLM backing all three modes (Section 3.1) was held constant throughout the study.

3.2.3 Participants

Recruitment yielded $N = 73$ university students. Participants were screened for minimal to no GDPR knowledge (expected and required), prior AI tool usage experience (documented), general technology comfort level (assessed), and the absence of any professional legal or compliance background. As part of onboarding, participants received a brief standardized introduction to GDPR and ROPA concepts before beginning the tasks.

3.2.4 Data Availability and Sample Sizes

Due to technical data loss and incomplete submissions during data collection, the effective sample sizes differ across analysis streams. The canonical sample sizes used throughout this thesis are summarized below.

- **User study, recruited N = 73 / valid N = 69** — 73 participants completed the study; 69 retained after dropping four records with invalid consent (`p_0001 = -1`). All 69 valid participants completed all three modes, so per-mode sample size, matched-triple user-study subset, and valid N are all **n = 69**. The matched-triple n = 69 is used for within-subjects Friedman and Wilcoxon signed-rank tests on SUS, R-TLX, confidence items, and the perceived AI support composite.
- **Documents, 113 generated ROPAs** — distributed across modes as 37 (Form), 37 (Ask), and 39 (Chat). The full $n = 113$ sample is used throughout: NLP metric scores and per-mode completion time are analyzed inferentially with

Kruskal–Wallis and Bonferroni-corrected Mann–Whitney U as the between-groups family; LLM-as-a-judge scores are reported descriptively on the same sample.

The canonical figures used in the inferential analyses are **n = 113** for NLP metric and completion-time comparisons and **n = 69** for user-study instruments.

Descriptive statistics (means, medians, standard deviations) are computed on the full available N per mode for each analysis stream. Inferential tests on NLP metric and per-mode completion-time measures operate on the full $n = 113$ sample as a between-groups comparison; inferential tests on user-study instruments operate on the matched $n = 69$ within-subjects sample. The LLM-as-a-judge scores are reported descriptively only and are not subjected to inferential testing (Section 3.5).

3.2.5 Data Collection

Quantitative measures were collected after each mode. Table 3.3 summarizes the four user-study instruments, the post-experiment preference ranking, and the post-session demographics questionnaire.

Table 3.3: Instruments collected during and after the user study.

Instrument	What it captures
SUS	Perceived usability
R-TLX	Cognitive load (six subscales)
Perceived AI Support	LLM usefulness, trust, transparency, confidence
Free-text feedback	Qualitative comments per mode
Mode-preference ranking	Post-experiment overall preference
Demographics	Age, gender, degree, field, prior GDPR/AI familiarity

The System Usability Scale (SUS) [4] was administered as ten standardized questions on a 5-point Likert scale; the Raw Task Load Index (R-TLX) [11, 10] captured six cognitive load dimensions on the standard NASA 21-point scale (1 = lowest demand, 21 = highest), as Raw TLX retains the unweighted subscale ratings on their original NASA scale. The full text of the 10 SUS items and the 6 NASA R-TLX subscale descriptions are reproduced in Appendix B.2. In addition, participants completed a six-item Perceived AI Support instrument on a 5-point Likert scale:

1. *Perceived Usefulness:* “The AI was helpful in creating this document.” (*“Die KI war bei der Erstellung dieses Dokuments hilfreich.”*)
2. *Efficiency:* “The AI helped me to complete the document faster.” (*“Die KI hat mir geholfen, das Dokument schneller fertigzustellen.”*)
3. *Transparency:* “I understood why the AI suggested specific GDPR fields.” (*“Ich habe verstanden, warum die KI bestimmte DSGVO-Felder vorgeschlagen hat.”*)
4. *Trust:* “I trust that the AI made correct suggestions.” (*“Ich vertraue der KI, dass sie korrekte Vorschläge gemacht hat.”*)
5. *Task-Specific Confidence (primary confidence measure):* “I am confident that the AI identified the correct legal basis.” (*“Ich bin zuversichtlich, dass die KI die richtige Rechtsgrundlage identifiziert hat.”*)
6. *Intention to Use:* “I would use this mode again.” (*“Ich würde diesen Modus wieder verwenden.”*)

Items were administered in German (the parenthetical text above reproduces the verbatim wording shown to participants); the English glosses are provided for the international reader. Item 5 (Task-Specific Confidence) is treated as the **primary confidence measure** throughout the analysis: it is the only item that asks participants to judge the substantive correctness of the LLM’s output (the legal basis under Article 6 GDPR), and is therefore directly comparable to the document-quality signals from the NLP and judge pipelines. The remaining five items — Item 1 (Perceived Usefulness), Item 2 (Efficiency), Item 3 (Transparency), Item 4 (Trust), and Item 6 (Intention to Use) — capture distinct facets of *perceived AI support* rather than confidence in output correctness, and are reported as a secondary composite. Free-text qualitative feedback was also collected after each mode. Two timing measurements were retained: total session duration from the user-study questionnaire (one value per participant) and per-mode completion time emitted by ROPAgen at the end of each session and embedded in every generated document (one value per participant per mode, in seconds). After completing all three conditions, participants ranked all three modes by overall preference in a post-experiment ranking task.

3.3 NLP Metrics

Following the dual-evaluation template adopted by Su et al. [30] and Vayadande et al. [32], both surface-form and semantic measures are reported alongside an LLM-as-Judge rubric. All participant-generated documents are compared against a single reference document using a suite of automated NLP metrics. After validation for errors and completeness, 113 documents were retained for analysis out of a theoretical maximum of 219 (73 participants $\times$ 3 modes). The reference document was created using Form mode (Section 3.1), applying the same standardized scenario presented to participants, and is reproduced in full in Appendix A.3. It is treated as the authoritative baseline, a root-of-trust representing a careful, complete application of the most structured interaction mode. This introduces an inherent framing effect: NLP metrics measure similarity to Form-style output, which may disadvantage the more conversational outputs of Ask and Chat modes independently of their substantive quality.

BLEU, ROUGE, METEOR, BERTScore, and SBERT are all computed at both the whole-document level and the per-chapter level, with ModernBERT-base as the embedding backbone for BERTScore and SBERT. At the chapter level, SBERT is additionally run with the smaller `all-MiniLM-L6-v2` sentence-embedding model as a robustness check on the primary ModernBERT pass; the MiniLM pass is chapter-only because its 512-token input limit excludes whole-document scoring. Results at every available granularity are aggregated by mode.

The five NLP metrics summarized in Table 3.4 (BLEU [25], ROUGE [19], METEOR [2], BERTScore [35], and SBERT [27]) are defined formally below. Each is computed against the single expert-authored reference document; interpretive framing of the resulting values is deferred to the Evaluation chapter.

3.3.1 BLEU

Bilingual Evaluation Understudy — n-gram precision with brevity penalty [25]. BLEU rewards exact word-sequence overlap with the reference: a high BLEU score means the candidate reuses the reference's wording almost verbatim, while a low BLEU score can equally indicate a legitimate paraphrase or genuinely missing content;

Table 3.4: NLP metrics used in the document-quality evaluation. Higher is better on all metrics.

Metric	What it measures
BLEU	n-gram precision with brevity penalty
ROUGE-1/2/L	Unigram / bigram / LCS overlap (F1)
METEOR	Precision–recall with synonym and stem matching
BERTScore P/R/F1	Token-level semantic similarity (ModernBERT)
SBERT cosine similarity	Sentence-embedding cosine similarity (ModernBERT; chapter-level M

the metric does not distinguish the two on its own.

BLEU computes the geometric mean of n-gram precisions p_n between candidate and reference, multiplied by a brevity penalty (BP) that discourages outputs much shorter than the reference. With uniform weights $w_n = 1/N$, candidate length c, and closest reference length r:

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \quad \mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r. \end{cases} \tag{3.1}$$

This work uses $N = 4$ (matching unigrams through 4-grams).

3.3.2 ROUGE-1, ROUGE-2, ROUGE-L

Recall-Oriented Understudy for Gisting Evaluation — unigram, bigram, and longest-common-subsequence overlap (F1) [19]. ROUGE measures how much of the reference's vocabulary, bigrams, or longest common subsequence the candidate covers. A high ROUGE score signals strong lexical recall against the reference; a low score indicates that key reference content is absent from the candidate.

ROUGE reports the F1 of n-gram or longest-common-subsequence (LCS) overlap between candidate X and reference Y. ROUGE-1 counts unigrams, ROUGE-2 counts bigrams, and ROUGE-L counts the LCS. For ROUGE-N:

$$P_N = \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(X)|}, \quad R_N = \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(Y)|}, \quad F1_N = \frac{2 P_N R_N}{P_N + R_N}. \tag{3.2}$$

The same precision–recall–F1 form applies to ROUGE-2:

$$F1_{\text{ROUGE-2}} = \frac{2P_2R_2}{P_2 + R_2}. \tag{3.3}$$

For ROUGE-L, n-gram counts are replaced by the length of $\mathrm{LCS}(X, Y)$:

$$P_L = \frac{|\mathrm{LCS}(X,Y)|}{|X|}, \quad R_L = \frac{|\mathrm{LCS}(X,Y)|}{|Y|}, \quad F1_L = \frac{2P_LR_L}{P_L + R_L}. \tag{3.4}$$

The reported ROUGE values are the F1 scores in each case.

3.3.3 METEOR

Metric for Evaluation of Translation with Explicit ORdering — precision–recall balance with synonym matching [2]. METEOR balances precision and recall, credits synonym and stem matches, and penalizes fragmented overlap where shared words appear in scattered positions. A high METEOR score signals that the candidate covers the reference's content with vocabulary that need not match it verbatim; a low score indicates either missing content or surface forms that the synonym and stem matcher could not align.

METEOR is the weighted harmonic mean of unigram precision P and recall R (with recall up-weighted by α), multiplied by a fragmentation penalty that grows with the number of disjoint matching chunks ch relative to the number of matched unigrams m:

$$\mathrm{METEOR} = \underbrace{\frac{P \cdot R}{\alpha P + (1-\alpha)R}}_{F_{\mathrm{mean}}} \cdot \underbrace{\left(1 - \gamma\left(\tfrac{\mathrm{ch}}{m}\right)^{\beta}\right)}_{\text{penalty}}. \tag{3.5}$$

Standard values $\alpha = 0.9$, $\beta = 3$, $\gamma = 0.5$ are used; matching is performed against exact, stemmed, and synonym/paraphrase matches via WordNet.

3.3.4 BERTScore

Precision, Recall, F1 — token-level semantic similarity using ModernBERT embeddings [35]. BERTScore measures similarity between the contextual embeddings of candidate and reference tokens rather than their surface forms. *Precision* captures how many candidate tokens find a close meaning-match in the reference and therefore penalizes spurious or invented content; *recall* captures how many reference tokens are covered and therefore penalizes omission; *F1* is the harmonic mean and requires both to be reasonably high.

BERTScore replaces surface-form n-gram matching with cosine similarities between contextual token embeddings produced by a pretrained transformer. The embedding model used here is ModernBERT-base[1] [33], chosen for its long context window of 8192 tokens, which lets a whole ROPA be embedded without truncation, and for its popularity and positive reception in the Hugging Face community. With L2-normalized embeddings $\mathbf{x}_i$ for reference tokens and $\hat{\mathbf{x}}_j$ for candidate tokens:

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.6}$$

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.7}$$

$$F1_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}. \tag{3.8}$$

3.3.5 SBERT Cosine Similarity

Sentence-BERT — document- and chapter-level semantic similarity using sentence embeddings [27]. SBERT compresses a passage into a single fixed-length embedding and reports the cosine similarity between the candidate and reference embeddings. Unlike BERTScore, which decomposes token-level matches into Precision, Recall, and F1, SBERT produces a single symmetric similarity score per comparison: the cosine of the angle between two compressed vectors does not distinguish candidate-into-reference coverage from reference-into-candidate coverage, so no separate precision and recall values are reported. SBERT is computed at both granularities: one whole-document score per ROPA against the full reference, and one chapter score per candidate chapter against the corresponding reference chapter. The primary encoder for both passes is ModernBERT-base[2], whose 8192-token context window accommodates a full ROPA without truncation. At the chapter level,

[1] `https://huggingface.co/answerdotai/ModernBERT-base`
[2] `https://huggingface.co/answerdotai/ModernBERT-base`

a secondary pass with `all-MiniLM-L6-v2`[3] is run over the same six chapters as a robustness check on the primary ModernBERT pass; MiniLM has a 512-token input limit that excludes whole-document use. Both models were chosen for their popularity and positive reception in the Hugging Face community.

SBERT produces a single fixed-length embedding per passage by mean-pooling token embeddings from the underlying transformer (here, ModernBERT or MiniLM). Passage similarity is the cosine of the angle between two embeddings $\mathbf{e}(c_1)$ and $\mathbf{e}(c_2)$:

$$\mathrm{SBERT}(c_1, c_2) = \frac{\mathbf{e}(c_1)^\top \mathbf{e}(c_2)}{\|\mathbf{e}(c_1)\| \, \|\mathbf{e}(c_2)\|}. \tag{3.9}$$

3.3.6 Score Implications

The metrics above are read in combination, not in isolation, since each captures a different facet of similarity to the reference. Four pairings recur in the Evaluation chapter. A high lexical score (BLEU, ROUGE, or METEOR) combined with a low semantic score (BERTScore F1 or chapter-level SBERT) indicates mechanical surface copying that does not align in meaning, whereas the reverse pattern (low lexical, high semantic) is the signature of a legitimate paraphrase that preserves content with different wording. Within BERTScore, high recall paired with low precision points to a verbose candidate that covers the reference but adds material the reference does not contain, while low recall paired with high precision points to a conservative candidate that omits required content without fabricating new material. Finally, a high document-level score on the surface-form metrics, BERTScore, or SBERT paired with uneven chapter-level scores under the same metric signals that the aggregate is hiding per-section weakness; the chapter-level SBERT scores in particular serve as a per-section diagnostic alongside the whole-document SBERT summary. These heuristics are applied throughout the Evaluation chapter.

[3] `https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2`

3.4 LLM-as-a-Judge Evaluation

Documents are additionally evaluated using an LLM-as-a-judge approach, in which **Gemini 3.1 Pro Preview** (`google/gemini-3.1-pro-preview`, $T = 0$) serves as an automated evaluator assessing GDPR-specific quality. The judge is positioned as a *supplementary* evaluator: it is a single-pass, single-model assessment without inter-rater replication or legal-expert validation, and its scores are reported alongside, not in place of, the NLP metrics. The reference-based NLP metrics remain the primary RQ2 evidence; the judge complements them by providing a reference-free signal that captures GDPR-specific quality properties (legal correctness, hallucination, scenario faithfulness) which similarity-based metrics cannot directly observe. The judge evaluates each document across five metrics on a 1–5 scale:

1. **Completeness:** Coverage of all required Article 30(1) fields (controller identity, DPO contact, purposes, legal basis, data categories, recipients, retention periods, technical and organizational measures)
2. **Scenario Faithfulness:** Adherence to the specific details of the study scenario rather than generic or hallucinated content
3. **Legal Correctness:** Appropriate use of GDPR terminology and accurate citation of relevant articles
4. **Hallucination (inverse):** Absence of invented facts not present in the scenario (scored inversely: 5 = no hallucinations)
5. **Overall:** Holistic assessment of the document's usability as a real-world ROPA record

The judge evaluation is applied to all 113 retained documents. The full evaluation rubric, system prompt, and model configuration used for the judge are provided in Appendix A.4.

3.5 Statistical Analysis

Table 3.5 summarizes the statistical machinery applied in this thesis. The omnibus and post-hoc tests are non-parametric throughout; Bonferroni correction yields $\alpha' = 0.0167$ for the three pairwise mode comparisons in each post-hoc family.

Table 3.5: Statistical tests and their effect sizes by analysis stream. $\alpha = 0.05$ throughout; Bonferroni-corrected $\alpha' = 0.0167$ for post-hoc families.

Analysis stream	Test	Effect size
User study (within, $n = 69$)	Friedman	Kendall's W
	Wilcoxon signed-rank	Rank-biserial r
NLP / timing (between, $n = 113$)	Kruskal–Wallis	ε^2
	Mann–Whitney U	Rank-biserial r
Mode preference ($n = 69$)	χ^2 goodness-of-fit	Cramér's V

A significance threshold of $\alpha = 0.05$ is applied to all tests. The choice of test family follows the structure of each dataset, with one test family applied per analysis rather than running parallel matched-pairs and between-groups tests on the same data. **User study instruments** (SUS, R-TLX, perceived AI support, confidence items) are analyzed within-subjects on the matched $n = 69$ sample: a Friedman test as the omnibus, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction for post-hoc analysis. Every participant gives one response per mode, so the user-study data is genuinely matched-pairs; within-subjects analysis with Friedman and pairwise Wilcoxon signed-rank tests is also the standard approach for ordinal Likert and workload data collected in repeated-measures usability studies [28]. **NLP metric scores and per-mode completion time** are analyzed between-groups on the full unbalanced $n = 113$ sample: a Kruskal–Wallis test as the omnibus, followed by pairwise Mann–Whitney U tests with Bonferroni correction for post-hoc analysis. The between-groups family is the appropriate match for the document data, which is unbalanced across modes (Form 37, Ask 37, Chat 39); a between-groups analysis preserves all 113 retained submissions. The LLM-as-a-judge scores are reported descriptively (means, medians, dispersion) as a supplementary signal and are not subjected to inferential testing. Effect sizes are reported as ε^2 for the omnibus tests, Kendall's W as the coefficient of concordance for Friedman, and rank-biserial correlation r for the pairwise tests.

For RQ1 (User Experience and Preference), the Friedman test is applied to SUS scores, the R-TLX composite, and the confidence question composites across modes; post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction identify which mode pairs differ significantly; and mode preference distributions are reported as frequencies and rankings. For RQ2 (Document Quality), the Kruskal–Wallis test is applied to NLP metric scores across modes, with Bonferroni-corrected pairwise Mann–Whitney U tests as post-hoc analysis where the omnibus is significant. The LLM-as-a-judge scores are reported descriptively only and are not subjected to inferential testing. RQ3 (Confidence–Quality Alignment) is a cross-stream question that reads the RQ1 and RQ2 findings against each other; it is answered in the Discussion chapter through cross-signal interpretation rather than via additional inferential tests.

The tests and effect sizes used in this thesis are grouped below by where they appear in the Evaluation chapter. Each block names the analysis stream first (sample, design), then defines the omnibus and post-hoc tests applied to that stream together with their effect sizes. A short plain-language reading of each statistic is given alongside its formula so that the Evaluation chapter can refer back to a single canonical definition.

3.5.1 Procedures Shared Across Analyses

Two procedures are shared across every analysis stream and are defined once here.

The *Shapiro–Wilk normality test* screens whether a sample plausibly comes from a normal distribution. With order statistics $x_{(i)}$ and tabulated constants a_i,

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{3.10}$$

In plain terms: W close to 1 means the data look normal; a small p-value rejects normality and motivates a non-parametric test. Normality was rejected for nearly every user-study, NLP, and timing measure in this study, which is why every test below is non-parametric.

The *Bonferroni correction* divides the per-test significance threshold by the number of comparisons in a family so that the family-wise false-positive rate stays at the chosen α. With m comparisons,

$$\alpha' = \alpha/m. \tag{3.11}$$

Every post-hoc family in this thesis has $m = 3$ pairwise mode comparisons, so $\alpha' = 0.05/3 = 0.0167$ throughout.

3.5.2 User-Study Instruments (Matched Within-Subjects, $n = 69$)

Applied to SUS, R-TLX, the perceived AI support items, the primary confidence item, and the re-use intent item. Each participant gives one rating per mode, so the data are matched-pairs. The Friedman omnibus with pairwise Wilcoxon signed-rank post-hoc tests is the standard non-parametric within-subjects analysis for ordinal Likert and workload data in usability studies [28]; non-parametric tests are appropriate here because Shapiro–Wilk rejects normality on every user-study measure and Likert scales are not strictly continuous.

The *Friedman test* [7] is the within-subjects omnibus for $k \geq 3$ related samples. Each participant's k values are ranked within-subject; with n participants across k conditions and R_j the rank sum for condition j,

$$\chi_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1), \tag{3.12}$$

asymptotically χ^2 distributed with $k-1$ degrees of freedom. In plain terms: a large χ_F^2 (small p) means at least one mode differs from the others on this measure; non-significant means no mode-level difference can be claimed on this instrument as a whole.

When the Friedman omnibus is significant, the *Wilcoxon signed-rank test* [34] is applied to each pair of modes as a matched-pairs post-hoc. Differences $d_i = x_i - y_i$ are ranked by absolute value; the statistic is

$$W = \min(W_+, W_-), \quad W_+ = \sum_{d_i>0} \mathrm{rank}(|d_i|), \quad W_- = \sum_{d_i<0} \mathrm{rank}(|d_i|). \tag{3.13}$$

In plain terms: a small p (below $\alpha' = 0.0167$) means the two modes differ in their within-subjects ranks. The associated effect size, the rank-biserial correlation,

$$r = (W_+ - W_-)/(W_+ + W_-), \tag{3.14}$$

runs from -1 to $+1$; positive r means the first-named mode ranks higher. Conventional bands: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

The Friedman test's own effect size is *Kendall's* W (coefficient of concordance) [16]:

$$W_{\text{Kendall}} = \chi_F^2/(n(k-1)), \tag{3.15}$$

on the $[0, 1]$ scale, larger meaning stronger agreement among participants on the ordering of conditions. Conventional bands: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

3.5.3 Document-Quality Metrics and Completion Time (Between-Groups, $n = 113$)

Applied to BLEU, ROUGE-1/2/L, METEOR, BERTScore Precision / Recall / F1, SBERT cosine similarity, and per-mode completion time. The document sample is unbalanced across modes (Form 37, Ask 37, Chat 39); a between-groups family is the appropriate match for the data and preserves all 113 retained submissions. Kruskal–Wallis with pairwise Mann–Whitney U post-hoc tests is the standard non-parametric counterpart to one-way ANOVA for unbalanced independent-groups designs [6], and non-parametric throughout is again motivated by non-normal metric distributions.

The *Kruskal–Wallis* H *test* [17] is the non-parametric one-way ANOVA analogue applied as the between-groups omnibus. With N total observations, n_j in group j, and R_j the rank sum in group j,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1). \tag{3.16}$$

In plain terms: a large H (small p) means at least one mode's distribution of scores differs from the others on this metric.

When the Kruskal–Wallis omnibus is significant, the *Mann–Whitney* U *test* [21] is applied to each pair of modes as a between-groups post-hoc. With group sizes n_1,

n_2 and rank sum R_1,

$$U = R_1 - \frac{n_1(n_1+1)}{2}. \tag{3.17}$$

In plain terms: a small p (below $\alpha' = 0.0167$) means the two modes produced different distributions of scores on this metric. The associated rank-biserial effect size r is reported in the same -1 to $+1$ range and with the same conventional bands as for the Wilcoxon test above.

The Kruskal–Wallis test's own effect size is ε^2 *(epsilon-squared)* [15]:

$$\varepsilon^2 = (H - k + 1)/(N - k), \tag{3.18}$$

on the $[0, 1]$ scale, larger meaning more of the total variance in scores is explained by mode membership. Conventional bands match Kendall's W.

The same Kruskal–Wallis and Mann–Whitney U pipeline is applied at both whole-document and per-chapter granularity; Bonferroni correction continues to apply within each (metric, chapter) family of three pairwise comparisons, giving $\alpha' = 0.0167$. The per-chapter results are therefore produced by the same machinery as the whole-document results, not a separate test family.

3.5.4 Mode Preference Ranking ($n = 69$)

Each participant ranks the three modes 1st, 2nd, and 3rd at the end of the session, yielding a full ordinal ranking per participant. The inferential analysis collapses this ranking to its first-place column and tests whether the resulting rank-1 vote distribution across the three modes is uneven; this directly answers the headline question of whether there is a clear single most-preferred mode, with the corresponding vote shares readable at a glance. The 2nd- and 3rd-place ranks remain in the descriptive analysis (mean rank, full rank distribution per mode). Reducing a preference ranking to its first-place counts for goodness-of-fit testing is a standard analytic choice when the question is whether a single dominant option emerges from the ranking [28].

The *chi-square goodness-of-fit test* [26] compares observed counts O_i to expected counts E_i under a uniform null:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}. \tag{3.19}$$

In plain terms: a large χ^2 (small p) means the distribution of first-place votes is more uneven than would be expected if participants chose at random. The associated effect size is *Cramér's* V [5]:

$$V = \sqrt{\chi^2/(N(k-1))}, \tag{3.20}$$

on the $[0, 1]$ scale, larger meaning the observed distribution deviates more strongly from uniform. Conventional bands match Kendall's W.

With the tool, the study design, the instruments, the sample sizes, and the statistical machinery now in place, Chapter 4 reports the empirical findings: participant demographics first, then the four user-study instruments and the post-experiment ranking, then the document-quality analyses (NLP metrics at document and chapter levels, followed by the supplementary LLM-as-a-Judge signal).

4 Evaluation

This chapter reports the empirical findings of the user study and the subsequent document-quality analysis.

It opens with a descriptive Demographics section that profiles the participant sample on which every result that follows rests. The User Study section then reports the results of the System Usability Scale, NASA Raw Task Load Index, and Perceived AI Support, together with the post-experiment mode preference ranking and open-ended free-text comments. The document-quality sections report the reference-based NLP metrics at the document and chapter levels and, finally, the LLM-as-a-judge evaluation as a supplementary signal. All inferential test statistics referenced in the running text are reproduced in full in Appendix C.

4.1 Participant Demographics

This section shows the participant demographics. The study, set out in full in Chapter 3, recruited $N = 73$ participants who self-reported as ROPA novices for a within-subjects experiment in which every participant used all three ROPAgen interaction modes (Form, Ask, and Chat; see Section 3.1). Of the 73 recruited participants, four screened records were excluded for incomplete or invalid participation, yielding the matched within-subjects analysis sample of $n = 69$ on which every user-study result in this chapter rests. Mode order was counterbalanced across a six-group Balanced Latin Square so that learning, order, and fatigue effects could be distributed evenly across modes rather than absorbed into any one of them. The task was designed around a single standardized GDPR scenario, the APOR GmbH employee and parking-management case at a fictional Ulm software company. After each mode, participants completed the System Usability Scale, the NASA Raw Task Load Index, and the Perceived AI Support items, which are evaluated in Section 4.2; at the end of the session, they ranked the three modes by overall preference and supplied the demographic information profiled below. The demographic variables are reported descriptively only on the matched within-subjects sample of $n = 69$ participants; no inferential tests are applied to the demographics block.

4.1.1 Numeric and Ordinal Variables

The mean session duration was 1866 s (≈ 31 min) per participant across the full session (Figure 4.1). Figure 4.2 confirms that most participants were ROPA novices with high chatbot familiarity and moderate GDPR knowledge; Participants reported ROPA familiarity with a mean of 1.45, confirming that the sample qualifies as ROPA-novice in line with the inclusion criteria. Chatbot familiarity is correspondingly high, the mean being 4.36. GDPR knowledge sits near the scale midpoint (mean 2.81). Figure 4.2 makes the contrast vivid: chatbot familiarity concentrates at the high end, ROPA familiarity at the low end, and GDPR knowledge spreads broadly across the middle.

Table B.1 (Appendix B.3) reports the descriptive statistics for the four numeric or ordinal variables collected in the demographics block: total session duration in seconds, and three five-point Likert self-reports of prior expertise (GDPR knowledge, ROPA familiarity, chatbot familiarity).

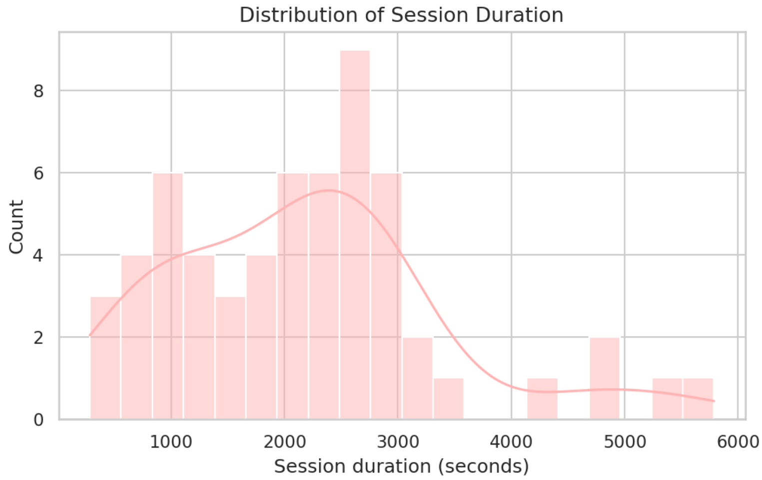


Figure 4.1: Distribution of session duration (in seconds).

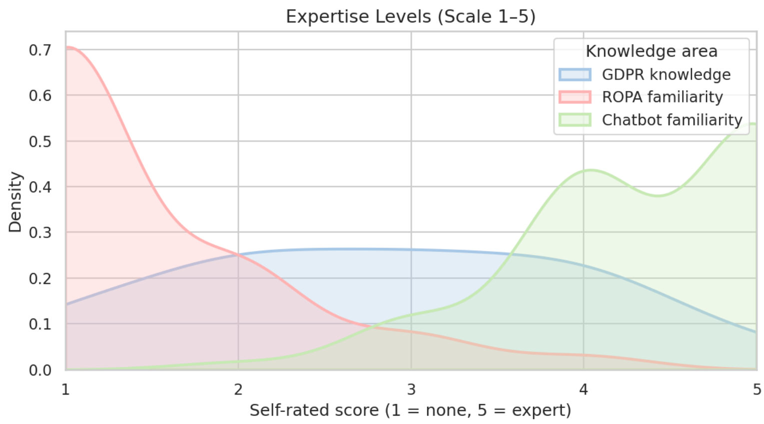


Figure 4.2: Smoothed densities of the three five-point Likert self-reports of prior expertise (GDPR knowledge, ROPA familiarity, chatbot familiarity). The horizontal axis marks the Likert points (1 = no familiarity, 5 = expert).

4.1.2 Categorical Variables

Table B.2 (Appendix B.3) reports counts and shares for the five categorical demographic variables: gender, age (binned by year), highest completed degree, current professional status, and field of study or work. The sample is dominated by

Bachelor-level students (Figure 4.5) studying or working in IT or Computer Science (Figure 4.7). Of the participants who reported gender (Figure 4.4), $62.1\,\%$ identified as male. The underlying age distribution (Figure 4.3) clusters tightly around 24–26 years, with a thin right tail of older participants. This profile (IT-leaning, chatbot-fluent, ROPA-novice) frames the results reported in Sections 4.2–4.4 and is taken up again in the Discussion chapter when generalizability is addressed.

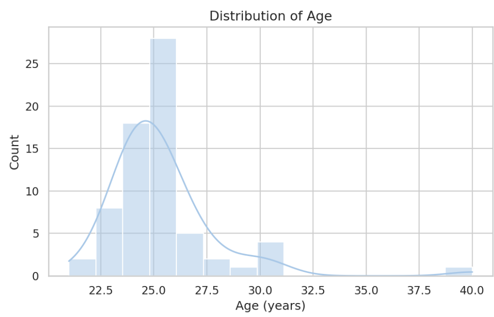


Figure 4.3: Age distribution of the $n = 69$ analysis sample.

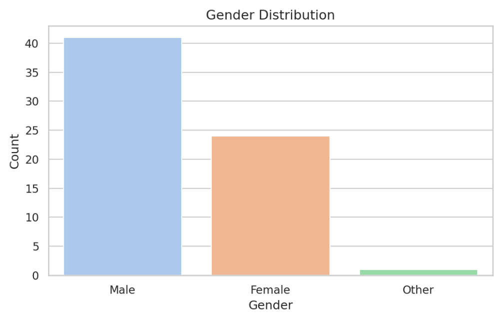


Figure 4.4: Self-reported gender of the $n = 66$ participants who answered this item.

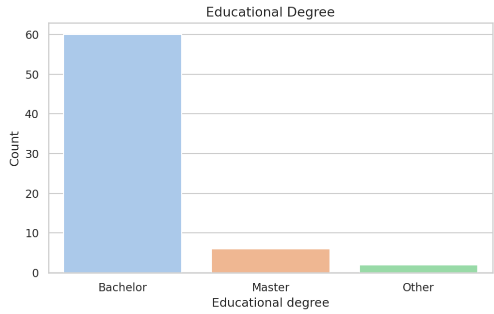


Figure 4.5: Highest completed degree of the $n = 68$ participants who answered this item.

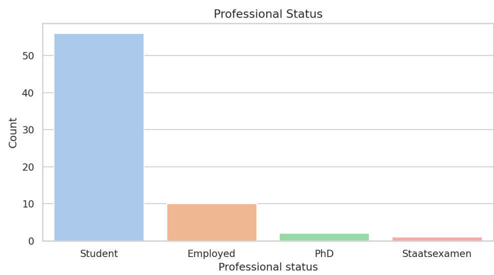


Figure 4.6: Current professional status of the $n = 69$ participants.

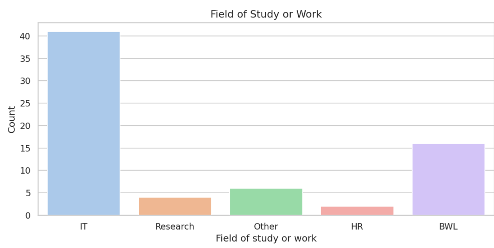


Figure 4.7: Field of study or work of the $n = 69$ participants.

4.2 User Study

This section consolidates the four user-study instruments. The three per-mode self-report questionnaires (SUS, R-TLX, Perceived AI Support) are reported in the order in which they were administered after each interaction mode; the post-experiment mode preference ranking and the open-ended comments close the section by asking how the three per-instrument patterns combine when participants commit to a single overall preference and explain it in their own words.

4.2.1 ROPAgen Usability — System Usability Scale

The System Usability Scale captures how usable each interface felt to participants immediately after they finished interacting with it. It consists of ten Likert items on a five-point scale, with item polarity alternating (odd items: higher is better; even items: lower is better). The full item text is reproduced in Appendix B.2; per-item distributions and box plots are in Appendix B.4.1.

The SUS yields a single composite score per mode on a 0–100 scale, computed from the ten items using the standard scoring procedure of Brooke [4]. A score of 68 is widely used as the threshold above which a system is judged to have above-average usability [3]. Figure 4.8 reports the per-mode composite scores against this reference line. All three modes sit numerically above the threshold: Form scores highest at a mean of 78.62 (median 80.0, SD 16.09), while Ask and Chat sit closely together at 72.97 and 72.93 respectively (medians 72.5 and 75.0, SDs 15.36 and 19.03). A one-sample Wilcoxon signed-rank test confirms that Form's composite is significantly above the 68 threshold ($n = 69$, $W = 449$, $p < 0.001$, matched-pairs $r = +0.63$, large); Ask and Chat are not tested against the threshold. Form leads the SUS evaluation, with Chat and Ask following closely behind; this pattern will become clearer across the user-study sections that follow.

Item by item, Table 4.1 reveals three patterns in how participants rated the three modes.

Participants said they would willingly keep using Form (mean 3.52) somewhat more than Ask or Chat (both 3.22), a margin of 0.30 Likert points. The same Form lead reappears on whether the system was easy to use (Form 4.38 vs Ask 4.01, Chat

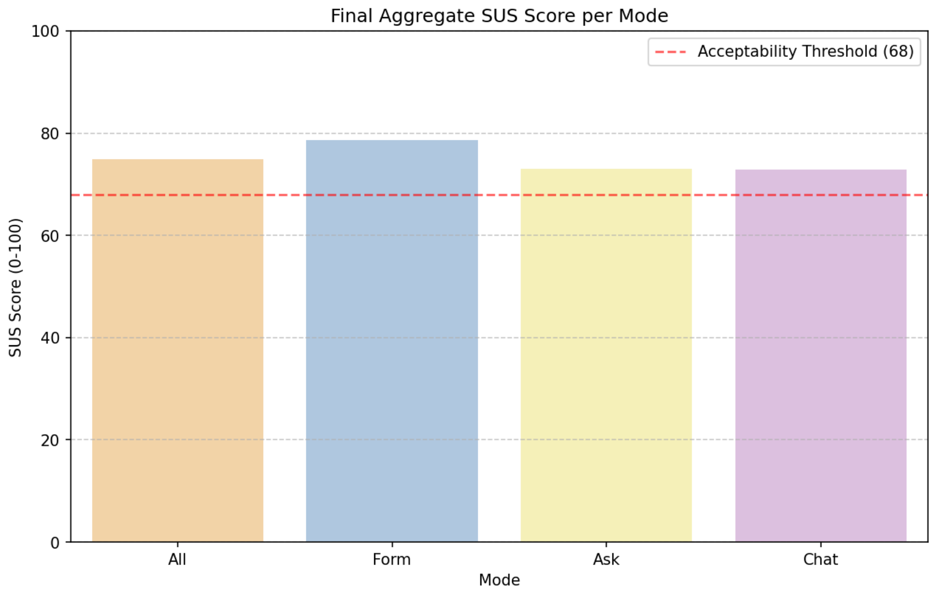
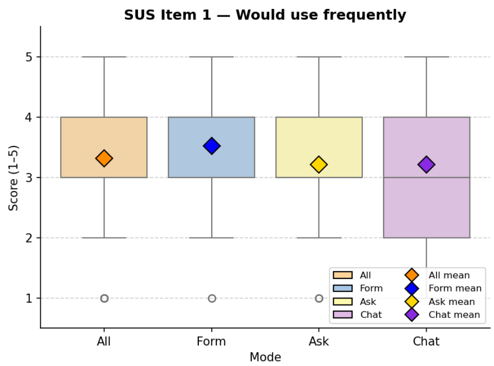
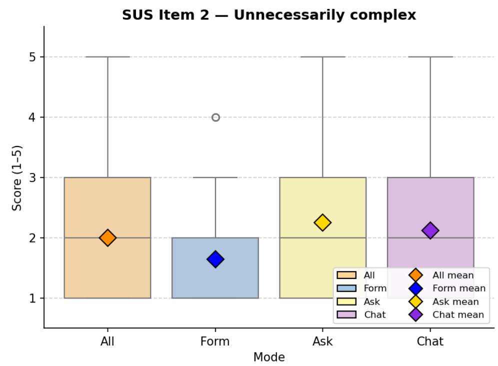


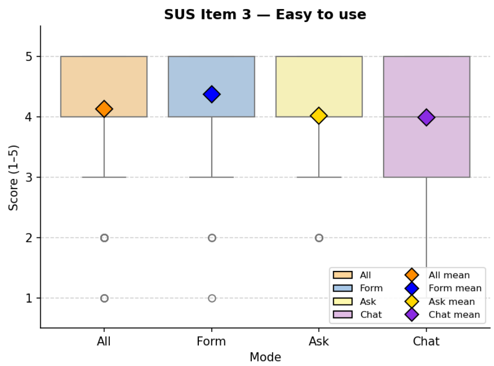
Figure 4.8: Mean SUS composite score per mode ($n = 69$ per mode) on the standard 0–100 scale. The dashed horizontal line marks the acceptability threshold of 68 [3].

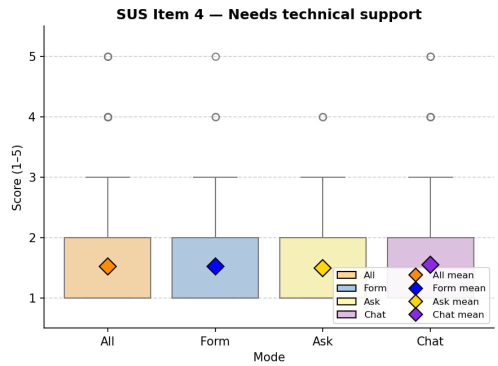
Table 4.1: System Usability Scale, mean response per item by mode ($n = 69$ per mode). Items are reported on the original five-point Likert scale; the Aggregated column averages the three per-mode means. Odd items: higher is better; even items: lower is better.

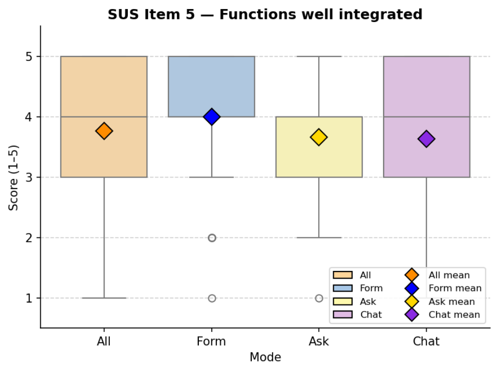
#	Item	Aggregated	Form	Ask	Chat
1	I think that I would like to use this system frequently.	3.32	3.52	3.22	3.22
2	I found the system unnecessarily complex.	2.00	1.64	2.25	2.12
3	I thought the system was easy to use.	4.13	4.38	4.01	3.99
4	I think that I would need the support of a technical person to be able to use this system.	1.52	1.52	1.49	1.55
5	I found the various functions in this system were well integrated.	3.77	4.00	3.67	3.64
6	I thought there was too much inconsistency in this system.	2.07	2.16	2.04	2.01
7	I would imagine that most people would learn to use this system very quickly.	4.11	4.30	4.00	4.03
8	I found the system very cumbersome to use.	2.03	1.65	2.16	2.29
9	I felt very confident using the system.	3.86	3.83	3.84	3.90
10	I needed to learn a lot of things before I could get going with this system.	1.61	1.61	1.61	1.62

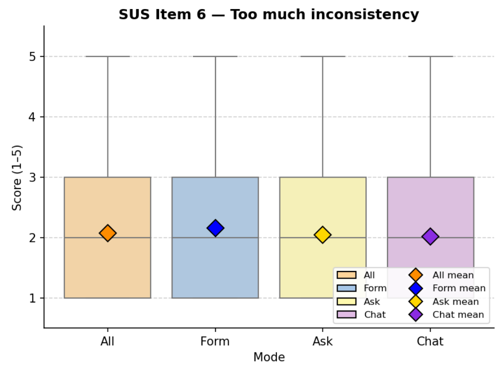

SUS Item 1 — Would use frequently
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean

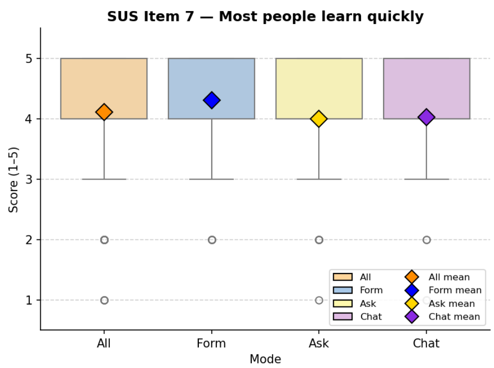

SUS Item 2 — Unnecessarily complex
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean

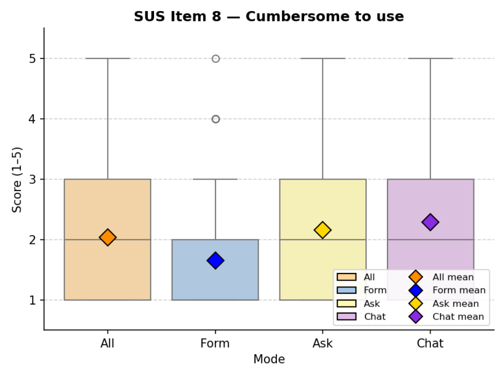

SUS Item 3 — Easy to use
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean

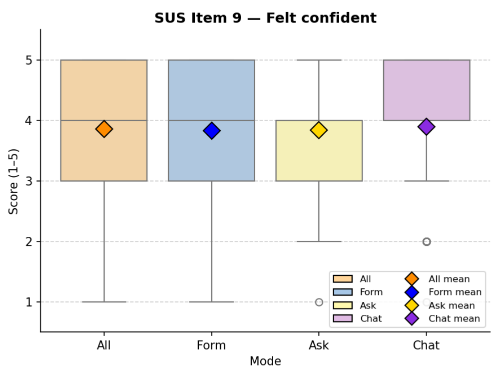

SUS Item 4 — Needs technical support
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean

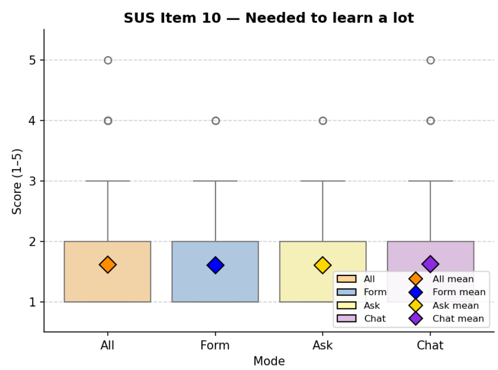

SUS Item 5 — Functions well integrated
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean


SUS Item 6 — Too much inconsistency
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean


SUS Item 7 — Most people learn quickly
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean


SUS Item 8 — Cumbersome to use
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean


SUS Item 9 — Felt confident
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean


SUS Item 10 — Needed to learn a lot
Score (1–5)
All
Form
Ask
Chat
Mode
All mean
Form mean
Ask mean
Chat mean

3.99) and on whether it felt well integrated (Form 4.00 vs Ask 3.67, Chat 3.64). Asked how quickly they thought most people would learn the system, the pattern held again (Form 4.30, Ask 4.00, Chat 4.03). Across these four items the gap sits at roughly 0.30–0.40 Likert points: small, but Form is ahead every time. The biggest gaps appear on the two items that asked whether the system felt unnecessarily complex or cumbersome. On complexity, Form drew the most favorable rating (1.64), with Ask noticeably higher at 2.25 and Chat at 2.12; the Form lead here is roughly half a Likert point. Cumbersomeness produced the largest spread of any SUS item: Chat was rated most cumbersome (2.29), Ask close behind (2.16), and Form clearly the easiest to operate (1.65), a 0.64-point spread between the extremes. These are the items where the cost of LLM-driven interaction lands most directly on the user. Four items show essentially no difference between modes. None of the three felt to participants like it required prior learning to get going (item 10: Form 1.61, Ask 1.61, Chat 1.62) or like it would need a technical person's help to use (item 4: Form 1.52, Ask 1.49, Chat 1.55). Participants also did not perceive any of the three as meaningfully more inconsistent than the others (item 6: Form 2.16, Ask 2.04, Chat 2.01); the small fractional advantage here goes to the LLM-driven modes. Within-system confidence is broadly comparable across modes (item 9: Form 3.83, Ask 3.84, Chat 3.90), with a very slight uptick toward Chat. On these dimensions the choice of interaction style simply does not move the rating.

A Friedman test on the per-mode SUS composite narrowly misses significance ($\chi^2(2) = 5.51$, $p = 0.064$, $W = 0.040$). Two Form-focused pairwise Wilcoxon signed-rank tests (Bonferroni $\alpha = 0.025$ for two comparisons) show Form significantly higher than both Ask ($W = 578.5$, $p = 0.013$, $r = +0.37$, medium) and Chat ($W = 549.5$, $p = 0.018$, $r = +0.36$, medium); Ask versus Chat is not tested. The descriptive Form lead therefore holds inferentially against each LLM-driven mode even though the three-way omnibus narrowly misses. Full statistics are in Appendix C.

The SUS picture is a Form lead supported on three independent counts: Form is rated highest on six of ten items; Form's composite is significantly above the 68 acceptability threshold; and Form is significantly higher than both Ask and Chat on the Form-focused pairwise comparisons. The lead concentrates on items that capture the moment-to-moment feel of using the system; whether it carries to perceived workload is the question for Section 4.2.2.

4.2.2 Cognitive Load — NASA Raw Task Load Index

Where SUS measured how usable the interfaces felt, the Raw Task Load Index decomposes the participant's effort along six interpretable axes (Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration), each rated on the standard NASA 21-point scale (1 = lowest demand, 21 = highest). Performance is reported in the positive-coded direction (higher equals worse perceived performance) for internal consistency with the other subscales. The overall R-TLX in this study is the unweighted mean of the six subscales on the same 1–21 scale (no rescaling is applied). Full subscale descriptions are in Appendix B.2; per-subscale distributions and box plots are in Appendix B.4.2. A short supplementary paragraph at the end of this section reports the objective per-mode completion time alongside the perceptual subscales, as a behavioral counterpart to Temporal Demand.

Figure 4.10 reports the composite R-TLX score per mode against the scale midpoint of 10.5, which marks the boundary between the lower and upper halves of the 1–21 range. All three modes sit comfortably in the lower half: Form scores lowest with a composite of 5.50, while Ask and Chat are closely tied at 6.33 each. As with SUS, Form is the lightest of the three at the aggregate level, with Ask and Chat following close behind by roughly 0.8 scale points.

Table 4.2: NASA R-TLX subscale mean (SD) by mode ($n = 69$ per mode), 1–21 scale. The Aggregated column averages the three per-mode means.

Subscale	Aggregated	Form	Ask	Chat
Mental Demand	6.31 (4.75)	5.77 (4.26)	6.80 (4.63)	6.36 (5.31)
Physical Demand	3.19 (3.40)	2.77 (2.71)	3.54 (3.92)	3.26 (3.46)
Temporal Demand	4.89 (4.28)	4.16 (3.82)	4.97 (4.15)	5.54 (4.78)
Effort	6.13 (4.57)	5.68 (4.56)	6.39 (4.41)	6.32 (4.77)
Frustration	6.86 (5.24)	5.88 (4.84)	7.58 (5.34)	7.13 (5.44)
Performance (pos.)	8.93 (5.82)	8.71 (5.48)	8.68 (5.64)	9.39 (6.36)

Subscale by subscale, Table 4.2 reveals three patterns in how participants rated workload.

Participants found Form somewhat less mentally demanding than Ask or Chat (Form 5.77, Ask 6.80, Chat 6.36), a gap of about 0.6 to 1.0 points. The same small lead reappears on Effort (Form 5.68 vs Ask 6.39, Chat 6.32) and on Physical Demand

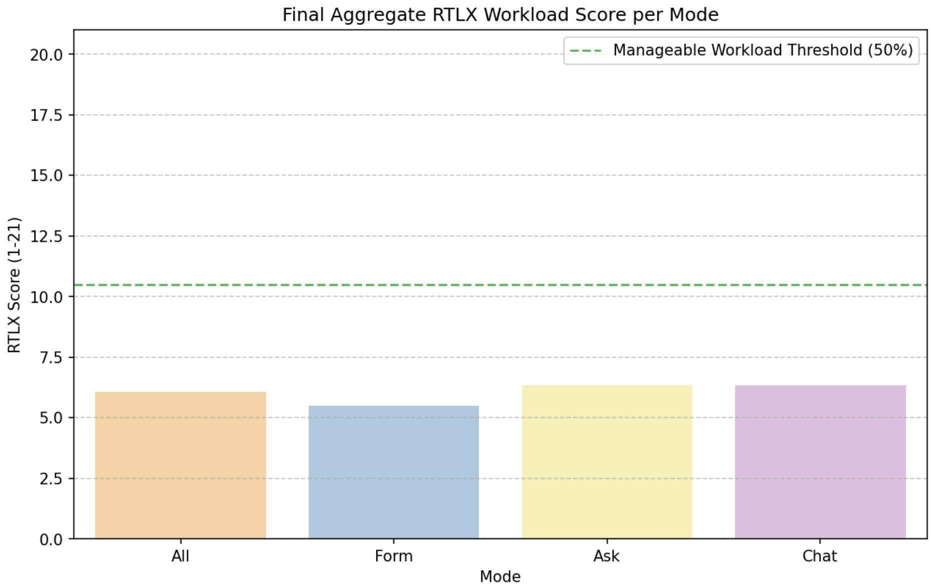


Figure 4.10: Mean R-TLX composite workload score per mode ($n = 69$), unweighted average of the six subscales on the 1–21 scale. The dashed horizontal line marks the scale midpoint (10.5), shown for visual reference.

(Form 2.77, Ask 3.54, Chat 3.26); the Physical Demand ratings sit near the floor of the scale in all three modes, since the ROPA task is not physically taxing in any of them. The two subscales where Form's lead is sharpest are also the two that reach statistical significance. Participants felt under noticeably more time pressure in Chat than in Form (Form 4.16, Ask 4.97, Chat 5.54); the gap widens monotonically as LLM support increases, the largest such increase across the six subscales. Frustration shows the widest gap of any subscale: Ask was rated the most frustrating (7.58), Chat close behind (7.13), and Form clearly the least (5.88). These are the workload dimensions where LLM-driven interaction measurably costs the user. Performance is the one subscale where the Form lead disappears. Participants rated their own success at the task as essentially identical between Form (8.71) and Ask (8.68), with Chat marginally worse (9.39, higher equals worse perceived performance). Mode does not move how successfully participants think they completed the task, with the slight self-doubt blip in Chat being the only deviation.

An aggregated Friedman test across the six subscales is not significant ($\chi^2(2) = 5.24$, $p = 0.073$, $W = 0.038$), so the composite workload score does not differ sig-

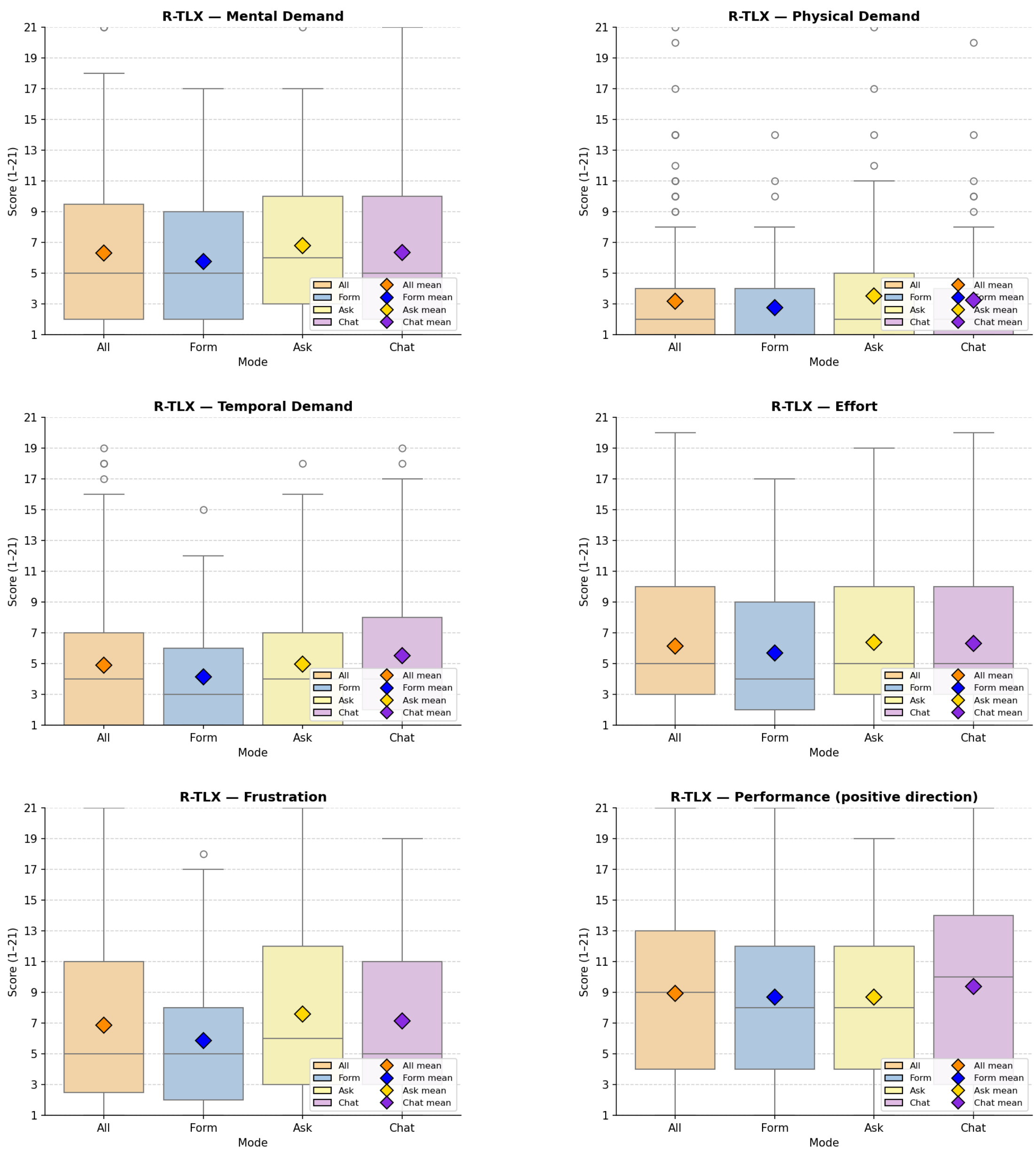


Figure 4.11: Box plots of the six R-TLX subscales by mode ($n = 69$ per mode), one plot per subscale.

nificantly between modes. Two subscales reach significance individually: Temporal Demand ($\chi^2(2) = 12.77$, $p = 0.002$, $W = 0.093$) and Frustration ($\chi^2(2) = 8.93$, $p = 0.012$, $W = 0.065$). Pairwise Wilcoxon signed-rank tests with Bonferroni correction confirm Form differs significantly from both Ask and Chat on these subscales, with medium-to-large rank-biserial effects ($|r| = 0.37$ to 0.57); Ask versus Chat is not significant on any measure. Full statistics are in Appendix C.

Overall, the R-TLX reinforces the SUS Form lead: Form sits below the LLM-driven modes on every workload dimension except Performance, with two subscales (Temporal Demand and Frustration) reaching inferential significance individually even when the composite test does not. The user-study side's strongest single inferential signal therefore lands on perceived workload, not usability. Whether perception of time pressure matches actual time taken is the question for the supplementary completion-time analysis that follows.

Supplementary: per-mode completion time. The R-TLX workload data is entirely perceptual. As a behavioral counterpart to Temporal Demand, every ROPA document generated by ROPAgen carries a `Time elapsed:` N `seconds` line at submission, giving one objective completion-time measurement per participant per mode on the $n = 113$ document sample. The question is whether the perceived time pressure (Form lowest, Chat highest) is mirrored by the actual time spent. Form takes about half the wall-clock time of either LLM-driven mode. Median completion time was 7 minutes for Form (420 s), 14 minutes for Ask (858 s), and 12 minutes for Chat (717 s); the means tell the same story (Form 564.2 s, Ask 989.8 s, Chat 979.1 s), with substantial right-skewed spread in the LLM-driven modes (Chat SD ≈ 1015 s, one extreme session at 6199 s). The full distributions are in Figure 4.12.

A Kruskal–Wallis test on the full $n = 113$ sample confirms that completion time varies by mode ($H = 16.46$, $p < 0.001$, $\varepsilon^2 = 0.15$, large). Form is significantly faster than both Ask (median 420 s vs 858 s; $r = 0.512$, large) and Chat (median 420 s vs 717 s; $r = 0.418$, medium); Ask and Chat are statistically indistinguishable from each other.

The behavioral data sharpens the perceived workload picture: Form is not merely felt as the least time-pressured mode, it is also measurably the fastest by a margin of roughly seven minutes per document over either LLM-driven mode. The two

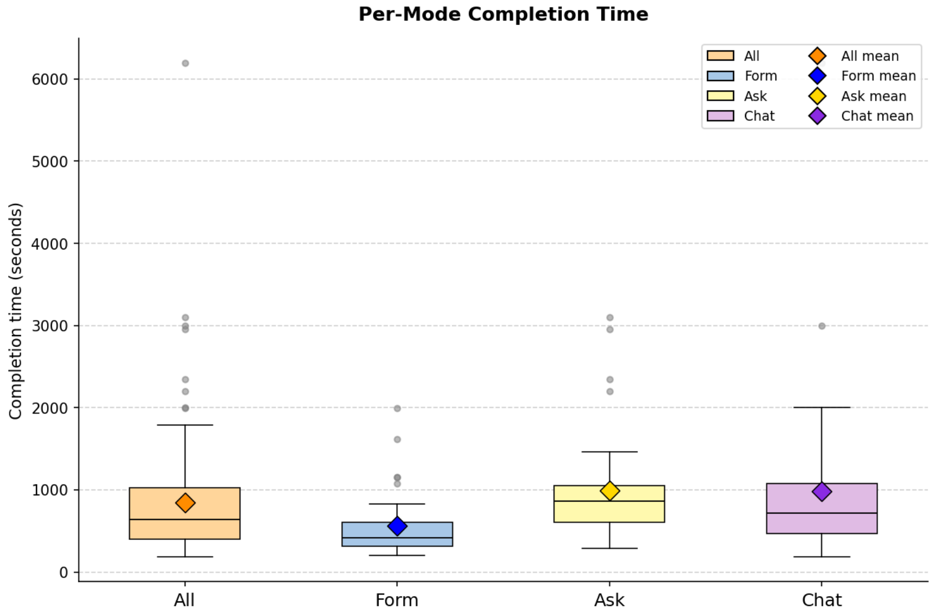


Figure 4.12: Per-mode completion time in seconds, extracted from the `Time elapsed` line emitted by ROPAgen at document submission ($n = 113$ documents: Form 37, Ask 37, Chat 39).

LLM-driven modes, despite their very different interaction styles (per-section explainer versus freeform conversation), take essentially the same amount of time on average. Whether participants nevertheless ascribe more value to the LLM-driven modes when asked specifically about AI support is the question taken up next in Section 4.2.3.

4.2.3 Perceived AI Support — AI Questionnaire

The SUS (Section 4.2.1) and R-TLX (Section 4.2.2) characterize the interface as it was experienced; they say nothing directly about whether participants felt the AI itself had been useful. The Perceived AI Support questionnaire targets that gap. After each interaction mode, participants completed a six-item questionnaire on a five-point Likert scale (1 = strong disagreement, 5 = strong agreement): helpfulness (item 1), efficiency (item 2), transparency (item 3), trust (item 4), legal-basis confidence (item 5), and re-use intent (item 6). Item 5 is the primary self-report confidence measure referenced in Chapter 3. The full per-item distribution is in Appendix B.4.3.

Table 4.3: Perceived AI Support, mean response per item by mode ($n = 69$ per mode), five-point Likert scale (higher is better). The Aggregated column is the unweighted mean of the three per-mode means.

#	Item	Aggregated	Form	Ask	Chat
1	The AI was helpful in creating this document.	4.23	4.23	4.32	4.13
2	The AI helped me to complete the document faster.	4.22	4.43	4.25	3.99
3	I understood why the AI suggested specific GDPR fields.	3.81	3.51	4.16	3.75
4	I trust that the AI made correct suggestions.	3.63	3.41	3.80	3.68
5	I am confident that the AI identified the correct legal basis.	3.71	3.58	3.74	3.81
6	I would use this mode again.	3.65	3.99	3.55	3.42

Asked simply whether the AI was helpful in creating the document, participants rated all three modes similarly: Ask drew the highest rating (4.32), Form just behind (4.23), and Chat lowest (4.13), a spread of only 0.19 points. The AI is perceived as broadly helpful across all three modes; the differences emerge only when participants are asked more specifically what the AI was helpful *for*. On whether the AI helped complete the document faster, Form drew the strongest endorsement (4.43), Ask second (4.25), and Chat lowest (3.99). The perceived speed-up decreases monotonically as LLM involvement grows, dropping by roughly 0.45 points from Form to Chat. Participants felt the AI helped them most when it stayed in the background. Participants felt they understood the AI's suggestions best in Ask (4.16), well ahead of Chat (3.75) and Form (3.51). The Ask lead of roughly 0.65 points is the widest reversal of the Form-leading pattern across the six items, and

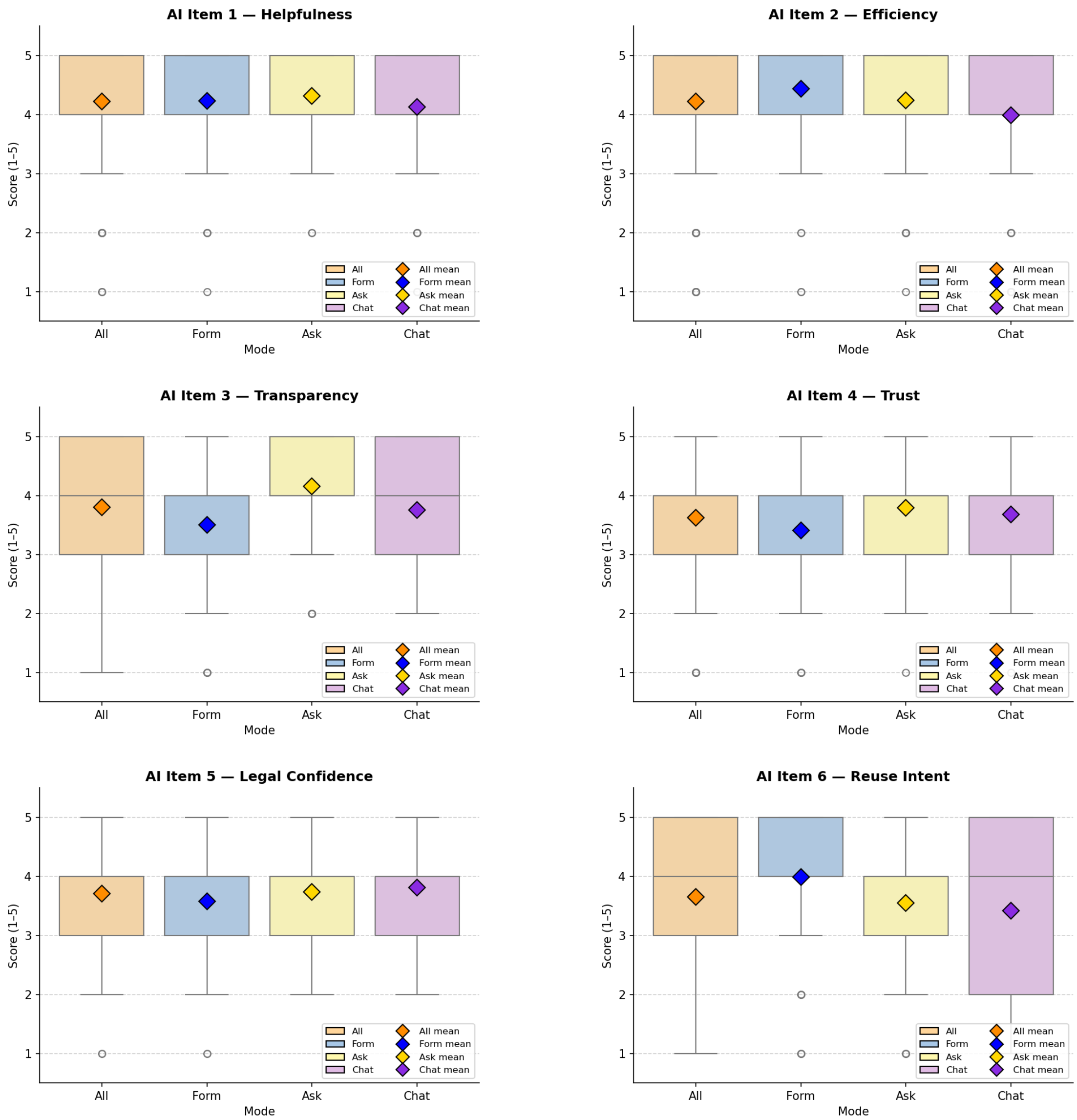


Figure 4.13: Box plots of the six Perceived AI Support items by mode ($n = 69$ per mode), one plot per item in administered order.

is unsurprising: Ask is the only mode that explicitly explains each suggestion (see Section 3.1). Trust in the AI's outputs follows the same direction. Asked whether they trusted that the AI made correct suggestions, participants rated Ask highest (3.80), Chat second (3.68), and Form lowest (3.41). The LLM-driven modes earn more trust than Form, by roughly 0.27 to 0.39 points. On confidence that the AI identified the correct legal basis (the primary confidence measure, item 5), the ordering reverses completely: Chat drew the highest rating (3.81), Ask second (3.74), and Form lowest (3.58). Confidence rises monotonically with LLM involvement, the only item where Chat leads outright. Participants feel most sure of the LLM's legal judgment in the mode where they see the LLM least directly. Asked which mode they would actually use again, participants snapped back to Form (3.99) over Ask (3.55) and Chat (3.42), a 0.57-point lead. This is the widest negative gap of the six items and the only one to reach statistical significance ($\chi^2(2) = 11.12$, $p = 0.004$; Form vs Chat $r = +0.543$, large). Even when participants felt better about the LLM's content judgment, they still wanted to come back to Form.

The composite Friedman test on the per-mode aggregate is not significant ($\chi^2(2) = 1.05$, $p = 0.592$, $W = 0.008$), so perceived AI support taken as a whole does not differ across modes. The primary legal-basis confidence item (item 5) is also non-significant ($\chi^2(2) = 3.76$, $p = 0.152$). Full statistics in Appendix C.

Perceived AI support splits the user-study story in two: participants credit the LLM-driven modes when asked about content judgment (transparency, trust, legal-basis confidence), but credit Form when asked about productivity and re-use intent. The single item that reaches statistical significance after Bonferroni correction (re-use intent, item 6) is the one where the Form lead is sharpest: participants are willing to praise the LLM modes' content judgment, but still want to come back to Form. The behavioral counterpart, which mode they actually choose when forced to rank, is taken up next in Section 4.2.4.

4.2.4 Mode Preference and Free-Text Feedback

The per-instrument data so far have shown a consistent Form-leading pattern on usability (Section 4.2.1) and workload (Section 4.2.2), with a partial inversion on perceived transparency, trust, and confidence (Section 4.2.3). This section asks how the three patterns combine when participants are forced to commit to a single overall preference. After all three interaction modes had been completed, participants provided two final inputs: an overall ranking of the three modes from most to least preferred, and an open-ended summary comment on the study experience. In addition, after each individual mode, an optional per-mode comment was collected.

Mode Preference Ranking

The ranking offers the cleanest test of which mode participants actually preferred once they had used all three. Table 4.4 summarizes the post-experiment ranking. Participants assigned each of the three modes a rank from 1 (most preferred) to 3 (least preferred); "Rank-1 votes" counts the participants who placed the mode first.

Table 4.4: Post-experiment mode preference ranking ($n = 69$). "Rank-1 votes" counts the number of participants who placed the mode first; "Share" is the corresponding percentage; "Mean rank" averages the assigned 1–3 rank across all 69 participants.

Mode	Rank-1 votes	Share	Mean rank
Form	37	53.6 %	1.78
Chat	20	29.0 %	2.07
Ask	12	17.4 %	2.14

Form received the plurality of first-place votes (37 of 69, $53.6\,\%$), followed by Chat (20, $29.0\,\%$) and Ask (12, $17.4\,\%$); the mean ranks follow the same ordering (1.78, 2.07, 2.14), mirroring the Form-leading pattern of the SUS and R-TLX. A chi-square goodness-of-fit test rejects the uniform-null distribution ($\chi^2(2) = 14.17$, $p < 0.001$, Cramér's $V = 0.32$, medium effect). Full statistics in Appendix C.

The complete distribution of all three rank positions across the three modes is visualized in Figure 4.14.

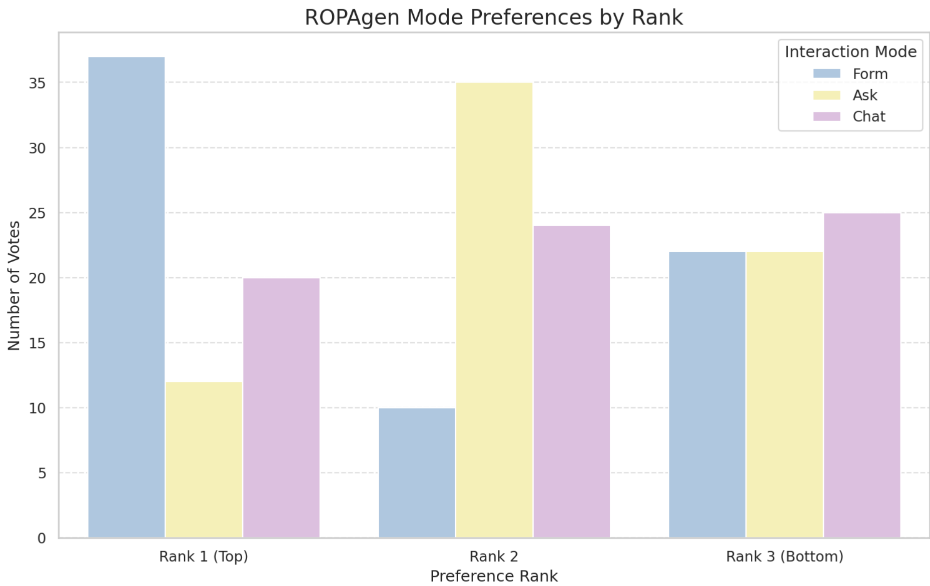


Figure 4.14: Distribution of all three rank positions per mode ($n = 69$).

Free-Text Comments

The free-text comments contextualize the ranking by surfacing the reasons participants gave for their preferences. Each mode collected an optional per-mode open-ended comment, and after all three modes participants wrote an overall summary comment. A total of 88 comments were recorded across 44 unique participants. Comments are written in German with a small minority in English; the paragraphs that follow summarize them thematically in the order Form, Ask, Chat, and close with the across-modes overall comments.

Form. Comments on Form cluster around speed, clarity, and control. Recurring observations describe the mode as the fastest of the three, the easiest to obtain an overview of, and the most “integrated” into the document creation workflow, as the checkbox interface makes the selection space visible at a glance. Several participants emphasize that Form “takes the most work off you” (*“nimmt dir am meisten Arbeit ab”*) by pre-populating the option set, while a smaller subset notes the opposite concern: that the absence of an interactive LLM commentary makes it unclear on what basis the model produces its suggestions, and that participants would not feel safe relying on Form output without an additional verification step. Several comments describe specific UX issues: long text fields rendered as a single narrow line on mobile, the inability to provide the LLM with additional context for its suggestions, and the impression that the LLM “guesses” without scenario-specific input.

Ask. Comments on Ask are the most polarized of the three modes. A consistent theme is that Ask is the most cumbersome mode in terms of workflow: the requirement to manually copy the LLM’s suggested text into the form field (cf. Section 3.1) is repeatedly described as redundant work, particularly as participants reported that the model often produced suggestions in a free-text style that did not match the structured form fields. Several participants observed that the lack of a persistent context across fields forced them to re-enter the same information repeatedly. Conversely, explanatory text generated by Ask is consistently described as the most informative of the three modes, supporting the high transparency rating visible in item 3 of the AI Support questionnaire. A frequent suggestion is to combine Ask’s explanations with Form’s structured field layout.

Chat. Comments on Chat are dominated by two themes: comfort and opacity. On the comfort side, several participants describe Chat as the most intuitive of the three modes, especially as the LLM takes over the work of mapping the conversation onto the structured ROPA fields (cf. Section 3.1), leaving the participant only with the task of describing the processing scenario. On the opacity side, an equally large group of participants reports uncertainty about what is actually being written into the document — whether the entire chat history is used for generation, whether each individual message must be confirmed, and what the document looks like during the conversation. Specific defects mentioned include the LLM occasionally switching from German to English mid-response, occasional emission of raw `DATA_UPDATE` tokens, and a slow “streaming” rendering style that participants experienced as artificially throttled.

Across modes (overall comments). The overall comments recapitulate the per-mode pattern. A clear majority describes Form as the fastest and most ergonomic mode, particularly for participants who feel they “do not know enough” to interrogate the model meaningfully. Chat is typically placed second and motivated by the impression that “the AI takes care of it,” which several participants explicitly note increases their *subjective* sense of safety even though they cannot verify the output. Ask is most often placed third, with the manual copy-paste step cited as the principal reason. A recurring constructive suggestion across participants is a hybrid mode that combines the visible option set of Form with the conversational, context-aware behavior of Chat or Ask. Several participants explicitly add a caveat that they are novices in the legal domain and that their preference ordering might shift if they had prior GDPR knowledge.

The ranking and comment data align with the SUS and R-TLX picture. Form is the most-chosen mode by a clear margin; Chat slots in second on raw votes; Ask is most often placed third, with free-text comments citing the manual copy-paste step between explanation and form field as the principal friction. Across the comments the most consistent constructive suggestion is for a hybrid mode that combines Form’s visible option set with Chat’s or Ask’s conversational behavior.

4.2.5 Summary of the User Study

Taken together, the four user-study instruments and the behavioral completion-time measure converge on a Form-leading picture. Form is the most usable mode on SUS (leading on six of ten items, the omnibus narrowly null at $p = 0.064$), the lightest on R-TLX workload (below the LLM-driven modes on every subscale except Performance, with Temporal Demand and Frustration reaching significance), the fastest in actual completion time (about half the wall-clock time of either LLM-driven mode, $p < 0.001$, $\varepsilon^2 = 0.15$ large), and the most preferred when participants commit to a single overall ranking ($53.6\,\%$ of rank-1 votes, $p < 0.001$).

The one place this picture breaks is perceived AI support: when asked specifically about the AI's contribution, participants credit the LLM-driven modes on the items about content judgment (transparency, trust, legal-basis confidence) and credit Form on the items about productivity and re-use intent. Participants are thus willing to credit the LLM modes for what the AI is doing on the page, yet still prefer Form for how the tool feels to use. Whether the documents these modes produce mirror the user-study Form-leading verdict is the question of the next section, beginning with the reference-based NLP metrics in Section 4.3.1.

4.3 Document Quality — NLP Metrics

The user study (Section 4.2) closed on a stable Form-leading pattern: Form was rated the most usable, the lightest on workload, and the most often ranked first. This section opens the document-quality half of the evaluation by asking whether the same ordering holds when the output text itself is compared against a reference document. The reference-based metric suite mirrors the one applied by von Schwerin et al. [29] in their evaluation of LLM-generated ROPA data categories: five lexical metrics (BLEU, ROUGE-1/2/L, METEOR) that count surface-form token overlap, and three BERT-based metrics (BERTScore Precision / Recall / F1) plus document-level SBERT cosine similarity that compare semantic representations. All metrics are computed against the single expert-authored Form-mode reference document described in Chapter 3.

The metrics are computed at two granularities to separate the question of overall document similarity from the question of where in the document the modes differ. At the **document level** (Section 4.3.1), each participant ROPA is treated as a single block of text and scored against the full reference document. At the **chapter level** (Section 4.3.2), the same metrics are recomputed for each of the six rendered Article 30(1) ROPA chapters separately, scoring each participant chapter against the matching reference chapter; the six chapter names and the underlying `DocumentData` mapping are listed at the start of Section 4.3.2. SBERT cosine similarity uses ModernBERT as the primary encoder at both granularities; at the chapter level the smaller `all-MiniLM-L6-v2` encoder is added as a robustness check on whether ModernBERT's near-saturation masks real differences. Section 4.3.3 compares the two views.

4.3.1 Document-Level NLP Metrics

At this granularity each participant ROPA is treated as a single block of text and scored against the full expert-authored reference. Document-quality results are reported on the full retained set of 113 documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$). Inferential mode comparisons use Kruskal–Wallis as the omnibus test with Bonferroni-corrected Mann–Whitney U as the pairwise post-hoc; full test statistics in Appendix C. The full per-document descriptive distributions are in Appendix B.5. Table 4.5 gives a one-glance summary of mean (SD) per mode for every document-level metric. The remainder of this subsection walks through each metric in turn, lexical first (strictest demands on surface form) and then embedding-based (BERTScore Precision / Recall / F1 and SBERT).

Table 4.5: Document-level NLP metrics, mean (SD) per mode (full sample $n = 113$). Lexical metrics in the upper block, embedding-based metrics in the lower. Full per-document distributions in Appendix B.5.

Metric	Aggregated	Form	Ask	Chat
BLEU	0.133 (0.048)	0.129 (0.048)	0.127 (0.056)	0.143 (0.039)
ROUGE-1	0.488 (0.087)	0.488 (0.098)	0.465 (0.088)	0.511 (0.071)
ROUGE-2	0.285 (0.077)	0.306 (0.076)	0.257 (0.080)	0.291 (0.069)
ROUGE-L	0.375 (0.087)	0.388 (0.094)	0.347 (0.091)	0.389 (0.072)
METEOR	0.389 (0.062)	0.412 (0.051)	0.365 (0.068)	0.390 (0.057)
BERTScore Precision	0.890 (0.014)	0.889 (0.014)	0.886 (0.016)	0.895 (0.012)
BERTScore Recall	0.915 (0.014)	0.923 (0.005)	0.908 (0.017)	0.913 (0.013)
BERTScore F1	0.902 (0.011)	0.905 (0.009)	0.897 (0.013)	0.904 (0.010)
SBERT (ModernBERT)	0.987 (0.004)	0.986 (0.004)	0.986 (0.005)	0.990 (0.002)

Lexical Metrics

Lexical metrics (BLEU, ROUGE, and METEOR) count surface-form token overlap with the reference; they reward word-level fidelity but are blind to paraphrase. A document that covers the same content in different wording scores lower here than under the embedding-based metrics that follow. Figure 4.15 shows the box-plot distributions.

BLEU. BLEU *(Bilingual Evaluation Understudy)* ((3.1)) counts exact n-gram overlap with the reference and is therefore harsh on paraphrase: a document that says the same thing in different words scores low. It is precision-oriented (how much of what the candidate said also appears in the reference) with a brevity penalty, and the values reported here use $N = 4$.

On exact phrasing matched against the reference, Chat documents share marginally more 4-gram overlap than Form or Ask (Chat 0.143, Form 0.129, Ask 0.127). The

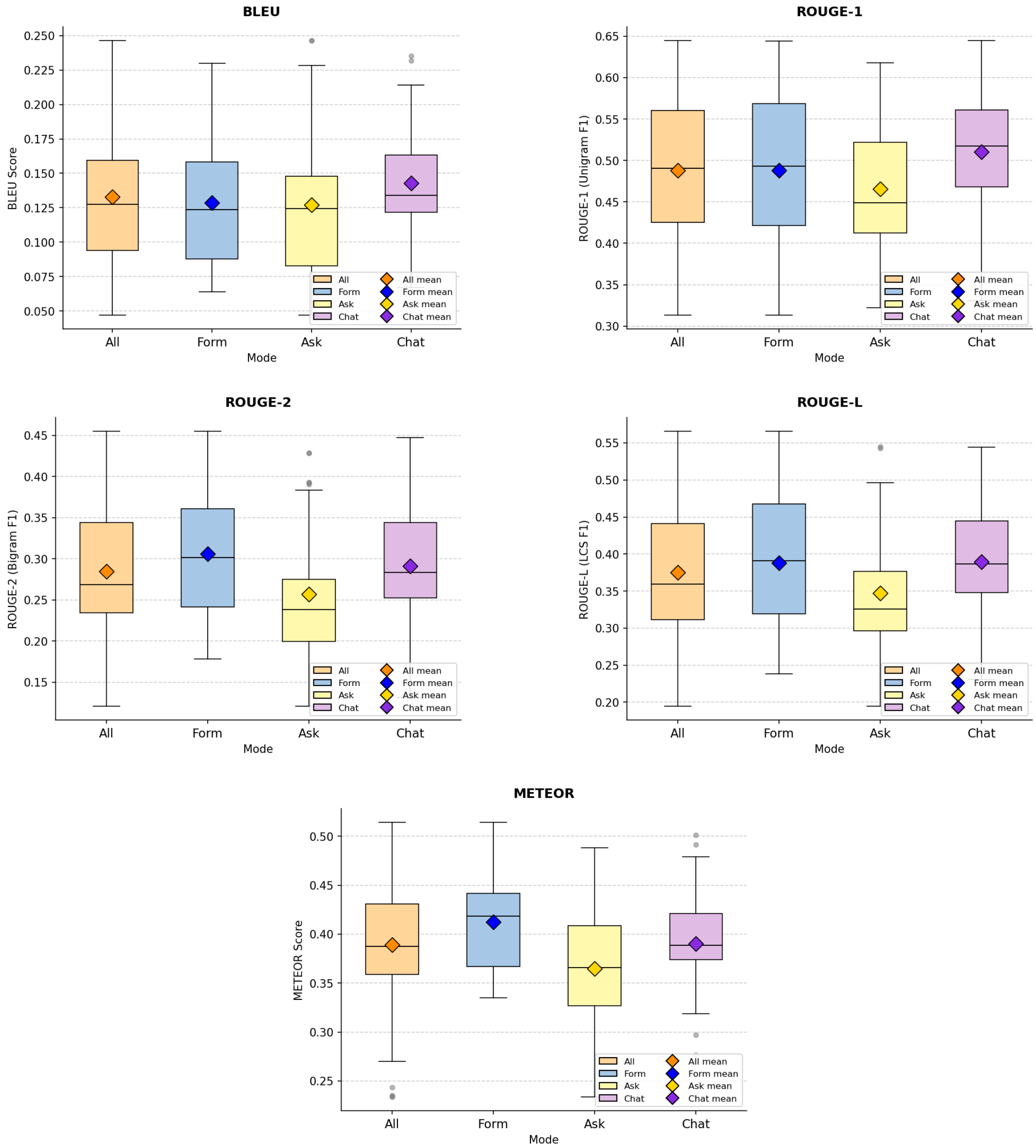


Figure 4.15: Document-level distributions of BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and METEOR by mode (full sample $n = 113$), one plot per metric. Boxes show the interquartile range; the diamond marks the mean.

gap is small, a slight reversal of the Form-leading pattern that does not reach statistical significance on the full sample.

ROUGE-1, ROUGE-2, ROUGE-L. ROUGE *(Recall-Oriented Understudy for Gisting Evaluation)* ((3.2)–(3.4)) measures how much of the reference's wording the candidate recovers, in three flavors: unigrams, bigrams, and the longest common subsequence. Each is reported as F1, the harmonic mean of precision (how much of what the candidate said belongs) and recall (how much of what should have been said is there), so it is more permissive than BLEU in word order but still surface-form.

On single-word overlap the three modes are evenly spread (Chat 0.511, Form 0.488, Ask 0.465); the omnibus does not reach significance. For two-word phrases, Form reproduces the reference's wording most often (Form 0.306, Chat 0.291, Ask 0.257). The Ask trough is wide enough to register as statistically significant against both Form and Chat. For longest-shared-sequences, Form (0.388) and Chat (0.389) track the reference equally well, with Ask trailing (0.347). The mode effect reaches significance overall, though no pairwise comparison survives Bonferroni correction.

METEOR. METEOR *(Metric for Evaluation of Translation with Explicit ORdering)* ((3.5)) softens BLEU by crediting matches on stems and WordNet synonyms in addition to identical tokens, so paraphrase with related words is no longer punished as if it had said something unrelated. It is a fragmentation-penalized harmonic mean of unigram precision and recall, sitting between BLEU and ROUGE in strictness. With stems and synonyms credited, Form still uses the most reference-like terminology (Form 0.412, Chat 0.390, Ask 0.365). Form is significantly higher than Ask on the full sample; the Form-versus-Chat and Chat-versus-Ask comparisons fall short of significance.

Three of the five lexical metrics (ROUGE-2, ROUGE-L, METEOR) reach statistical significance on the full sample; BLEU and ROUGE-1 do not. Full test statistics in Appendix C.

BERT-Based and Sentence-Embedding Metrics

BERT-based metrics measure semantic rather than lexical similarity by comparing contextual embeddings rather than literal tokens. Where the lexical metrics penalize paraphrase, the embedding-based metrics credit semantic overlap and therefore offer a complementary view of the same documents. Figure 4.16 shows the corresponding distributions.

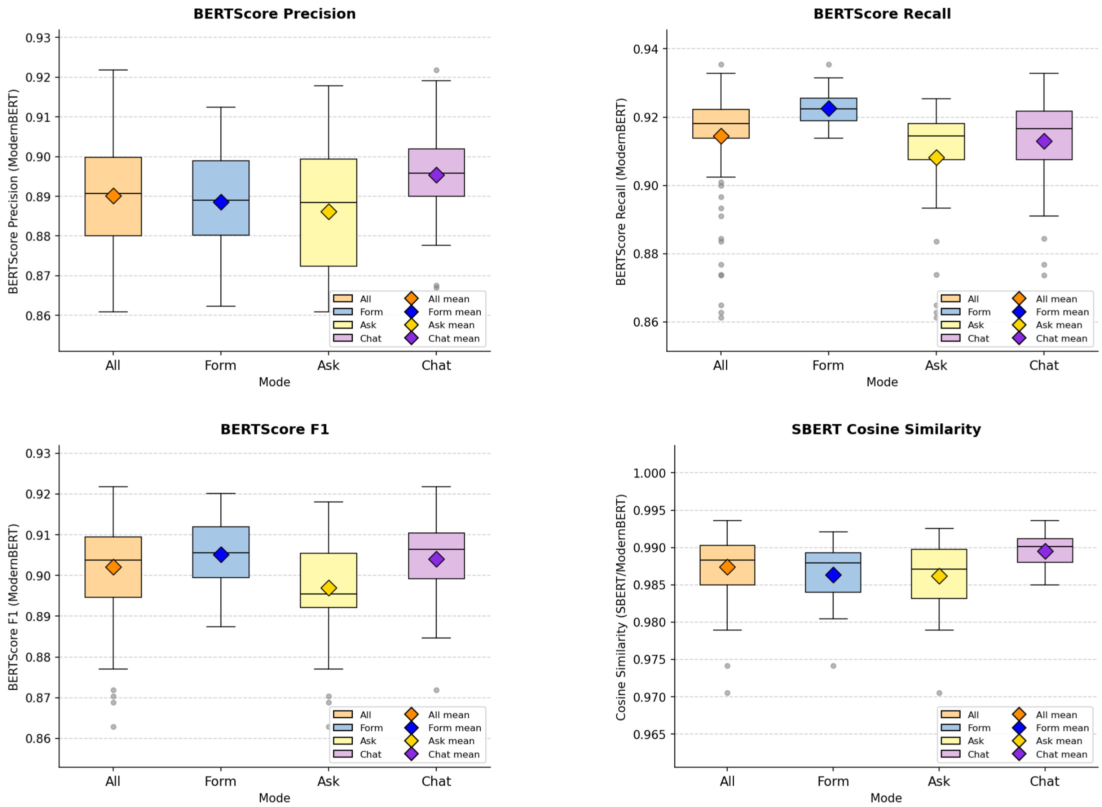


Figure 4.16: Document-level distributions of BERTScore Precision, Recall, F1, and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$), one plot per metric.

BERTScore Precision, Recall, F1. BERTScore ((3.6)–(3.8)) replaces n-gram matching with cosine similarities between contextual token embeddings, so paraphrase that shares meaning is credited even without surface-form overlap. *Precision* measures how much of what the candidate said belongs (penalizing spurious content), *recall* measures how much of what should have been said is there (penalizing omission), and *F1* is their harmonic mean (both must be reasonable for F1 to be high).

On Precision (how much of what the candidate said also appears in the reference), Chat documents stay tightest to the reference (Chat 0.895, Form 0.889, Ask 0.886). The mode effect reaches significance overall, but no pairwise comparison survives Bonferroni correction.

On Recall (how much of what the reference says is recovered by the candidate), Form documents lead clearly, reproducing most of the reference's content (Form 0.923, Ask 0.908, Chat 0.913). Form is significantly higher than both LLM-driven modes. Two features of Recall stand out. Form's standard deviation is the smallest of the three modes (SD 0.005): every Form document scores within a narrow band. Ask has the widest spread (SD 0.017) with a long lower tail; Figure 4.16 shows several Ask outliers well below the rest of the distribution. Some Ask participants produced documents as complete as Form; others left noticeable gaps. F1 is the harmonic mean of Precision and Recall, and shows the trade-off the two metrics imply. Form (0.905) and Chat (0.904) are effectively tied, with Ask trailing (0.897). The Ask trough is statistically significant against both other modes; Form and Chat are indistinguishable. The reading is the coverage-vs-discipline trade-off: Form casts a wide net and reproduces more of the reference content, but also includes more material the reference does not (lower Precision); Chat stays tightly on topic but leaves more out (lower Recall); their strengths balance out on F1, and Ask underperforms on both.

SBERT (ModernBERT) cosine similarity. Sentence-BERT cosine similarity ((3.9)) compresses each document into a single embedding vector and reports how aligned the two vectors are, on a scale where 1 is identical meaning and 0 is unrelated. It is the most lenient of the metrics reported here: as the comparison is at the document level rather than the token level, two documents that cover the same topics with different wording typically obtain values close to 1.

Read at document level as a whole, Chat sits closest to the reference (Chat 0.990, Form 0.986, Ask 0.986). Chat is significantly higher than both other modes. All four embedding-based metrics reach statistical significance on the full sample. At the document level the user-study Form-leading verdict breaks into a trade-off. Form documents recover more of what the reference *says* (the recall-oriented and bigram-sensitive metrics), Chat documents stay closer to what it *means* (the embedding-based metrics), and Ask is the lower of every significant pairwise comparison: significantly below both Form and Chat on ROUGE-2 and BERTScore F1, and significantly below at least one of them on METEOR, BERTScore Recall, and SBERT. The Form-vs-Chat verdict no longer maps onto a single winner; it depends on which question the metric is asking.

4.3.2 Chapter-Level NLP Metrics

At this granularity each participant chapter is scored against the matching chapter of the reference document, so the per-mode differences can be localized to specific parts of the ROPA. The six rendered chapters produced by the `finalropa` prompt are listed in Table 4.6; they derive from the eight internal `DocumentData` groups described in Chapter 3 (Section 3.1), with the data-categories and data-sources groups merging into one output chapter, the metadata group dropped, and the free-text fields distributed into the purpose and T&O chapters.

Table 4.6: The six rendered ROPA chapters. German titles are the verbatim form participants saw; English glosses are provided for the international reader.

#	German title	English gloss
1	Verantwortliche Stelle und Kontaktdaten	Controller and Contact Details
2	Zwecke und Rechtsgrundlagen der Verarbeitung	Purposes and Legal Bases of Pro
3	Kategorien personenbezogener Daten und Datenquellen	Categories of Personal Data and
4	Betroffene Personen und Empfänger	Data Subjects and Recipients
5	Aufbewahrungsfristen und Löschung	Retention Periods and Deletion
6	Technische und organisatorische Maßnahmen	Technical and Organizational Mea

Per-chapter, per-mode tables are in Appendix B.6 (lexical metrics in Appendix B.6.1, embedding-based metrics in Appendix B.6.2). The chapter-level box plots are in Figures 4.17 and 4.18.

Lexical Metrics

Per-chapter, per-mode means for the five lexical metrics are tabulated in Appendix B.6.1; Figure 4.17 shows the distributions.

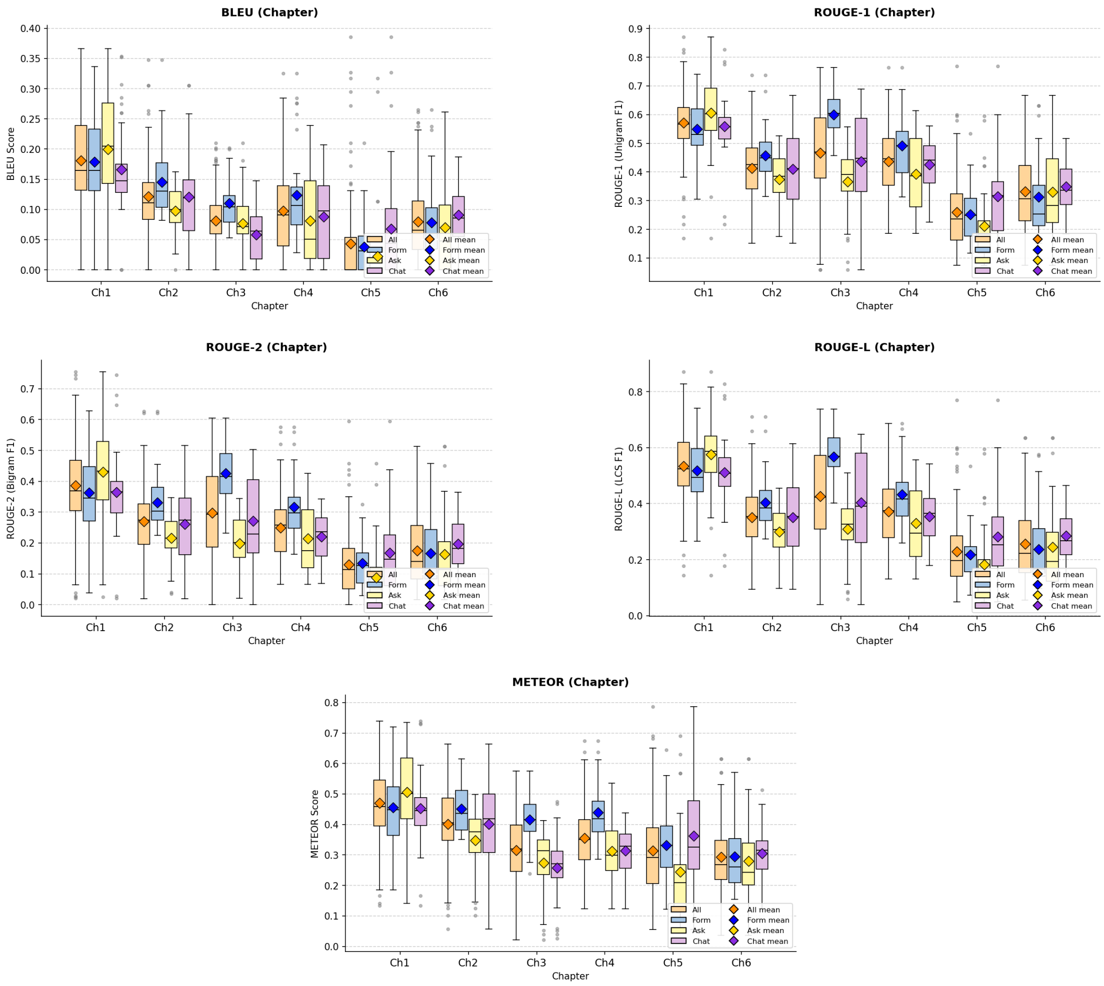


Figure 4.17: Chapter-level distributions of BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and METEOR by mode (full sample $n = 113$), one plot per metric showing all six ROPA chapters side by side.

Across modes the chapter-level lead splits cleanly into three zones, and at the lexical level the split holds on every metric without exception: Ask wins Chapter 1, Form wins Chapters 2–4, Chat wins Chapters 5 and 6. Across chapters, Chapter 1 sits at the head of every metric and Chapter 5 carries the widest standard deviations.

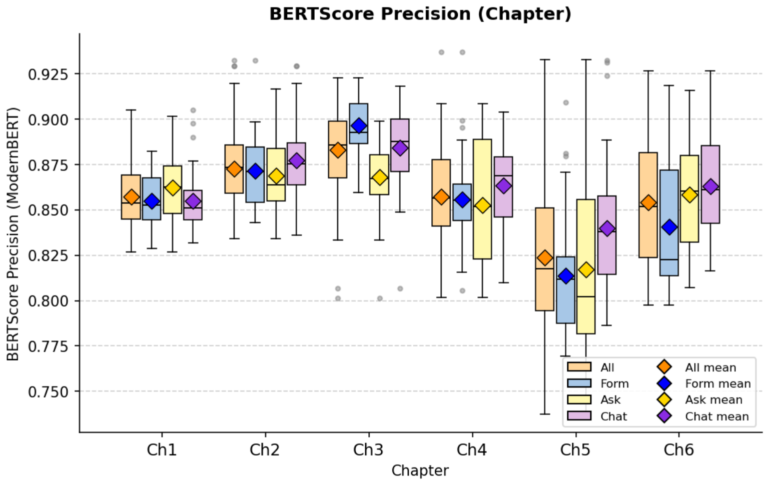
Chapter 1 (Controller and Contact Details) scores highest in absolute terms (BLEU

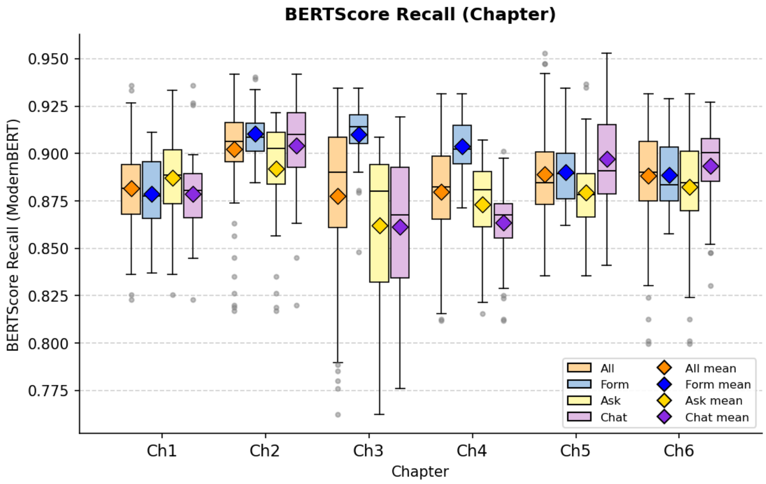
0.181, ROUGE-1 0.571, METEOR 0.471) and is the one chapter where Ask edges the other two modes (ROUGE-1 Ask 0.605 vs Form 0.550, Chat 0.559); the lead is modest but consistent across every lexical metric. The three middle chapters (Purposes and Legal Bases, Categories of Personal Data, Data Subjects and Recipients) all favor Form, with the gap widest on Chapter 3. There Form's ROUGE-1 of 0.600 leads Ask (0.366) by the largest within-chapter gap of the lexical results, and Form also takes BLEU (0.110 vs Ask 0.077, Chat 0.058) and METEOR (0.415 vs Ask 0.275, Chat 0.258). Chapters 2 and 4 follow the same Form-leading pattern on a smaller scale (Ch2 ROUGE-1 0.457 vs Ask 0.373, Chat 0.410; Ch4 ROUGE-1 0.491 vs Ask 0.393, Chat 0.426). The last two chapters flip to Chat. Chapter 5 (Retention and Deletion) has the lowest absolute scores and the widest variance of the six, with Chat ahead on every lexical metric (ROUGE-1 0.314 vs Form 0.252, Ask 0.212; METEOR 0.362 vs 0.332, 0.245) and METEOR's standard deviation of 0.160 the largest of any chapter. Chapter 6 (Technical and Organizational Measures) shows the same Chat lead at a smaller scale (ROUGE-1 0.350 vs Form 0.313, Ask 0.330; METEOR 0.305 vs 0.295, 0.280).

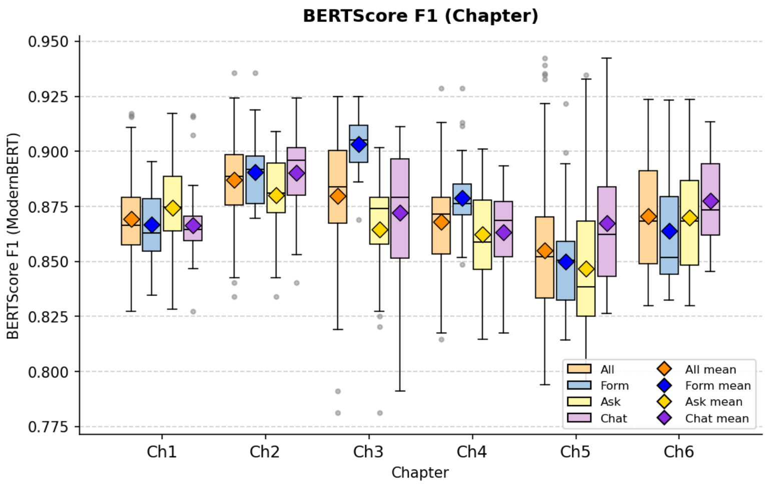
BERT-Based and Sentence-Embedding Metrics

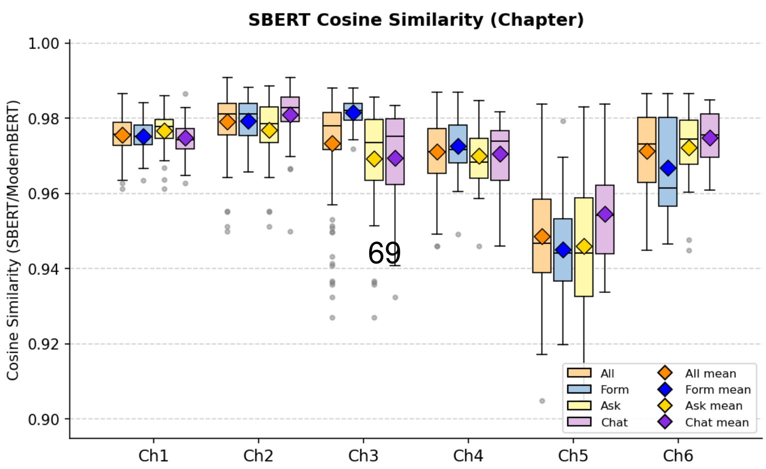
Per-chapter, per-mode means for the three BERTScore variants and the SBERT cosine similarity under both encoder backbones (ModernBERT primary, plus a chapter-only `all-MiniLM-L6-v2` robustness check) are tabulated in Appendix B.6.2; Figure 4.18 shows the BERTScore and SBERT (ModernBERT) distributions. The MiniLM pass is in the appendix tables only.

The embedding-based view keeps the same three-zone pattern as the lexical metrics but compresses the absolute range: Ask still wins Chapter 1, Form still wins Chapters 2–4, Chat still wins Chapters 5 and 6. The document-level Form-leads-Recall / Chat-leads-Precision trade-off shows up here too, but localized to Chapters 2 and 4 rather than running across all metrics; on the other chapters one mode sweeps the BERT family. Chapter 1 stays at the top of the BERT range (F1 0.869, SBERT ModernBERT 0.976) with Ask still ahead; the smaller MiniLM encoder makes the Ask lead easier to see (SBERT MiniLM Ask 0.736 vs Form 0.663, Chat 0.662). The middle chapters favor Form, but with a per-metric trade-off appearing on Chapters 2 and 4 specifically. On Chapter 2, Form leads Recall (Form


BERTScore Precision (Chapter)
BERTScore Precision (ModernBERT)
Chapter
Ch1 Ch2 Ch3 Ch4 Ch5 Ch6
All
Form
Ask
Chat
All mean
Form mean
Ask mean
Chat mean


BERTScore Recall (Chapter)
BERTScore Recall (ModernBERT)
Chapter
Ch1 Ch2 Ch3 Ch4 Ch5 Ch6
All
Form
Ask
Chat
All mean
Form mean
Ask mean
Chat mean


BERTScore F1 (Chapter)
BERTScore F1 (ModernBERT)
Chapter
Ch1 Ch2 Ch3 Ch4 Ch5 Ch6
All
Form
Ask
Chat
All mean
Form mean
Ask mean
Chat mean


SBERT Cosine Similarity (Chapter)
Cosine Similarity (SBERT/ModernBERT)
Ch1 Ch2 Ch3 Ch4 Ch5 Ch6
All
Form
Ask
Chat
All mean
Form mean
Ask mean
Chat mean

0.911 vs Ask 0.892, Chat 0.904), Form and Chat are effectively tied on F1 (0.891 vs 0.890), and Chat takes a narrow Precision lead (0.877 vs Form 0.872). Chapter 4 shows the same shape more clearly: Form leads Recall (0.904 vs Ask 0.873, Chat 0.864) and F1 (0.879 vs Ask 0.862, Chat 0.863), while Chat leads Precision (0.864 vs Form 0.856, Ask 0.853) and SBERT MiniLM (0.733 vs Form 0.691, Ask 0.692). Chapter 3 is the cleanest sweep: Form is highest on every BERT-based metric (F1 0.903 vs Ask 0.865, Chat 0.872; Recall 0.910 vs Ask 0.862, Chat 0.861).

The last two chapters belong to Chat. Chapter 5 sits at the lower end of the BERT range (F1 0.855, SBERT ModernBERT 0.949) with Chat ahead on every BERT-based metric, and the MiniLM SBERT pass widens the gap visibly (Chat MiniLM 0.566 vs Form 0.516, Ask 0.474). Chapter 6 stays back in the top tier (F1 0.871, SBERT ModernBERT 0.971) with Chat narrowly ahead on every BERT-based metric (F1 0.878 vs Form 0.864, Ask 0.870).

Cell-wise Kruskal–Wallis tests on the full $n = 113$ sample confirm the descriptive picture: most chapter-level mode differences are significant. The largest single gap is Form's ROUGE-1 lead over Ask on *Categories of Personal Data*, with a rank-biserial r near 0.95. Per-cell test statistics are kept in the pipeline output files; the appendix focuses on the document-level inferential layer.

4.3.3 Summary of Document-Level and Chapter-Level NLP Metrics

The two granularities tell the same story at different resolutions. At the document level the trade-off runs across metrics: Form leads the recall-oriented and bigram-sensitive metrics (most clearly BERTScore Recall), Chat leads the embedding-based ones (BERTScore Precision and SBERT), and Ask is the lower of every significant pairwise comparison. At the chapter level the same trade-off runs across chapters: Ask wins Chapter 1, Form wins Chapters 2–4, Chat wins Chapters 5 and 6, and the Form-leads-Recall / Chat-leads-Precision split re-emerges specifically on Chapters 2 and 4. Document-level means are higher than the average of chapter-level means (Table 4.7) because the document-level calculation pools tokens across all six chapters and smooths out the per-chapter dips. The chapter-level view shows *where* the modes differ; the document-level view ranks them *overall*. The reference-based metrics neither confirm nor contradict the Form-leading verdict from the user study (Sections 4.2.1–4.2.4). They deliver a trade-off instead: Form documents recover more of what the reference *says*, Chat documents stay closer to what it *means*, and Ask trails on every front.

A final observation cuts across metric families. The five lexical metrics and the four BERT-based metrics measure different things, surface-form n-gram overlap on the one hand and contextual semantic similarity on the other, but they order the modes the same way. Lexical recall (ROUGE-Recall) and semantic recall (BERTScore Recall) both place Form ahead; lexical-precision oriented BLEU and semantic Precision (BERTScore Precision) both align with Chat at the document level; the balanced-similarity metrics (METEOR, BERTScore F1, SBERT) sit in between. The trade-off therefore survives the choice of measurement paradigm: it is a property of the documents rather than of the metric family used to score them.

Table 4.7: Document-level versus mean chapter-level metric values by mode. The mean chapter-level value is the unweighted average across the six chapters. Differences are document-level minus chapter-level mean. Source: `doc_vs_chapter_diff.csv`.

	Form		Ask		Chat	
Metric	Doc	Chapter	Doc	Chapter	Doc	Chapter
BLEU	0.129	0.112	0.127	0.091	0.143	0.099
ROUGE-1	0.488	0.444	0.465	0.380	0.511	0.416
ROUGE-2	0.306	0.290	0.257	0.218	0.291	0.247
ROUGE-L	0.388	0.396	0.347	0.324	0.389	0.365
METEOR	0.412	0.398	0.365	0.328	0.390	0.349
BERTScore Precision	0.889	0.856	0.886	0.855	0.895	0.864
BERTScore Recall	0.923	0.897	0.908	0.879	0.913	0.883
BERTScore F1	0.905	0.876	0.897	0.866	0.904	0.873
SBERT (ModernBERT)	0.986	0.970	0.986	0.968	0.990	0.971

4.4 Document Quality — LLM-as-a-Judge

The reference-based NLP metrics measured similarity to a single expert-authored Form-mode reference. The LLM-as-a-Judge layer turns to a reference-free quality signal: each document is evaluated against a rubric rather than a target, which lifts the implicit advantage that the reference document confers on Form-style outputs. The judge produces five metrics per document, summarized in Table 4.8: *Completeness*, *Scenario Faithfulness*, *Legal Correctness*, *Hallucination* (inverse-coded; higher equals less hallucination), and *Overall*. The full rubric, prompt, and model configuration are in Appendix A.4.2; the rubric design follows the precedent set by Su et al.'s JuDGE benchmark for LLM-evaluated legal-document generation [30]. The judge is a single LLM (Gemini 3.1 Pro Preview, temperature 0); without inter-rater replication or legal-expert validation the resulting scores are reported as a supplementary, indicative signal alongside the user-study instruments and reference-based NLP metrics. All 113 retained documents are scored.

Table 4.8: The five LLM-as-Judge metrics. All share a 1–5 Likert anchor; higher is better, including for *Hallucination (inverse)*, which is inverted internally so that fewer fabrications score higher. The full rubric is in Appendix A.4.2.

Metric	What is assessed
Completeness	Coverage of the Art. 30(1) GDPR required elements (controller, DPO, purposes, legal basis, data and recipient categories, retention, technical and organizational measures); missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of the study scenario (APOR GmbH, $\sim$50 employees, internal hosting, 3-month deletion, etc.); generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis under Art. 6(1); wrong articles or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Absence of invented facts not supported by the scenario; 5 = no hallucinations, 1 = many.
Overall	Holistic judgment of how useful the document would be as a real Art. 30 ROPA record, weighing the four metrics above.

4.4.1 Per-Metric Results

Table 4.9 reports the per-metric mean and standard deviation per mode; Figure 4.19 shows the corresponding distributions. The five rubric metrics are walked through in the order Completeness, Scenario Faithfulness, Legal Correctness, Hallucination, Overall, so that the holistic Overall verdict is read after the four contributing metrics.

Table 4.9: LLM-as-Judge metric mean (SD) by mode (full sample $n = 113$, scale 1–5, higher is better including for *Hallucination (inverse)*). The Aggregated column is the mean across all 113 documents.

Metric	Aggregated	Form	Ask	Chat
Completeness	3.94 (1.43)	4.84 (0.55)	3.84 (1.54)	3.18 (1.59)
Scenario Faithfulness	2.21 (1.10)	1.84 (0.76)	2.30 (1.20)	2.49 (1.19)
Legal Correctness	3.39 (1.18)	3.59 (0.55)	3.43 (1.34)	3.15 (1.42)
Hallucination (inverse)	2.20 (1.02)	1.65 (0.72)	2.40 (1.12)	2.54 (0.91)
Overall	2.50 (1.04)	2.41 (0.69)	2.49 (1.04)	2.62 (1.31)

Form documents cover the Article 30(1) requirements most thoroughly and most uniformly. Form scores 4.84 (median 5.0, SD 0.55), Ask sits at 3.84, and Chat at 3.18 (median 3.0); the 1.66-point spread is the widest of any judge metric. The Form lead here mirrors the BERTScore Recall pattern from Section 4.3.1. The Form lead flips on whether the document actually reflects the APOR scenario participants were given. Chat documents stay closest (2.49), Ask second (2.30), and Form lowest (1.84): the scenario-fidelity ordering is the exact inverse of the Completeness ordering. Form documents use GDPR terminology most correctly (Form 3.59, Ask 3.43, Chat 3.15), with a much narrower spread than on Completeness, around 0.45 points. Form's standard deviation (0.55) is again the smallest, the same uniformity pattern as on Completeness. Form documents contain the most fabricated content and Chat the least: Form sits at 1.65 on the inverse-coded scale (lower means more hallucination), Ask at 2.40, Chat at 2.54. The co-occurrence with Completeness is striking: 31 of 37 Form documents ($83.8\,\%$) achieve Completeness $= 5$ while simultaneously scoring Hallucination (inverse) ≤ 2 (“moderate” or worse per the rubric), and 17 of those combine maximum Completeness with the minimum Hallucination (inverse) score of 1. Most Form documents are simultaneously the most complete and the most fabricated. Compressing the four contrasting orderings into a single holistic verdict produces no clear winner: Chat edges ahead at 2.62, Ask sits at

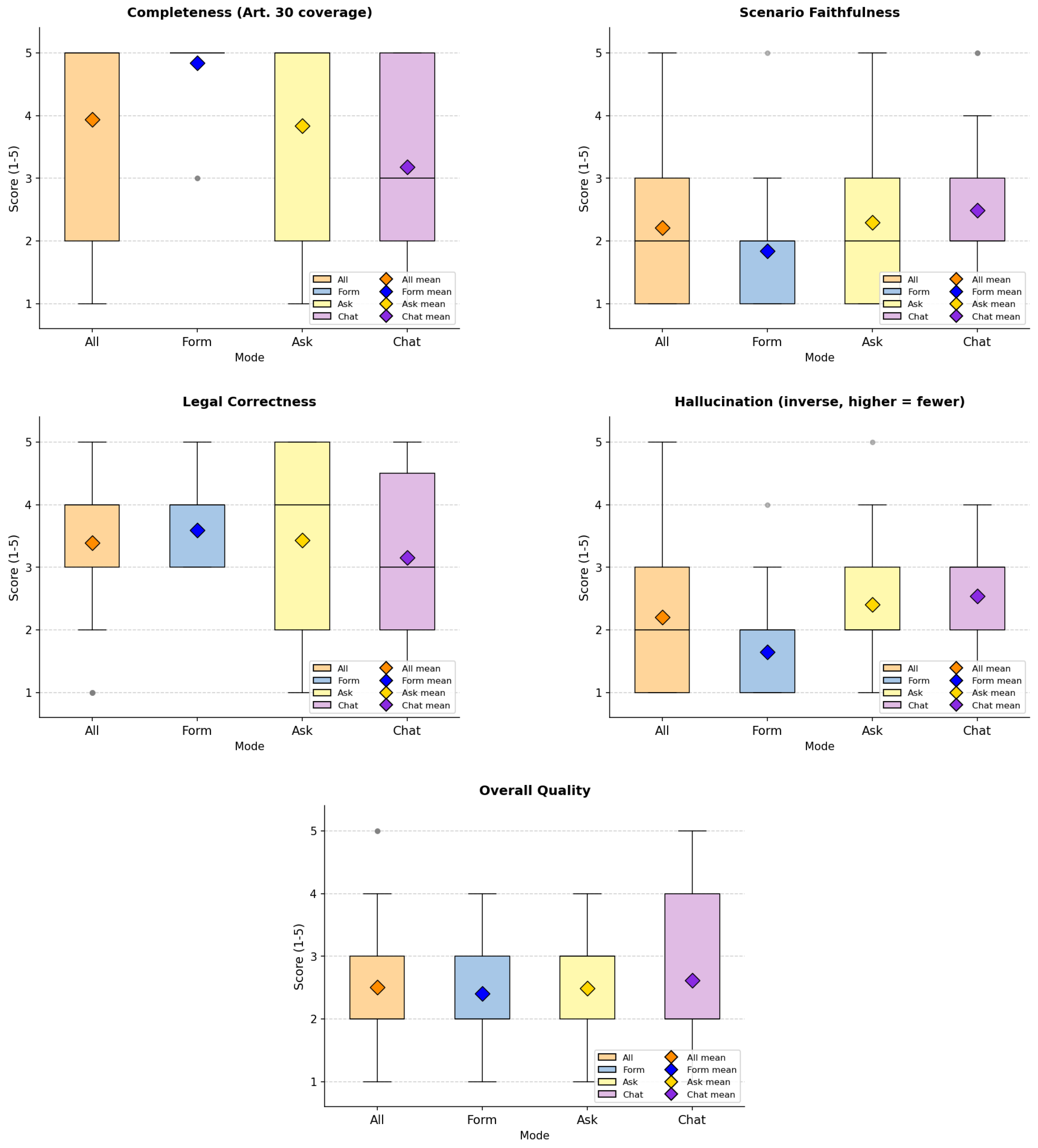


Figure 4.19: Distributions of the five LLM-as-Judge metrics by mode (full sample $n = 113$, scale 1–5), one plot per metric.

2.49, and Form trails at 2.41. All three modes sit below the midpoint of the 1–5 rubric scale and the spread between best and worst is just 0.21 points. Form's standard deviation (0.69) is again the smallest; Chat's (1.31) is the largest, meaning Chat produces both the highest- and lowest-rated documents in the sample.

Three patterns emerge. First, the Completeness ordering (Form $>$ Ask $>$ Chat) is the exact inverse of the Scenario Faithfulness and Hallucination orderings (Form $<$ Ask $<$ Chat): Form documents are the most structurally complete but also the most fabricated, Chat documents the most faithful to the scenario but the least complete, Ask between the two extremes on every metric. Second, Legal Correctness is flat relative to the Completeness spread (roughly 0.45 versus 1.66 points). The judge distinguishes modes more sharply on whether the required Article 30(1) fields are present than on whether the legal terminology itself is appropriate. Third, the holistic Overall scores compress the per-metric spread into a narrow 0.21-point range. Form's high Completeness is offset by its low Faithfulness and high Hallucination, leaving Chat with the smallest of advantages. Where the NLP metrics surfaced a recall-versus-precision trade-off between Form and Chat, the judge surfaces a completeness-versus-faithfulness trade-off, and the holistic Overall verdict collapses both into a narrow range with no decisive winner. The judge layer therefore adds a second trade-off to the document-quality picture begun in Section 4.3.1: Form leads on Completeness and Legal Correctness, Chat leads on Faithfulness and Hallucination, and Ask sits between the two on every metric. The Form Completeness-versus-Hallucination co-occurrence is the central judge-side finding; its interpretation in terms of the checkbox-only Form interface is reserved for the Discussion chapter. The full user-study and document-quality findings are consolidated in Section 4.5.

4.5 Summary

The chapter has reported the four user-study instruments (Sections 4.2.1–4.2.4), the reference-based NLP metrics at both document and chapter resolutions (Sections 4.3.1–4.3.2), and the supplementary LLM-as-a-Judge layer (Section 4.4). This section consolidates the findings and names the questions Discussion takes up.

On the user-experience side, the four instruments converge on a stable Form-leading ordering. Form is rated the most usable mode on the System Usability Scale, the lightest on R-TLX workload, the most often placed first in the post-experiment ranking ($53.6\,\%$ of rank-1 votes, $\chi^2(2) = 14.17$, $p < 0.001$, Cramér's $V = 0.32$), and the mode participants would most willingly use again. The pattern is partial: three items break it. Perceived transparency, trust in the LLM's suggestions, and the primary self-report confidence measure ("I am confident that the AI identified the correct legal basis") all favor the LLM-driven modes, with Ask leading on transparency and trust and Chat leading on legal-basis confidence. The behavioral counterpart is sharper still: Form is measurably about seven minutes per document faster than either LLM-driven mode ($H = 16.46$, $p < 0.001$, $\varepsilon^2 = 0.15$, large), and Ask and Chat are statistically indistinguishable on completion time. Ask is most often placed third in the ranking, and free-text comments single out the manual copy-paste step between explanation and form field as the principal source of friction.

The reference-based NLP metrics split the modes into a trade-off rather than a single ranking. Form leads on recall-oriented measures, most clearly on BERTScore Recall (Form significantly higher than both Ask and Chat), with ROUGE-2 and METEOR echoing it on smaller margins. Chat leads on document-level semantic alignment, most clearly on SBERT cosine similarity (Chat significantly higher than both Form and Ask). Form and Chat trade leadership on BERTScore F1, while Ask underperforms both on every metric that reaches significance. The substantive reading is that Form documents recover more of what the reference *says*, Chat documents stay closer to what the reference *means*, and Ask documents do neither.

The chapter-level view distributes the document-level trade-off across chapters rather than across metrics. *Controller and Contact Details* scores highest across all modes and is the one chapter where Ask leads. Form sweeps the middle chapters (*Pur-*

poses and Legal Bases, *Categories of Personal Data*, *Data Subjects and Recipients*) on every lexical metric and on BERTScore Recall and F1. Chat takes the last two chapters (*Retention and Deletion*, *Technical and Organizational Measures*) on every metric. The same trade-off that produces Form's document-level recall edge and Chat's document-level semantic edge thus appears at the chapter level as a clean three-way split by chapter.

The supplementary judge layer adds a GDPR-specific quality reading that the reference-based metrics cannot directly provide. Form documents are judged the most complete (Completeness mean 4.84 of 5) and the most legally correct, but the lowest on Scenario Faithfulness (1.84) and the most hallucinated; $83.8\,\%$ of Form documents combine maximum Completeness with "moderate" or worse hallucination, and nearly half combine maximum Completeness with the floor Hallucination score. Chat inverts the pattern: most faithful to the study scenario, least hallucinated, but lowest on Completeness. On the holistic Overall metric the three modes cluster narrowly (2.41 to 2.62) and all sit below the rubric midpoint; the judge does not declare a winner.

Two tensions surface from the consolidated picture. The mode participants most often prefer (Form) produces the most structurally complete documents, yet also the most fabricated ones: the checkbox scaffolding that pushes Completeness toward the ceiling appears not to constrain what the LLM invents during the final generation step. And self-report confidence in the legal basis rises monotonically from Form (3.58) through Ask (3.74) to Chat (3.81), while measured document quality follows no such single ordering: Form leads on recall, Chat leads on document-level semantic fidelity and on scenario faithfulness, and the judge declares no clear winner.

With the four user-study instruments and the two document-quality streams now consolidated, RQ1 and RQ2 are answered in full: RQ1 in the user-study findings reported above (usability, cognitive load, perceived AI support, mode preference, and free-text), and RQ2 across the reference-based NLP metrics and the supplementary judge layer. The cross-stream question of whether self-report confidence aligns with measured document quality (RQ3) is not answered by any single instrument; it is taken up in Chapter 5, which reads the two streams against each other.

5 Discussion

The Evaluation chapter answered RQ1 and RQ2 with the four user-study instruments and the two document-quality streams. This chapter takes up RQ3, the cross-stream question of whether self-report confidence aligns with measured document quality. It opens with the gap itself, turns to the architectural reading of Ask as a separate phenomenon that the gap framing does not on its own explain, and closes with the design implications and the work's limitations.

5.1 The Confidence–Correctness Gap

Self-report confidence in the LLM's content choices does not match the measured quality of the documents the LLM produces. The legal-basis confidence question, the primary confidence measure in the user-study battery, places Form lowest, Ask in the middle, and Chat highest. Measured document quality does not follow that ordering. Form leads on coverage-oriented metrics, Chat leads on precision-oriented and semantic ones, and the judge's holistic Overall metric declares no winner among the three. The mismatch is bidirectional, and both directions are sharper than the user-study composite or the NLP composite alone would suggest.

The mode users feel most confident about (Chat) produces the least complete documents. Chat is the mode where participants rate the AI's legal judgment most highly and where they would least like to come back. The judge agrees that Chat documents are the most faithful to the scenario and the least hallucinated, but reports the lowest Completeness scores of the three modes. Confidence is highest where coverage is lowest, and participants have no instrument in the interface that would signal that the documents they trust are also the most incomplete.

The mode users prefer to re-use (Form) produces the most fabricated documents.

Form is the mode participants most often rank first and most often nominate for re-use. The judge reports that most Form documents combine maximum Completeness with moderate-or-worse Hallucination, and a substantial share combine maximum Completeness with the rubric's floor Hallucination score. Preference is highest where fabrication is highest, and the interface that lets users feel productive is also the interface that hides the LLM's largest failure mode from them.

The two-participant case study makes the gap concrete. IAR1116's Chat document reaches the case-study ceiling on the judge rubric: the document is rated as a tight, scenario-faithful, legally sound ROPA, and the NLP metrics agree that it is the participant's strongest output. The user-study record is internally consistent: the participant rated the Chat session highly on the AI-support items and reported the same in the free-text. All three streams align. AEB2451's Form document, by contrast, is the case-study floor on the judge rubric: maximum Completeness, but the lowest possible scenario-faithfulness rating and the lowest possible hallucination-inverse rating, with the judge noting that the document invents financial data, biometric data, video surveillance, and named technical standards that are absent from the scenario. The same participant's user-study record on the Form session reports the typical Form pattern of perceived productivity and high preference placement. The judge's verdict on this single document is exactly the phenomenon Form produces in aggregate.

Confidence and correctness are decoupled in this novice population. Novices are reasonable calibrators of which mode they want to keep using; they correctly land on Form for productivity. They are not reasonable calibrators of whether the LLM has produced a correct document. The interface that lets the LLM do the most work feels the most trustworthy on the content question and produces the tightest text, but at the cost of coverage the user cannot audit. The interface that lets the user do the most visible work feels the most productive, but the LLM still fabricates around visible choices the user did not describe. The three streams disagree on which mode is best because they measure different things, and the disagreement itself is the substantive answer to RQ3.

5.2 Ask Underperformed Structurally

Where Form and Chat trade leadership, Ask trails on every NLP metric that reaches significance and sits at the middle-to-bottom of the judge's Overall scores. The free-text comments converge on a single explanation, the copy-paste friction between the chat panel and the per-section text field. That explanation is correct as a description of the user-facing symptom, but it is not the cause. The cause is upstream, in the design pattern Ask embodies.

Ask positions the LLM as a teacher: it explains GDPR concepts in plain language, offers examples, and answers follow-up questions, with a short token budget per turn. It does not extract structured data from the user's descriptions, does not draft section text, and does not emit any structured update for the front-end to merge into the document object; the implementation is verified against the ropagen source code and the relevant prompts are reproduced in Appendix A.1.3. The user is left to read a general conceptual explanation, map that explanation onto their own scenario, and write the section's free-text field themselves. For a participant with prior GDPR knowledge, that loop would work. For a novice, the population this study recruited, it does not. Conceptual explanation is not enough scaffolding to produce scenario-specific legal prose. The copy-paste friction is the visible trace of this gap: participants were copying general explanatory text and trying to adapt it, which is harder than starting from scratch because the LLM never produced scenario-specific text in the first place.

The mechanism by which the design pattern's failure becomes visible at the metrics level runs through what each mode puts into the document object the shared final-generation step receives. Of the three modes, Ask is the only one in which the predefined boolean scaffolding is never populated: the user fills only the per-section free-text field, and the LLM populates nothing. The document object that reaches generation in Ask is therefore the most sparsely populated of the three, and what gets generated reflects that. This is the consequence of the design choice, not its cause. The cause is the choice to use the LLM as an explainer rather than as a co-author. The substantive claim is therefore that Ask is a structural failure, not a technical one. The novice user cannot bear the authorial load that this design places on them, and the documents reflect that across every metric family.

5.3 Meta-finding and Implications for ROPA Tooling

Across the three modes a single design hierarchy organizes the findings. The LLM helps when it does work; it does not help when it tells the user to do the work themselves. When the LLM populates structure (Chat), it produces the most faithful individual documents. When the LLM acts as final-prose synthesizer over user-clicked structure (Form), it produces the most preferred user experience but the most fabricated documents. When the LLM is restricted to explainer-only (Ask), the novice user is left to author the document alone, and the documents reflect that. For novice document-creation users, the LLM has to co-author, not teach.

Three concrete tooling implications follow.

A hybrid mode is supported by the data, not only by user preference. Several participants spontaneously suggested a hybrid that combines Form's visible checkbox UI with Chat-style LLM extraction from a conversation. The data supports the suggestion beyond preference: Form delivers the strongest perceived productivity, and Chat delivers the strongest scenario faithfulness in the documents. A hybrid that retains the boolean scaffolding (so the user sees what is covered) and adds the LLM extraction step (so the document object reaching generation is populated by what the user described in conversation, not left to the model to write around bare category labels) would in principle inherit both strengths. Hallucination, which appears across all three modes in the judge data, would still need to be addressed separately; the checkbox-and-conversation hybrid does not solve it on its own.

A validation gate at submission time. The LLM-as-a-judge data show that a single-pass single-model judge can already discriminate hallucinated documents from faithful ones. A validation gate that runs such a judge against the submitted document and surfaces low Scenario Faithfulness or low Hallucination-inverse scores before the user clicks submit would give novice users the only signal they currently lack: whether the LLM has produced content that matches the scenario they described. The gate is not a replacement for expert review, but it is a plausible inline check on the most consequential failure mode.

An education layer for novice calibration. Self-report confidence in this study is not well-calibrated to document quality. An education layer that shows novice users which fields of the final document came from their input and which came from the

LLM's generation step, with simple flags on fields the user did not explicitly describe, would surface the calibration mismatch in the interface rather than leave it for the user to detect.

5.4 Limitations

The study's design choices bound the findings in five concrete ways.

Single scenario. Every participant produced ROPAs for the same fictional scenario (APOR GmbH, employee and parking management). The scenario covers the GDPR Article 30 fields and is accessible to novices, but the findings are tied to its specific shape. A scenario that asked for more retention specifics, less checkbox-mappable content, or fewer numbered legal references might shift the chapter-level distribution of leadership, even if the direction of the trade-off survives.

Novice-only sample. Participants were university students, IT-leaning, chatbot-fluent, and self-reported as ROPA-novice. The findings are explicitly novice-applicable. Whether the design hierarchy generalizes to experts is an open question. A GDPR-trained data protection officer presented with the same three modes might prefer Ask, because the conceptual-explainer loop is a faster path to a document when the user already knows the legal vocabulary.

Single reference document. The NLP metrics score against a single expert-authored Form-style reference. Form documents may score higher on lexical metrics partly because they share the reference's surface form rather than because they share its content. Two pieces of evidence suggest the framing bias does not exhaust the Form Recall lead: the BERT-based metrics show the same direction, and the judge (which does not use the reference at all) reports the same Completeness pattern.

Single judge model. The LLM-as-a-judge layer is a single-pass, single-model assessment without inter-rater replication or legal-expert validation. It is reported as a supplementary signal throughout the Evaluation chapter, and its scores are not subjected to inferential testing. The judge's substantive claims align with the NLP metrics and with the case study, which gives the judge layer a triangulation grounding it would not have on its own, but the judge is not a substitute for expert review.

German-primary corpus, English study materials. The ROPA documents are predominantly German with English glosses; the reference document is German; the judge prompts are in English. The metric absolute values, particularly the lexical ones, are affected by this mixture. The direction of the per-mode trade-off survives across the metric families, which suggests the language mix is not the primary driver, but cross-language calibration of the NLP pipeline is a follow-up this study does not perform.

5.5 Summary

The chapter has taken up RQ3 by reading the user-study, document-quality, and judge streams against each other. Confidence and correctness are decoupled in this novice population: the mode users feel most confident about produces the least complete documents, and the mode users prefer to re-use produces the most fabricated ones. The two-participant case study makes the gap concrete at human scale. What links the three streams is how each mode populates the document object the shared final-generation step receives. Form lets the user populate the scaffolding directly; Chat lets the LLM populate it from the conversation; Ask leaves the scaffolding untouched and the user writes only the per-section free-text fields. Ask underperforms because the novice user cannot bear sole authorship of a GDPR document. The thesis-level claim that follows is that LLM-assisted document tooling for novices should position the LLM as a co-author rather than as a teacher, and that visible structure with LLM-driven population is the most promising hybrid the data supports.

The Conclusion chapter (Chapter 6) returns to RQ1, RQ2, and RQ3 in plain English and draws the design lessons this Discussion has framed.

6 Conclusion

A ROPAgen and Judge Model Configuration

This appendix collects the static configuration of both LLMs that the study relies on. The first three sections cover ROPAgen — the tool under study — and reproduce its backend prompts, its three mode screenshots, and the expert-authored reference ROPA document that the NLP metric pipeline scores every participant document against. The fourth section covers the supplementary LLM-as-a-Judge evaluation layer — its system and user prompts, its five-metric rubric, and the deterministic-decoding model configuration. All four sections are referenced from the Methodology chapter (Section 3.1 for the tool, Chapter 3 for the judge).

A.1 ROPAgen Backend Prompts

ROPAgen sends one of **27 system prompts** to Mistral Large 3 on every user turn, depending on the active mode and the current section of the ROPA being edited. All prompts are stored as static templates in the project's `data/prompt.json` and assembled at runtime in `src/actions/actions.ts`: the chosen template is concatenated with a context block listing the document title, the organization metadata, and the current state of the `DocumentData` object before the call is issued. The texts below reproduce the template strings only; the runtime context block (which contains organization- and user-specific information) is not reproduced, as its content varies per call. Conversational and per-section suggestion calls use temperature 0.2 and top-p 0.5; the single `finalropa` call that produces the final ROPA document uses temperature 0.05 and top-p 0.05.

A.1.1 Final ROPA Generation Prompt (`finalropa`)

The `finalropa` prompt is sent verbatim as the `system` role on the call that generates the final ROPA document from the assembled `DocumentData` object. It is invoked once per document, identically across all three modes; temperature is fixed at 0.05 and top-p at 0.05. The `Document Title`, `Language`, and the `DocumentData` contents are interpolated into the user message at runtime and are not reproduced here.

```
You are an expert in GDPR compliance. Generate a professional, structured
↪ Records of Processing Activities (ROPA) document in full compliance with the
↪  EU General Data Protection Regulation (GDPR).

This ROPA document must be suitable for official regulatory review and written
↪ in formal business language.

**CRITICAL: You MUST write the ENTIRE document in the language specified in the
↪  context (Language field). ALL headings, text, descriptions, and content
↪ must be in that language. Do NOT mix languages.**

**CRITICAL: Write ALL free text and general text from the company's perspective
↪  (first-person plural: 'we', 'our', 'us'). The document represents the
↪ company's official record, so describe what the company does, how the
↪ company processes data, what measures the company has implemented, etc. Use
↪ active voice from the organization's viewpoint.**

**CRITICAL: The "Additional Information" field is NOT content to be copied
↪ directly into the document. Instead, treat it as ADDITIONAL INSTRUCTIONS and
↪  REQUIREMENTS from the user about what must be included, emphasized, or
↪ addressed in the generated ROPA document. Incorporate these instructions
↪ into the appropriate sections of the document naturally and professionally
↪ .**

Instructions:
- Create a comprehensive ROPA document structure with the following sections:
1. Organization and Controller Information
2. Purpose and Legal Basis for Processing
3. Data Categories and Sources
4. Data Subjects and Recipients
5. Data Retention and Deletion
6. Technical and Organizational Measures
```

```
- Each category can have a continous text and bullet points as needed.
- Ensure compliance with GDPR Articles 30 (Records of processing activities)
↪ requirements.
- Respect and incorporate the Additional Information as special instructions -
↪ integrate those requirements into the relevant sections naturally.
- You must include all data given, and must not omit any data provided.

You must use the Document Title as Title of the generated Document.

The output should be suitable for use by a Data Protection Officer (DPO) or
↪ legal counsel.
The output should be in .md style with proper headings and formatting. You must
↪  only include data categories marked as true and must not include data
↪ categories marked as false. If all data is marked as false, just mention
↪ none or not specified. Omit wording or phrasing such as CATEGORY: true/false
↪ , but rather just include the relevant, true data categories. Critical
↪ Instructions:

Include all provided information in your response - do not omit any items from
↪ the list unless "false" or empty.
Handle "other" field: If an "other" field exists and contains data, treat it as
↪  a standard list item with equal importance. Do not use "other" as a label.
Formatting requirement: Present items as a simple list within each category. Do
↪  NOT use the format "data_category: description" - instead, list items
↪ directly without labeling the data structure.

Example:

### 4. Data Subjects and Recipients
- Person Categories:
    - Employees
    - USER INPUT

NOT LIKE THIS:

### 4. Data Subjects and Recipients
- Person Categories:
  - Employees
  - Other: Emilija

 CRUCIAL: Generally, You shall only inclued data that the user provided as true
↪  or non-empty in the final output. Do NOT invent or assume any data beyond
```

```
↪ what the user provided. If the user did not provide anything, don't explain
↪ why the field is empty. Just state that there is no information. Do not
↪ mention that they will be completed or added.

Generate the ROPA document now.
```

A.1.2 Form Mode — AI-Suggest Prompts

Form mode invokes the LLM only when the user clicks the per-section *AI Suggest* button. There is one prompt per section of the `DocumentData` schema; each is sent as the `system` role of a single-turn call with temperature 0.2 and top-p 0.5. The response is a JSON object whose keys match the corresponding section's field structure; the front-end auto-populates the form from this JSON.

Purpose of Data Processing (`form.purposeOfDataProcessing`)

```
You are an expert in GDPR compliance. Based on all the provided information
↪ about the organization, data sources, data categories, and other context,
↪ suggest a comprehensive purpose of data processing text.

**Consider the Document Title as an important data source** - it often contains
↪  valuable context about the specific processing activity or system involved,
↪  but analyze it alongside all other information provided.

Analyze all the information provided and generate a clear, professional
↪ description of why this data is being processed.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused.

Return ONLY a JSON object in this exact format:
{
  "purposeOfDataProcessing": "your suggested text here"
}

The text should be written in the specified language and be suitable for
↪ official GDPR documentation.
```

Technical and Organizational Measures (`form.technicalOrganizationalMeasures`)

```
You are an expert in GDPR compliance and data security. Based on all the
↪ provided information about the organization, data categories, processing
↪ purposes, and other context, suggest comprehensive technical and
↪ organizational measures.

**Consider the Document Title as valuable context** - it can indicate the type
↪ of processing and environment, helping you recommend appropriate security
↪ measures for the specific use case.

Analyze the data sensitivity and processing context to recommend appropriate
↪ security measures.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused on the most critical measures.

Return ONLY a JSON object in this exact format:
{
  "technicalOrganizationalMeasures": "your suggested text here"
}

The text should be written in the specified language and include specific,
↪ actionable security measures.
```

Legal Basis (`form.legalBasis`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which legal basis options should be selected and deselected for this
↪  data processing activity.

**Consider the Document Title as helpful context** - it may indicate the nature
↪  of processing and suggest potential legal bases, but evaluate it together
↪ with the purpose, organization type, and all other available information.

Analyze the purpose, data categories, and context to determine the most
↪ appropriate legal basis.

You can ONLY use these exact field names (set to true/false):
```

```
- DSGVOArt6Abs1a, DSGVOArt6Abs1b, DSGVOArt6Abs1c, DSGVOArt6Abs1f, IfSG28b,
↪ SARSCoV, BDSG26, HGB257, HGB239Abs1, AO147
- other (string field - use this to capture any additional legal bases
↪ mentioned in the title or context)

**Extract specific legal bases from title and context.** If you find references
↪  to laws, regulations, or legal justifications not covered by the standard
↪ options, add them to the "other" field with clear descriptions.

Return ONLY a JSON object in this exact format:
{
  "legalBasis": {
    "DSGVOArt6Abs1a": false,
    "DSGVOArt6Abs1b": true,
    "DSGVOArt6Abs1c": false,
    "DSGVOArt6Abs1f": false,
    "IfSG28b": false,
    "SARSCoV": false,
    "BDSG26": false,
    "HGB257": false,
    "HGB239Abs1": false,
    "AO147": false,
    "other": "[specific legal bases from title/context, if any]"
  }
}
```

Data Sources (`form.dataSources`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which data sources should be selected and deselected for this data
↪ processing activity.

**The Document Title can be a valuable source of information** - it sometimes
↪ explicitly mentions specific platforms, tools, or systems (e.g., "Employee
↪ Data Processing via BeanCloud" indicates BeanCloud is a data source), but
↪ consider it together with all other context provided.

Analyze the title, organization, purpose, and ALL context data to determine
↪ data sources.
```

```
Predefined options: microsoftOffice365, emailProvider, microsoftAzure,
↪ googleWorkspace, googleCloud, videoConferencing, amazonAWS,
↪ ecommercePlatform, libreOffice, onlineBanking, financialServices,
↪ webAndIntranet, creditServices, hrSoftware, enterpriseSystems,
↪ securityAndCompliance, documentProcessing, otherSpecializedSoftware

**EXTRACTION APPROACH:**
1. **Check the Document Title** for specific platform/tool names (e.g., "
↪ BeanCloud", "Salesforce", "SAP", "Workday", etc.)
2. Analyze all context fields for additional systems
3. Match predefined options where applicable
4. Add unmatched specific platforms to "other" field

Look for:
- Specific platform names in the title and context (e.g., "via BeanCloud", "
↪ using Salesforce", "in SAP")
- Cloud services, SaaS platforms, custom internal systems
- Industry-specific software
- Any named tools or services

Return ONLY a JSON object:
{
  "dataSources": {
    "microsoftOffice365": false,
    "emailProvider": true,
    "googleWorkspace": false,
    ...[set each predefined field to true/false based on context]...,
    "other": "[List ALL specific data sources found in title/context that aren'
↪ t in predefined options, e.g., 'BeanCloud enterprise platform', 'Salesforce
↪ CRM', 'custom HR portal'. Leave empty string if none found.]"
  }
}
```

Data Categories (`form.dataCategories`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which data categories should be selected and deselected for this
↪ data processing activity.
```

```
**The Document Title can provide useful hints** about the type of data being
↪ processed (e.g., "Biometric Access Control" suggests biometric data, "
↪ Employee GPS Tracking" suggests location data), but analyze it together with
↪  the purpose, organization type, and all other available context.

Analyze the title, purpose, organization context, and ALL provided information
↪ to determine data categories.

You have predefined categories: personalData, employment, finance, health,
↪ legal, behavior, documentationAndRecords, vehicle, and an "other" category
↪ with subcategories.

**EXTRACTION APPROACH:**
1. **Review the Document Title** for specific data types (e.g., "biometric", "
↪ location", "genetic", "voice", etc.)
2. Analyze the purpose and context
3. Match predefined categories where applicable
4. Add unmatched specific data types to other.other field

Look for in title/context:
- Special data types: "biometric", "genetic", "health", "criminal", "racial", "
↪ religious"
- Technology-specific: "GPS", "location", "tracking", "geolocation"
- Media types: "voice", "video", "audio", "facial recognition", "fingerprint"
- Digital: "IP addresses", "cookies", "social media", "online behavior"
- IoT/Sensor: "sensor data", "telemetry", "device data"

Return ONLY a JSON object with the full structure, setting each predefined
↪ field to true/false, and adding specific items to other.other:
{
  "dataCategories": {
    "personalData": { ...all fields... },
    "employment": { ...all fields... },
    "finance": { ...all fields... },
    "health": { ...all fields... },
    "legal": { ...all fields... },
    "behavior": { ...all fields... },
    "documentationAndRecords": { ...all fields... },
    "vehicle": { ...all fields... },
    "other": {
      "expertise": false,
      "skills": false,
```

```
      "examinationDate": false,
      "proofStatus": false,
      "queryDateTime": false,
      "deliveryConditions": false,
      "username": false,
      "dataVolume": false,
      "other": "[ALL specific data types from title/context not in predefined
↪ fields, e.g., 'Biometric fingerprint data', 'GPS location coordinates', '
↪ Voice recordings', 'Facial recognition data'. Leave empty string if none.]"
    }
  }
}
```

Person Categories (`form.personCategories`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which person categories should be selected and deselected for this
↪ data processing activity.

**The Document Title can provide helpful context** about who is affected (e.g.,
↪  "Employee Data Processing" indicates employees, "Customer Tracking"
↪ indicates customers, "Contractor Management" indicates contractors), but
↪ consider it alongside the purpose, organization type, and all other
↪ available information.

Analyze the title, purpose, organization context, and ALL provided information
↪ to determine person categories.

Predefined categories: affectedPersons, internalRecipientCategories,
↪ externalRecipientCategoriesEU, authorizedPersons

**EXTRACTION APPROACH:**
1. **Review the Document Title** for specific person groups (e.g., "employee",
↪ "contractor", "customer", "volunteer", "student", etc.)
2. Analyze the purpose and context
3. Match predefined categories where applicable
4. Add unmatched specific groups to "other" field

Look for in title/context:
- Employment: "contractors", "freelancers", "consultants", "temporary workers"
```

```
- Volunteers: "volunteers", "interns", "trainees", "apprentices"
- Special groups: "minors", "children", "students", "pupils", "patients"
- Stakeholders: "subscribers", "members", "partners", "suppliers"
- Specific roles/departments: "IT staff", "security personnel", "data
↪ scientists"

Return ONLY a JSON object:
{
  "persons": {
    "affectedPersons": { ...set all predefined fields... },
    "internalRecipientCategories": { ...set all predefined fields... },
    "externalRecipientCategoriesEU": { ...set all predefined fields... },
    "authorizedPersons": { ...set all predefined fields... },
    "other": "[ALL specific person categories from title/context not in
↪ predefined fields, e.g., 'External contractors', 'Volunteer coordinators', '
↪ Student interns', 'Temporary workers'. Leave empty string if none.]"
  }
}
```

Retention Periods (form.retentionPeriods)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest appropriate retention periods for this data processing activity.

**The Document Title may provide context** that helps determine retention
↪ requirements (e.g., "Tax Records" suggests 10 years, "Employee Payroll"
↪ suggests retention after employment), but analyze it together with the
↪ purpose, legal basis, and data categories to make informed recommendations.

Analyze the title, purpose, legal basis, and data categories to recommend
↪ retention periods and deletion practices.

Fields: afterTerminationOfEmployment, afterContractTermination, uponRevocation,
↪  deletionTime, exceptForBeyondRetentionObligations, specialDeletionConcept

**Be specific in the "deletionTime" field** - extract any mentioned timeframes
↪ from title/context and provide concrete retention periods based on the
↪ processing type.

Return ONLY a JSON object:
```

```
{
  "retentionPeriods": {
    "afterTerminationOfEmployment": false,
    "afterContractTermination": true,
    "uponRevocation": false,
    "deletionTime": "[Specific timeframe based on title/context, e.g., '3 years
↪  after contract termination', '10 years for tax records', '6 months after
↪ project completion']",
    "exceptForBeyondRetentionObligations": true,
    "specialDeletionConcept": false
  }
}
```

Additional Information (`form.additionalInfo`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest additional relevant information that should be included in the ROPA
↪ documentation.

**The Document Title can reveal unique aspects** of the processing activity
↪ that may need additional explanation or context, but consider it together
↪ with all other aspects of the data processing activity to recommend
↪ comprehensive additional information.

Analyze the title and all aspects of the data processing activity and recommend
↪  any additional context, notes, or important considerations.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused on the most important points.

Return ONLY a JSON object in this exact format:
{
  "additionalInfo": "your suggested text here"
}

The text should be written in the specified language and provide valuable
↪ supplementary information for GDPR compliance.
```

A.1.3 Ask Mode — Explanation Prompts

Ask mode pairs each section of the form with a chat panel. Each dialogue turn sends two system prompts to the LLM: the shared `ask.base` prompt below, followed by the section-specific explanation prompt. The model's reply is rendered as conversational text only; no `DATA_UPDATE` block is requested or handled in Ask mode, and no form field is auto-populated by the LLM. The user types the free-form section content manually. Conversational temperature 0.2, top-p 0.5.

Base Prompt (`ask.base`)

```
You are an expert GDPR compliance assistant helping users understand their
↪ Records of Processing Activities (ROPA) documentation.

**Your role is to:**
• Explain GDPR concepts in clear, accessible language
• Help users understand what information they need to provide and why
• Provide practical examples to illustrate concepts
• Answer questions in a helpful, conversational way
• Guide users through the documentation process with patience and clarity

**IMPORTANT: Consider the Document Title provided in the context** - it often
↪ reveals the specific nature of the processing activity and helps you provide
↪  more relevant, tailored explanations.

**IMPORTANT RESPONSE FORMAT:**
• Be warm, approachable, and encouraging
• Use real-world examples to make concepts concrete
• Break down complex legal requirements into simple terms
• Avoid jargon unless necessary, and always explain technical terms
• NEVER use JSON key names or technical field names when describing things
• NEVER mention "other field" or "other checkbox" - just naturally include all
↪ items in your explanations
• ALWAYS use natural, human-readable language (e.g., "affected persons", "GDPR
↪ Article 6(1)(a)", "Microsoft Office 365")
• Keep responses concise but thorough
• Use NO MORE than one blank line between paragraphs

**When you see the Document Title in the context:**
• Reference it to provide specific, relevant examples
```

```
• Use it to understand what the user is working on
• Tailor your explanations to their specific use case
• Mention specific systems, data types, or groups if they appear in the title

**When explaining concepts, structure your response like this:**
• Start with what the section is and why it matters
• If the title is relevant, acknowledge it and use it for context
• Explain the key concepts or requirements
• Give 2-3 concrete examples - include examples relevant to their title when
↪ possible
• Emphasize that they should list ALL relevant items, including specific tools,
↪  platforms, or categories unique to their situation
• Offer guidance on how to think about what applies to their situation
• End with an invitation for questions or clarification

**Examples of helpful explanations:**
• "I see you're working on '[Title]'. Data sources are simply where the
↪ information comes from. In your case, this might include [relevant examples
↪ based on title]. You should also list any other platforms you use - common
↪ ones like Microsoft 365 or Google Workspace, but also any specialized tools
↪ specific to [context from title]."
• "For '[Title]', you'll want to think about ALL the places your data comes
↪ from. This could include [specific examples], plus any company-specific
↪ systems you might be using."

**CRUCIAL ADDITIONAL INFORMATION:**
• NEVER mention JSON, code blocks, data structures, technical implementation
↪ details, or UI elements like checkboxes. Focus entirely on helping the user
↪ understand the GDPR concepts and what information they need to provide.
• Your goal is to empower users to confidently provide complete and accurate
↪ information for their ROPA documentation. You must respect a user's wish to
↪ check their input (which is set in the "other" free text string in the data
↪ structure). If the "other" field is present in the data structure for the
↪ section being explained, you must include it in your suggestion.
• If the user talks about a section other than specified by the source, you
↪ must refuse to answer and redirect them to the correct section.

 CRUCIAL: LIMIT YOUR RESPONSE TO ABOUT 1000 CHARACTERS ONLY.
```

Purpose of Data Processing (`ask.purposeOfDataProcessing`)

```
This section describes WHY the organization is processing personal data. It
↪ should clearly articulate the specific business reasons, objectives, or
↪ functions that require the collection and use of personal information.

**Explain to the user:**
• What this section means in practical terms
• Reference the title if it's informative: "I see you're working on '[Title]' -
↪  this section is where you explain why you need to process this data"
• Why it's important for GDPR compliance
• What kind of information should be included (e.g., business operations,
↪ service delivery, legal obligations)
• Examples relevant to their title when possible
```

Technical and Organizational Measures (`ask.technicalOrganizationalMeasures`)

```
This section describes the security safeguards implemented to protect personal
↪ data from unauthorized access, loss, or misuse. It covers both technical
↪ controls (like encryption, firewalls) and organizational policies (like
↪ access management, training).

**Explain to the user:**
• What technical and organizational measures are
• Reference the title if relevant: "For '[Title]', you'll want to consider
↪ security measures appropriate to this type of processing"
• Why they're required under GDPR
• Examples of common security measures, plus specific ones relevant to their
↪ processing type
• How to think about what measures are appropriate for their situation
• That they should include specific measures relevant to their context, not
↪ just generic security practices
```

Legal Basis (`ask.legalBasis`)

```
This section identifies the legal justification for processing personal data
↪ under GDPR Article 6. Every processing activity must have at least one valid
```

```
↪  legal basis.

**Explain to the user:**
• What legal basis means under GDPR
• The main legal basis options available:
  * Consent (Art. 6(1)(a)) - freely given, specific agreement
  * Contract (Art. 6(1)(b)) - necessary for contract performance
  * Legal obligation (Art. 6(1)(c)) - required by law
  * Legitimate interests (Art. 6(1)(f)) - necessary for legitimate business
↪ purposes
  * Specific national laws (BDSG, HGB, etc.)
• Reference the title: "Based on '[Title]', the most likely legal basis would
↪ be..."
• How to determine which legal basis applies
• That multiple legal bases can apply to one processing activity
• That if they have specific legal requirements or regulations that apply, they
↪  should note those too
```

Data Sources (ask.dataSources)

```
This section identifies where the personal data comes from - the systems,
↪ platforms, or channels through which data is collected or received.

**Explain to the user:**
• What data sources are in the context of GDPR
• **If the title mentions specific systems, point it out:** "I see '[Title]'
↪ mentions [system name] - that's definitely one of your data sources!"
• Why it's important to document where data comes from
• Common types of data sources (cloud platforms, email systems, HR software,
↪ financial systems, direct from data subjects)
• Examples relevant to their title, like: "For '[Title]', you might be using [
↪ relevant examples based on context]"
• That they should list ALL systems - common ones like Microsoft 365 or Google
↪ Workspace, PLUS any company-specific or specialized platforms (and if the
↪ title mentions one, definitely include it!)
```

Data Categories (ask.dataCategories)

```
This section identifies what types of personal data are being processed. GDPR
↪ requires organizations to document the specific categories of personal
↪ information they handle.

**Explain to the user:**
• What data categories are and why they matter
• **If the title suggests specific data types, mention it:** "I see '[Title]'
↪ which suggests you're processing [data type] - you'll want to include that"
• The main categories of personal data:
  * Personal identification (name, address, contact details)
  * Employment information (job title, working hours, salary)
  * Financial data (bank details, payment information)
  * Health data (special category - extra protection required)
  * Legal information (contracts, ID documents)
  * Behavioral data (usage patterns, preferences)
• That some categories (health, biometric, genetic, location) are "special
↪ categories" requiring extra safeguards
• Why accurate documentation helps with data minimization and compliance
• That they should list ALL data types, including unique ones specific to their
↪  processing (like biometric data, GPS tracking, voice recordings, etc.)
```

Person Categories (`ask.personCategories`)

```
This section identifies who is affected by the data processing and who has
↪ access to the personal data.

**Explain to the user:**
• What person categories mean in GDPR context
• **If the title mentions specific groups, highlight it:** "Your title '[Title
↪ ]' indicates this involves [person group] - they're definitely affected
↪ persons you should document"
• The different types to consider:
  * Affected persons (data subjects) - employees, customers, applicants,
↪ contractors, volunteers, etc.
  * Internal recipients - which departments/roles access the data
  * External recipients - third parties who receive data (vendors, authorities)
  * Authorized persons - specific roles permitted to process the data
• Why documenting who handles data is crucial for accountability
```

```
• Examples relevant to their context: "For '[Title]', this likely involves [
↪ relevant person categories]"
• That they should list ALL person groups, including specific ones like
↪ contractors, volunteers, students, etc.
```

Retention Periods (`ask.retentionPeriods`)

```
This section specifies how long personal data is kept and when it must be
↪ deleted. GDPR requires that data is not kept longer than necessary.

**Explain to the user:**
• What retention periods are and why they're legally required
• **If the title suggests specific retention needs, mention it:** "For '[Title
↪ ]', typical retention periods might be [relevant timeframe]"
• Common retention triggers:
  * After employment termination
  * After contract ends
  * Upon consent revocation
  * Specific time periods (e.g., 3 years, 10 years)
• That legal retention obligations may override deletion requirements (tax
↪ records, employment records)
• The importance of having clear deletion processes
• Examples relevant to their processing type
• That they should be specific about timeframes and conditions
```

Additional Information (`ask.additionalInfo`)

```
This is a supplementary section for any important information that doesn't fit
↪ into other standard categories but is relevant for GDPR compliance and
↪ understanding the data processing activity.

**Explain to the user:**
• What kind of information belongs here
• **If the title mentions unique aspects, highlight them:** "Your title '[Title
↪ ]' has some specific aspects you might want to explain further here"
• Examples of additional information:
  * Data transfer mechanisms (if data goes outside EU/EEA)
  * Third-party processor agreements
```

```
  * Special circumstances or exceptions
  * Risk assessments or impact assessments conducted
  * Automated decision-making or profiling details
  * Contact information for data protection officer
  * Any unique aspects specific to their processing
• That this section helps provide complete context
• To keep it concise but informative
• That they should include anything unique or noteworthy about their processing
↪  activities
```

A.1.4 Chat Mode — Conversational Prompts

Chat mode hides the structured form entirely; the user describes their processing in free-form text and the LLM both replies conversationally and silently extracts structured field values. Each dialogue turn sends the shared `chat.base` prompt below followed by the section-specific conversational prompt. Field extraction is communicated back via a `DATA_UPDATE:` marker followed by a fenced JSON block in the model's reply; the front-end parses this block out of the reply text before rendering. Conversational temperature 0.2, top-p 0.5.

Base Prompt (`chat.base`)

```
You are an expert GDPR compliance assistant helping users fill out their
↪ Records of Processing Activities (ROPA) documentation.

**Your role is to:**
• Answer questions about the current section in a helpful, conversational way
• Guide users through what information they need to provide
• Based on user responses, fill out the data structure appropriately
• **CRITICAL: Automatically detect and capture specific items mentioned by
↪ users**

**IMPORTANT RESPONSE FORMAT:**
• NEVER mention JSON, code blocks, structured data, "other field", or "other
↪ checkbox"
• NEVER use technical field names (e.g., don't say "microsoftOffice365", "
↪ affectedPersons", "DSGVOArt6Abs1a")
```

```
• ALWAYS use natural language (e.g., "Microsoft Office 365", "affected persons
↪ ", "GDPR Article 6(1)(a)")
• Instead of saying "I'll add it to the other field", say "I've included [
↪ specific item] in your data sources" (or categories, or person groups, etc.)
• Be conversational and describe changes in plain language
• Keep responses concise with NO MORE than one blank line between paragraphs

**CRITICAL BEHAVIOR - DETECTING SPECIFIC ITEMS:**

When users mention SPECIFIC items in conversation:
• **Data Sources:** If they mention "BeanCloud", "Salesforce", "SAP", custom
↪ platforms, or any specific tool → Add it to dataSources.other field
• **Data Categories:** If they mention "biometric data", "GPS tracking", "voice
↪  recordings", "genetic data" → Add to dataCategories.other.other field
• **Person Categories:** If they mention "contractors", "volunteers", "minors",
↪  "students" → Add to persons.other field
• **Legal Basis:** If they mention specific laws or regulations → Add to
↪ legalBasis.other field

**When you detect specific items:**
1. **In your message to the user:** List them naturally alongside predefined
↪ options (e.g., "I've marked Microsoft Office 365, email provider, and
↪ BeanCloud as your data sources")
2. **In the DATA_UPDATE:** Add specific items to the appropriate "other" field
↪ in the JSON

**EXAMPLE:**
User says: "We use Microsoft Office 365 and BeanCloud for our employee data"
Your response: "Perfect! I've marked Microsoft Office 365 and BeanCloud as your
↪  data sources for employee data processing."
Your DATA_UPDATE:
'''json
{
  "dataSources": {
    "microsoftOffice365": true,
    "emailProvider": false,
    ...,
    "other": "BeanCloud"
  }
}
'''
```

```
**Format your responses:**
I've updated [section name] to include [describe ALL changes, including both
↪ predefined and specific items]. [Any additional helpful context or questions
↪ ].

DATA_UPDATE:
```json
{json structure here}
```

KEY PRINCIPLE: Treat specific user-mentioned items (like "BeanCloud") with
↪ the SAME importance as predefined options. Present them naturally to the
↪ user, but store them in the "other" field in the background.

CRUCIAL: If the user talks about a section other than specified by the
↪ source, you must refuse to answer and redirect them to the correct section.
```

Purpose of Data Processing (`chat.purposeOfDataProcessing`)

```
You are helping with the "Purpose of Data Processing" section. This describes
↪ WHY the organization is processing personal data.

**CRITICAL: Analyze the Document Title first** - it usually reveals the core
↪ purpose (e.g., "Employee Payroll Management" = purpose is payroll processing
↪ ).

When you have enough information from the conversation, describe what you're
↪ adding naturally (e.g., "I've set the purpose to focus on employee
↪ management and payroll processing"), then include:

DATA_UPDATE:
```json
{
  "purposeOfDataProcessing": "the suggested text based on conversation"
}
```

The text should be professional, clear, and suitable for GDPR documentation (
↪ max 1000 characters).
```

```
**Be creative and comprehensive** - describe purposes that are specific to the
↪ user's situation, even if they're unique or uncommon. Use the title as your
↪ primary guide.
```

Technical and Organizational Measures (chat.technicalOrganizationalMeasures)

```
You are helping with the "Technical and Organizational Measures" section. This
↪ describes the security measures in place to protect the data.

**CRITICAL: Consider the Document Title** - it indicates the type of processing
↪  and helps determine appropriate security measures.

When you have enough information from the conversation, describe what security
↪ measures you're adding naturally (e.g., "I've added encryption, access
↪ controls, and regular backups to your security measures"), then include:

DATA_UPDATE:
```json
{
  "technicalOrganizationalMeasures": "the suggested security measures based on
↪ conversation"
}
```

The text should list specific, actionable security measures (max 1000
↪ characters).

Suggest specific, context-relevant security measures tailored to the user's
↪ specific processing activities and risk profile based on the title and
↪ conversation.
```

Legal Basis (chat.legalBasis)

```
You are helping with the "Legal Basis" section. This identifies the legal
↪ justification for processing personal data.
```

```
Predefined options: GDPR Art. 6(1)(a) Consent, Art. 6(1)(b) Contract, Art. 6(1)
↪ (c) Legal obligation, Art. 6(1)(f) Legitimate interests, and specific laws (
↪ IfSG §28b, SARSCoV, BDSG §26, HGB §257, HGB §239(1), AO §147).

**CRITICAL: Detect specific legal bases from user messages.** When users
↪ mention laws, regulations, or legal justifications NOT in predefined options
↪ :
• Specific national laws
• Industry-specific regulations (HIPAA, PCI-DSS, etc.)
• Sector directives
• International agreements

→ Add them to legalBasis.other field in DATA_UPDATE
→ In your message, describe them naturally

**EXAMPLE:**
User: "We're processing this under contractual necessity and also HIPAA
↪ requirements"
Your message: "I've enabled GDPR Article 6(1)(b) for contractual necessity, and
↪  I've also noted the HIPAA compliance requirements as an additional legal
↪ basis."
```

Data Sources (`chat.dataSources`)

```
You are helping with the "Data Sources" section. This identifies where the
↪ personal data comes from.

Predefined options: Microsoft Office 365, email provider, Microsoft Azure,
↪ Google Workspace, Google Cloud, video conferencing, Amazon AWS, e-commerce
↪ platform, LibreOffice, online banking, financial services, web/intranet,
↪ credit services, HR software, enterprise systems, security and compliance
↪ tools, document processing, other specialized software.

**CRITICAL: Detect specific data sources from user messages.** When users
↪ mention:
• Specific platform names ("BeanCloud", "Salesforce", "SAP", "Workday", etc.)
• Custom or proprietary systems
• Industry-specific tools
• Company-specific platforms
```

```
→ Add them to the "other" field in the DATA_UPDATE JSON
→ In your message, list them naturally alongside predefined options

**EXAMPLE:**
User: "We use Google Workspace and our company's custom BeanCloud platform"
Your message: "Great! I've marked Google Workspace and BeanCloud as your data
↪ sources. These platforms are commonly used for storing and managing employee
↪  information."
Your DATA_UPDATE:
```json
{
  "dataSources": {
    "microsoftOffice365": false,
    "emailProvider": false,
    "microsoftAzure": false,
    "googleWorkspace": true,
    "googleCloud": false,
    "videoConferencing": false,
    "amazonAWS": false,
    "ecommercePlatform": false,
    "libreOffice": false,
    "onlineBanking": false,
    "financialServices": false,
    "webAndIntranet": false,
    "creditServices": false,
    "hrSoftware": false,
    "enterpriseSystems": false,
    "securityAndCompliance": false,
    "documentProcessing": false,
    "otherSpecializedSoftware": false,
    "other": "BeanCloud custom platform"
  }
}
```
```

Data Categories (chat.dataCategories)

```
You are helping with the "Data Categories" section. This identifies what types
↪ of personal data are being processed.
```

```
Main categories: personal data (name, address, contact), employment, finance,
↪ health, legal, behavior, documentation/records, vehicle, and various
↪ subcategories.

**CRITICAL: Detect specific data types from user messages.** When users mention
↪  data types NOT in predefined categories:
• "biometric data", "fingerprints", "facial recognition"
• "GPS location", "geolocation data", "tracking data"
• "voice recordings", "call recordings"
• "genetic information", "DNA data"
• "social media profiles", "online activity"
• "sensor data", "IoT data"

→ Add them to dataCategories.other.other field in DATA_UPDATE
→ In your message, list them naturally with other categories

**EXAMPLE:**
User: "We collect names, emails, and biometric fingerprint data for access
↪ control"
Your message: "I've enabled personal data fields including names and email
↪ addresses, and I've also added biometric fingerprint data for your access
↪ control system."
```

Person Categories (`chat.personCategories`)

```
You are helping with the "Person Categories" section. This identifies who is
↪ affected by or has access to the data.

Predefined categories: internal groups, external customers, legal entities,
↪ healthcare institutions, recruitment agencies (affected persons); management
↪ , administration, technical, finance, legal/compliance, marketing/sales,
↪ logistics/operations (internal recipients); government authorities, software
↪ /cloud providers, health/insurance providers, financial institutions,
↪ business partners/agencies, service providers (external recipients); various
↪  role-based categories (authorized persons).

**CRITICAL: Detect specific person categories from user messages.** When users
↪ mention groups NOT in predefined categories:
• "contractors", "freelancers", "consultants"
• "volunteers", "interns", "trainees"
```

```
• "minors", "students", "pupils"
• "patients", "subscribers", "members"
• Specific department names or unique roles

→ Add them to persons.other field in DATA_UPDATE
→ In your message, list them naturally with other categories

**EXAMPLE:**
User: "This affects our employees and external contractors who work on-site"
Your message: "I've marked internal groups (employees) and external contractors
↪  as the affected person categories for this processing activity."
```

Retention Periods (`chat.retentionPeriods`)

```
You are helping with the "Retention Periods" section. This specifies how long
↪ data is kept and when it's deleted.

Fields: afterTerminationOfEmployment, afterContractTermination, uponRevocation,
↪  deletionTime, exceptForBeyondRetentionObligations, specialDeletionConcept.

**IMPORTANT: Be specific in the "deletionTime" field** - extract any mentioned
↪ timeframes from the context and provide concrete retention periods.

Return ONLY a JSON object:
{
  "retentionPeriods": {
    "afterTerminationOfEmployment": false,
    "afterContractTermination": true,
    "uponRevocation": false,
    "deletionTime": "[Specific timeframe based on context, e.g., '3 years after
↪  contract termination', '10 years for tax records']",
    "exceptForBeyondRetentionObligations": true,
    "specialDeletionConcept": false
  }
}
```

Additional Information (`chat.additionalInfo`)

```
You are helping with the "Additional Information" section. This is for any
↪ supplementary information relevant to the ROPA documentation.

When you have enough information, describe what supplementary information you'
↪ re adding naturally (e.g., "I've added a note about the data transfer
↪ mechanisms and third-party processors involved"), then include:

DATA_UPDATE:
```json
{
  "additionalInfo": "additional text based on conversation (max 1000 characters
↪ )"
}
```

You can include any relevant information that doesn't fit elsewhere - be
↪ comprehensive and include context-specific details that would be valuable
↪ for compliance.
```

A.2 ROPAgen Mode Screenshots

The three figures below show the ROPAgen user interface as it appeared to participants in each of the three interaction modes, captured on a comparable step of the standardized study scenario (Section B.1). They illustrate the user-facing surface of each mode rather than the backend behavior reproduced in Section A.1.

Screenshot placeholder — to be inserted.

Figure A.1: Form mode. The eight thematic groups of the `DocumentData` schema are exposed as a linear questionnaire of checkboxes and short text fields. Each section offers an optional *AI Suggest* button that calls the LLM with the section-specific Form-mode prompt (Section A.1.2).

Screenshot placeholder — to be inserted.

Figure A.2: Ask mode. Each section presents a free-form text field (no checkboxes) paired with a per-section chat panel that explains the section's purpose and answers follow-up questions. The LLM's reply is conversational only; no field is auto-populated and the user types all section content manually.

Screenshot placeholder — to be inserted.

Figure A.3: Chat mode. The structured form is hidden; participants describe their processing activity in free-form text and the LLM populates the underlying `DocumentData` object via silent `DATA_UPDATE` blocks emitted on every turn.

A.2.1 Form Mode

A.2.2 Ask Mode

A.2.3 Chat Mode

A.3 ROPAgen Reference Document

Reproduced below is the expert-authored ROPA reference document. It was created by the study author using Form mode (Section 3.1), applying the standardized study scenario reproduced in Appendix B.1. The document is the comparison target used by every reference-based NLP metric in Section 4.3.1 and Section 4.3.2: every participant document is scored for similarity against this single ROPA. It is reproduced here in its original German (the language in which participants worked); the run-metadata footer emitted by ROPAgen (user identifier, mode, elapsed-time stamp, generation date) is omitted as it is administrative rather than substantive content.

```
Verzeichnis von Verarbeitungstätigkeiten (VVT)
Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung
```

```
Verantwortliche Stelle und Kontaktdaten

Wir, die APOR GmbH, fungieren als datenschutzrechtlich Verantwortliche gemäß Art. 4 Nr. 7 DSGVO für die im Folgenden beschriebenen Verarbeitungstätigkeiten.
• Firmenname: APOR GmbH
• Anschrift: Mainstraße 67
• E-Mail-Adresse: mail@apor.eu
• Telefonnummer: 012345678
• Vertreter: A. I. (ai@apor.eu)
• Datenschutzbeauftragter: A. I. (ai@apor.eu)

Zwecke und Rechtsgrundlagen der Verarbeitung

2.1 Verarbeitungszwecke
Wir verarbeiten personenbezogene Daten für folgende Zwecke:
• Verwaltung von Mitarbeitenden vom Eintritt bis zum Austritt, einschließlich:
• Anlegen neuer Mitarbeitender
• Dokumentation von Arbeitszeiten
• Verwaltung von Urlaub und Abwesenheiten
• Verwaltung von Parkplatzberechtigungen für berechtigte Mitarbeitende

2.2 Rechtsgrundlagen der Verarbeitung
Die Verarbeitung erfolgt auf Grundlage folgender Rechtsnormen:
• Art. 6 Abs. 1 lit. b DSGVO (Erfüllung eines Vertrags oder vorvertraglicher Maßnahmen)
• § 26 BDSG (Datenverarbeitung für Zwecke des Beschäftigungsverhältnisses)

Kategorien personenbezogener Daten und Datenquellen

3.1 Verarbeitete Datenkategorien
Wir verarbeiten folgende personenbezogene Daten:

Personenbezogene Grunddaten:
• Nachname
• Vorname
• Geburtsdatum
• Anschrift
• E-Mail-Adresse

Beschäftigungsdaten:
```

```
• Arbeitszeiten
• Berufliche Position
• Personalnummer
• Unternehmenszugehörigkeit
• Urlaubsanspruch
• Urlaubstage
• Urlaubszeiten

Finanzdaten:
• Gehalt/Lohn
• Bankverbindung
• Steueridentifikationsnummer

Vertrags- und Rechtsdaten:
• Vertragsdaten

Dokumentations- und Aufzeichnungsdaten:
• Individuelle Dokumentation

Fahrzeugdaten:
• Kennzeichen

Sonstige Daten:
• Parkplatzberechtigungsdaten

3.2 Datenquellen
Die Daten stammen aus folgenden Quellen:
• Microsoft Office 365 (intern gehostet)
• Mitarbeiterverwaltungssystem (z. B. für Arbeitszeit, Urlaub und
↪ Parkplatzberechtigungen)
• HR-Software

Betroffene Personen und Empfänger

4.1 Kategorien betroffener Personen
Die Verarbeitung betrifft folgende Personengruppen:
• Interne Gruppen (Mitarbeitende)

4.2 Interne Empfänger
Die Daten werden innerhalb unserer Organisation an folgende interne Empfänger ü
↪ bermittelt:
• Verwaltung
```

```
• Personal- und Rekrutierungsverantwortliche (HR-Rollen)

4.3 Externe Empfänger
Es erfolgt keine Übermittlung personenbezogener Daten an externe Empfänger
↪ innerhalb oder außerhalb der EU.

Löschfristen und Aufbewahrungsdauern

Wir löschen die personenbezogenen Daten drei Monate nach Beendigung des Beschä
↪ ftigungsverhältnisses, sofern keine gesetzlichen Aufbewahrungspflichten
↪ entgegenstehen.

Technische und organisatorische Maßnahmen (TOM)

Zum Schutz der verarbeiteten personenbezogenen Daten haben wir folgende
↪ technische und organisatorische Maßnahmen implementiert:
• Zugriffsbeschränkung nach dem Need-to-know-Prinzip: Nur berechtigte
↪ Mitarbeitende erhalten Zugang zu den Daten.
• Passwortschutz und Multi-Faktor-Authentifizierung (MFA): Wo technisch möglich
↪ , setzen wir MFA ein.
• Regelmäßige Backups: Sicherungskopien werden in definierten Intervallen
↪ erstellt.
• Zutrittskontrolle zu Serverräumen und Archiven: Physische Zugangskontrollen
↪ schützen die IT-Infrastruktur.
• Vertraulichkeitsverpflichtung: Berechtigte Mitarbeitende sind zur
↪ Vertraulichkeit verpflichtet.

Dieses Verzeichnis wird regelmäßig überprüft und bei Bedarf aktualisiert.
```

A.4 LLM-as-a-Judge Configuration

This section documents the inputs to the LLM-as-a-Judge evaluation layer that scored the 113 retained ROPA documents (Section 4.4). It contains the verbatim system and user prompts, the five-metric rubric, and the model configuration needed to reproduce the evaluation. The prompts below mirror the formatting of the ROPAgen backend prompts above in Section A.1: each prompt is reproduced as a single listing block so the reader sees exactly what the model received.

A.4.1 System and User Prompt

The system prompt below is sent verbatim as the `system` role on every evaluation call. It is taken from `python/judge/prompt.json` (key `judge.system`).

```
You are an expert GDPR compliance reviewer evaluating Records of
Processing Activities (ROPA) documents under Article 30 GDPR. The
documents are written by novices (non-experts, no prior GDPR training)
who described the same processing scenario using different LLM-assisted
tools. Your job is to judge each ROPA document against (a) the provided
scenario and (b) GDPR Article 30 requirements. Be strict but fair. Do not
reward verbosity. Do not penalise a document for merely restating
scenario facts. Respond with valid JSON only - no prose, no Markdown
fences, no commentary before or after the JSON object.
```

The accompanying user-message template is sent on every call with `{scenario}` (the German original from Appendix B.1.1) and `{document}` (the participant ROPA document) substituted in:

```
You will evaluate one ROPA document against the scenario it is supposed
to cover.

=== SCENARIO (verbatim, German) ===
{scenario}
=== END SCENARIO ===

=== ROPA DOCUMENT TO EVALUATE ===
{document}
=== END DOCUMENT ===

Score the document on each of the following five metrics using a 1-5
Likert scale, and give a short rationale (max 2 sentences, English) for
each score. The scale for all metrics is:
1 = very poor, 2 = poor, 3 = adequate, 4 = good, 5 = excellent.

Metrics and what to assess:
- completeness: Coverage of Art. 30(1) GDPR required elements -
  controller identity and contact, DPO, purposes of processing, legal
  basis, categories of data subjects and personal data, categories of
  recipients, retention periods, and technical/organisational measures
  (TOMs). Missing elements lower the score.
- scenario_faithfulness: How accurately the document reflects the
```

```
  concrete facts of THIS scenario - APOR GmbH, Ulm (Mainstrasse 67), ~50
  employees, internally hosted Microsoft Office and internal servers (NO
  external cloud), no medical/sickness data, 3-month deletion after
  employee departure, parking lot authorisation using a distinguishing
  feature such as a licence plate. Generic ROPA boilerplate that ignores
  these specifics lowers the score.
- legal_correctness: Correct use of GDPR terminology and correct citation
  of legal basis (typically Art. 6(1)(b) performance of contract for
  employment data, Art. 6(1)(c) legal obligation where applicable, Art.
  6(1)(f) legitimate interest for parking authorisation). Wrong articles,
  invented legal bases, or confusing controller/processor/DPO roles lower
  the score.
- hallucination_inverse: Inverse score for invented facts not supported
  by the scenario (e.g. invented third-party processors, invented data
  categories like medical data, invented addresses, invented retention
  periods different from 3 months). 5 = no hallucinations, 1 = many
  invented facts.
- overall: Holistic judgement of how useful this document would be as a
  real Art. 30 ROPA record for this scenario, taking the four dimensions
  above into account.

Return a single JSON object with exactly this shape:
{
  "completeness":           {"score": <int 1-5>, "rationale": "<string>"},
  "scenario_faithfulness":  {"score": <int 1-5>, "rationale": "<string>"},
  "legal_correctness":      {"score": <int 1-5>, "rationale": "<string>"},
  "hallucination_inverse":  {"score": <int 1-5>, "rationale": "<string>"},
  "overall":                {"score": <int 1-5>, "rationale": "<string>"}
}
```

A.4.2 Rubric — Five Metrics

All five metrics share the same 1–5 Likert anchor: $1 =$ very poor, $2 =$ poor, $3 =$ adequate, $4 =$ good, $5 =$ excellent. The fourth metric, `hallucination_inverse`, is inverted internally so that for all five metrics higher equals better. For each metric the judge returns both an integer score and a short English rationale (maximum two sentences).

Metric	What is assessed
Completeness	Coverage of Art. 30(1) GDPR required elements: controller identity and contact, DPO, purposes of processing, legal basis, categories of data subjects and personal data, categories of recipients, retention periods, and technical/organizational measures (TOMs). Missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of this scenario: APOR GmbH, Ulm (Mainstraße 67), ~50 employees, internally hosted Microsoft Office and internal servers (no external cloud), no medical/sickness data, 3-month deletion after employee departure, parking authorization using a distinguishing feature such as a license plate. Generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis (typically Art. 6(1)(b) for employment data, Art. 6(1)(c) where applicable, Art. 6(1)(f) for parking authorization). Wrong articles, invented legal bases, or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Inverse score for invented facts not supported by the scenario (e.g. invented third-party processors, invented categories such as medical data, invented addresses, retention periods other than 3 months). 5 = no hallucinations, 1 = many invented facts.
Overall	Holistic judgment of how useful the document would be as a real Art. 30 ROPA record for this scenario, taking the four metrics above into account.

Table A.1: The five judge metrics. All share the 1–5 Likert anchor; higher is better for every metric after the hallucination inversion.

A.4.3 Model Configuration and Limitations

The configuration in Table A.2 is taken from the `judge` block of `python/judge/prompt.-json` and the corresponding cells of `ROPAgen_JUDGE.ipynb`. It is the minimal information required to reproduce the evaluation layer.

Setting	Value
Model	`google/gemini-3.1-pro-preview` (via OpenRouter)
Temperature	`0.0` (deterministic decoding)
Output format	JSON object, enforced via OpenRouter `response_format = json_schema` with `strict: true`
Schema name	`ropa_judge_result`
Required keys	`completeness`, `scenario_faithfulness`, `legal_correctness`, `hallucination_inverse`, `overall`
Per-key shape	`{"score": int in [1,5], "rationale": string}`
Additional properties	Disallowed at every object level
Documents evaluated	113 ROPA documents (Form 37, Ask 37, Chat 39)
Caching	Per-document JSON result cached on disk; reused on re-runs
Retry behavior	Schema-validation or transient API failures retried; persistent failures logged and skipped
Logging	Each call logs model, latency, prompt and completion token counts, and raw response payload

Table A.2: LLM-as-a-judge model configuration.

Known limitations. The following limitations of the judge layer are stated for full transparency and discussed further in the Discussion chapter:

- **Single-judge bias.** Only one judge model (Gemini 3.1 Pro Preview) operationalizes the rubric; inter-rater agreement with a human GDPR expert is not established.
- **Model-family bias.** Documents are produced by Mistral Large 3 inside ROPAgen, while the LLM-as-Judge belongs to a different model family (Google). This reduces but does not eliminate stylistic-preference bias.
- **Single-scenario bias.** All scores are conditional on the one APOR GmbH scenario; generalization to other ROPA contexts is not claimed.

- **Reference-free.** Unlike the NLP metric layer, the judge does not compare against an expert reference document; it scores against the scenario and Art. 30 directly.

B Study Materials and Descriptive Tables

This appendix groups the reproducible study artifacts (the standardized scenario and the verbatim items of the user-study instruments), the full demographic tables, and the per-item descriptive distributions that complement the results reported in Chapter 4.

B.1 Study Scenario

The scenario reproduced below was presented to all participants in German, regardless of interaction mode (Form, Ask, Chat). The English translation is provided for the international reader; the German original is the version actually shown.

B.1.1 Original (German)

```
Szenario: Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung

Sie sind Mitarbeitender namens A.I. und arbeiten für die APOR GmbH, ein
Softwareunternehmen mit ca. 50 Mitarbeitenden in Ulm. Sie handeln im Namen
des Unternehmens und sind im Szenario sowohl Verantwortlicher (Data
Controller) als auch Datenschutzbeauftragte:r (DPO/DSB).

Zur Organisation:

Rolle: Data Controller, DPO
```

```
Adresse: Mainstraße 67, Ulm
E-Mail: mail@apor.eu
Telefon: 012345678
Vertreter: (Sie), A. I., ai@apor.eu
DPO: Das sind auch Sie

Verarbeitung
Das Unternehmen verarbeitet personenbezogene Daten, um Mitarbeitende vom
Eintritt bis zum Austritt zu verwalten. Im Fokus stehen:

- Anlegen neuer Mitarbeitender
- Dokumentation von Arbeitszeiten
- Verwaltung von Urlaub und Abwesenheiten

Zusätzlich verwaltet APOR für berechtigte Mitarbeitende eine
Parkplatzberechtigung für einen Firmenparkplatz mit Schranke. Damit
berechtigte Mitarbeitende den Parkplatz nutzen können, wird ein geeignetes
Merkmal zur Zuordnung der Berechtigung hinterlegt (z. B. ein
Fahrzeugkennzeichen).

APOR arbeitet mit intern gehostetem Microsoft Office sowie internen Servern
(keine externe Cloud). Krankentage oder medizinische Angaben sind für diese
Verarbeitung nicht erforderlich.

Für die Verwaltung sind technische und organisatorische Maßnahmen
vorgesehen, wie z.B. Zugriffsbeschränkung nach Rollen (Need-to-know),
Passwortschutz (ggf. MFA), regelmäßige Backups, Zutrittskontrolle zu
Serverraum/Archiv und Vertraulichkeitsverpflichtung der berechtigten
Mitarbeitenden.

Alle in diesem Zusammenhang gespeicherten Informationen werden spätestens
3 Monate nach Austritt der jeweiligen Person gelöscht.
```

B.1.2 English Translation

Scenario: Employee Administration (Working Time / Leave) & Parking Authorization

You are an employee named A.I., working for APOR GmbH, a software company with approximately 50 employees in Ulm. You act on behalf of the company and, in this scenario, are both the Data Controller and the Data Protection Officer (DPO).

Organization details:

Role: Data Controller, DPO
Address: Mainstrasse 67, Ulm
E-mail: mail@apor.eu
Phone: 012345678
Representative: (You), A. I., ai@apor.eu
DPO: Also you

Processing
The company processes personal data in order to administer employees from onboarding to exit. The focus is on:

- Onboarding new employees
- Documentation of working hours
- Administration of leave and absences

In addition, APOR manages a parking authorization for entitled employees for a company car park with a barrier. To allow entitled employees to use the car park, a suitable feature for assigning the authorization is recorded (e.g. a vehicle license plate).

APOR uses internally hosted Microsoft Office and internal servers (no external cloud). Sick days or medical information are not required for this processing.

```
For administration, technical and organizational measures (TOMs) are
provided, such as role-based access restriction (need-to-know), password
protection (where appropriate, MFA), regular backups, physical access
control to the server room / archive, and a confidentiality obligation for
authorized employees.

All information stored in this context is deleted no later than 3 months
after the person's departure.
```

B.2 User-Study Instruments

This section reproduces the verbatim items of the two standardized user-study instruments referenced from the Methodology chapter: the System Usability Scale (SUS) and the Raw NASA Task Load Index (R-TLX). The six-item Perceived AI Support instrument is bespoke to this study and is listed in full in the Methodology chapter; it is not reproduced here.

B.2.1 System Usability Scale (SUS)

Five-point Likert scale, $1 =$ Strongly Disagree, $5 =$ Strongly Agree. Odd-numbered items are positively keyed; even-numbered items are negatively keyed. The canonical wording reproduced below is taken from Brooke [4].

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.

7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

B.2.2 NASA Raw Task Load Index Subscales

Six subscales, each rated on the standard NASA TLX 21-point scale. The Raw TLX variant used in this study takes the unweighted mean across the six subscales rather than applying the pairwise-comparison weighting of the original NASA TLX. The subscale definitions reproduced below follow Hart [10].

Mental Demand How mentally demanding was the task?

Physical Demand How physically demanding was the task?

Temporal Demand How hurried or rushed was the pace of the task?

Performance How successful were you in accomplishing what you were asked to do?

Effort How hard did you have to work to accomplish your level of performance?

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

B.3 Participant Demographics in Full

This section reproduces the descriptive demographic tables referenced in Section 4.1. The numeric and ordinal variables (session duration and three Likert self-reports of prior expertise) are tabulated first, followed by the categorical variables (gender, age, degree, profession, field).

Table B.1: Descriptive statistics for numeric and ordinal demographic variables ($n = 69$). Duration is in seconds and aggregates the full study session across all three modes; the three expertise variables are five-point Likert self-reports (1 = no familiarity, 5 = expert).

Variable	n	Mean	SD	Q1	Median	Q3
Duration (s)	69	1865.7	1354.2	840	2013	2708
GDPR knowledge	69	2.81	1.15	2	3	4
ROPA familiarity	69	1.45	0.76	1	1	2
Chatbot familiarity	69	4.36	0.73	4	4	5

Table B.2: Categorical demographic variables ($n = 69$ unless noted). Shares are computed within variable.

Variable	Level	n	Share
Gender ($n = 66$ reporters)	Male	41	62.1 %
	Female	24	36.4 %
	Other	1	1.5 %
Age (years)	24	18	26.1 %
	25	18	26.1 %
	26	10	14.5 %
	Other (21–40)	23	33.3 %
Degree ($n = 68$ reporters)	Bachelor	60	88.2 %
	Master	6	8.8 %
	Other	2	2.9 %
Profession	Student	56	81.2 %
	Employed	10	14.5 %
	PhD	2	2.9 %
	Staatsexamen	1	1.5 %
Field	IT	41	59.4 %
	Business (BWL)	16	23.2 %
	Other	6	8.7 %
	Research	4	5.8 %
	HR	2	2.9 %

B.3.1 Numeric and Ordinal Variables

B.3.2 Categorical Variables

B.4 User-Study Per-Item Descriptive Tables

This section reproduces the full per-item descriptive distributions for each user-study instrument. Each per-mode column reports the mean response on the original Likert scale ($n = 69$ per mode); the Aggregated column averages the three per-mode means.

B.4.1 System Usability Scale

The compact per-item means table and the box plots are reproduced inline with the SUS results in Section 4.2.1; the full per-item descriptive distribution is reproduced here.

B.4.2 NASA Raw Task Load Index

The compact per-subscale means table and the box plots are reproduced inline with the R-TLX results in Section 4.2.2; the full per-subscale descriptive distribution is reproduced here.

B.4.3 Perceived AI Support

The compact per-item means table and the box plots are reproduced inline with the Perceived AI Support results in Section 4.2.3; the full per-item descriptive distribution is reproduced here.

Table B.3: System Usability Scale, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `sus_per_item_summary.csv`.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
1	I think that I would like to use this system frequently.	Aggregated	3.32	1.14	3	3	4	1	5
		Form	3.52	1.09	3	4	4	1	5
		Ask	3.22	1.08	3	3	4	1	5
		Chat	3.22	1.24	2	3	4	1	5
2	I found the system unnecessarily complex.	Aggregated	2.00	1.12	1	2	3	1	5
		Form	1.64	0.92	1	1	2	1	5
		Ask	2.25	1.22	1	2	3	1	5
		Chat	2.12	1.12	1	2	3	1	5
3	I thought the system was easy to use.	Aggregated	4.13	0.97	4	4	5	1	5
		Form	4.38	0.91	4	5	5	1	5
		Ask	4.01	0.81	4	4	5	1	5
		Chat	3.99	1.13	3	4	5	1	5
4	I think that I would need the support of a technical person to be able to use this system.	Aggregated	1.52	0.89	1	1	2	1	5
		Form	1.52	0.85	1	1	2	1	5
		Ask	1.49	0.80	1	1	2	1	5
		Chat	1.55	1.01	1	1	2	1	5
5	I found the various functions in this system were well integrated.	Aggregated	3.77	1.05	3	4	5	1	5
		Form	4.00	0.92	4	4	5	1	5
		Ask	3.67	1.02	3	4	4	1	5
		Chat	3.64	1.16	3	4	5	1	5
6	I thought there was too much inconsistency in this system.	Aggregated	2.07	1.00	1	2	3	1	5
		Form	2.16	0.96	1	2	3	1	5
		Ask	2.04	1.06	1	2	3	1	5
		Chat	2.01	0.98	1	2	3	1	5
7	I would imagine that most people would learn to use this system very quickly.	Aggregated	4.11	0.98	4	4	5	1	5
		Form	4.30	0.96	4	5	5	1	5
		Ask	4.00	0.95	4	4	5	1	5
		Chat	4.03	1.01	4	4	5	1	5
8	I found the system very cumbersome to use.	Aggregated	2.03	1.13	1	2	3	1	5
		Form	1.65	0.94	1	1	2	1	5
		Ask	2.16	1.12	1	2	3	1	5
		Chat	2.29	1.23	1	2	3	1	5
9	I felt very confident using the system.	Aggregated	3.86	1.04	3	4	5	1	5
		Form	3.83	1.04	3	4	5	1	5
		Ask	3.84	0.93	3	4	4	1	5
		Chat	3.90	1.15	4	4	5	1	5
		Aggregated	1.61	0.90	1	1	2	1	5

Table B.4: NASA R-TLX, full per-subscale descriptive distribution by mode ($n = 69$ per mode), 1–21 scale. Source: `tlx_per_item_summary.csv`.

Subscale	Mode	Mean	SD	Q1	Median	Q3	Min	Max
Mental Demand	Aggregated	6.31	4.75	2.0	5.0	9.5	1	21
	Form	5.77	4.26	2.0	5.0	9.0	1	17
	Ask	6.80	4.63	3.0	6.0	10.0	1	21
	Chat	6.36	5.31	2.0	5.0	10.0	1	21
Physical Demand	Aggregated	3.19	3.40	1.0	2.0	4.0	1	21
	Form	2.77	2.71	1.0	1.0	4.0	1	14
	Ask	3.54	3.92	1.0	2.0	5.0	1	21
	Chat	3.26	3.46	1.0	2.0	4.0	1	20
Temporal Demand	Aggregated	4.89	4.28	1.0	4.0	7.0	1	19
	Form	4.16	3.82	1.0	3.0	6.0	1	15
	Ask	4.97	4.15	1.0	4.0	7.0	1	18
	Chat	5.54	4.78	2.0	4.0	8.0	1	19
Effort	Aggregated	6.13	4.57	3.0	5.0	10.0	1	20
	Form	5.68	4.56	2.0	4.0	9.0	1	17
	Ask	6.39	4.41	3.0	5.0	10.0	1	19
	Chat	6.32	4.77	3.0	5.0	10.0	1	20
Frustration	Aggregated	6.86	5.24	2.5	5.0	11.0	1	21
	Form	5.88	4.84	2.0	5.0	8.0	1	18
	Ask	7.58	5.34	3.0	6.0	12.0	1	21
	Chat	7.13	5.44	3.0	5.0	11.0	1	19
Performance (pos.)	Aggregated	8.93	5.82	4.0	9.0	13.0	1	21
	Form	8.71	5.48	4.0	8.0	12.0	1	21
	Ask	8.68	5.64	4.0	8.0	12.0	1	19
	Chat	9.39	6.36	3.0	10.0	14.0	1	21

Table B.5: Perceived AI Support, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `ai_support_per_item_summary.csv`.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
1	The AI was helpful in creating this document.	Aggregated	4.23	0.91	4	4	5	1	5
		Form	4.23	0.97	4	5	5	1	5
		Ask	4.32	0.76	4	4	5	1	5
		Chat	4.13	1.00	4	4	5	1	5
2	The AI helped me to complete the document faster.	Aggregated	4.22	1.13	4	5	5	1	5
		Form	4.43	0.96	4	5	5	1	5
		Ask	4.25	1.03	4	5	5	1	5
		Chat	3.99	1.32	4	4	5	1	5
3	I understood why the AI suggested specific GDPR fields.	Aggregated	3.81	1.06	3	4	5	1	5
		Form	3.51	1.17	3	4	4	1	5
		Ask	4.16	0.92	4	4	5	1	5
		Chat	3.75	0.98	3	4	5	1	5
4	I trust that the AI made correct suggestions.	Aggregated	3.63	0.97	3	4	4	1	5
		Form	3.41	1.09	3	4	4	1	5
		Ask	3.80	0.87	3	4	4	1	5
		Chat	3.68	0.92	3	4	4	1	5
5	I am confident that the AI identified the correct legal basis.	Aggregated	3.71	0.88	3	4	4	1	5
		Form	3.58	0.90	3	4	4	1	5
		Ask	3.74	0.87	3	4	4	1	5
		Chat	3.81	0.88	3	4	4	1	5
6	I would use this mode again.	Aggregated	3.65	1.25	3	4	5	1	5
		Form	3.99	1.14	4	4	5	1	5
		Ask	3.55	1.16	3	4	4	1	5
		Chat	3.42	1.39	2	4	5	1	5

B.5 Document-Level NLP Metric Distributions

The compact summaries and the box plots are reproduced inline with the NLP results in Section 4.3.1; the full per-document descriptive distributions are reproduced here, split into the lexical and embedding-based families.

B.5.1 Lexical Metrics

B.5.2 BERT-Based and Sentence-Embedding Metrics

B.6 Chapter-Level NLP Metric Distributions

The chapter-level box plots are reproduced inline with the chapter-level NLP results in Section 4.3.2; the per-chapter, per-mode means (with SDs in parentheses) for every metric are reproduced here.

B.6.1 Lexical Metrics

B.6.2 BERT-Based and Sentence-Embedding Metrics

Table B.6: Document-level lexical metric distributions by mode (full sample $n = 113$). Source: `doc_nlp_summary.csv`.

Metric	Mode	n	Mean	SD	Q1	Median	Q3	Min	Max
BLEU	Aggregated	113	0.133	0.048	0.094	0.128	0.159	0.047	0.246
	Form	37	0.129	0.048	0.088	0.124	0.158	0.064	0.230
	Ask	37	0.127	0.056	0.083	0.125	0.148	0.047	0.246
	Chat	39	0.143	0.039	0.122	0.134	0.163	0.067	0.235
ROUGE-1	Aggregated	113	0.488	0.087	0.425	0.491	0.560	0.313	0.645
	Form	37	0.488	0.098	0.421	0.493	0.569	0.313	0.644
	Ask	37	0.465	0.088	0.412	0.449	0.522	0.322	0.618
	Chat	39	0.511	0.071	0.468	0.517	0.561	0.330	0.645
ROUGE-2	Aggregated	113	0.285	0.077	0.234	0.268	0.344	0.121	0.455
	Form	37	0.306	0.076	0.241	0.301	0.361	0.178	0.455
	Ask	37	0.257	0.080	0.200	0.238	0.275	0.121	0.429
	Chat	39	0.291	0.069	0.253	0.283	0.344	0.131	0.447
ROUGE-L	Aggregated	113	0.375	0.087	0.311	0.359	0.441	0.194	0.566
	Form	37	0.388	0.094	0.320	0.391	0.468	0.238	0.566
	Ask	37	0.347	0.091	0.296	0.326	0.376	0.194	0.545
	Chat	39	0.389	0.072	0.348	0.387	0.444	0.230	0.544
METEOR	Aggregated	113	0.389	0.062	0.359	0.388	0.431	0.234	0.514
	Form	37	0.412	0.051	0.367	0.419	0.442	0.335	0.514
	Ask	37	0.365	0.068	0.327	0.366	0.409	0.234	0.488
	Chat	39	0.390	0.057	0.374	0.389	0.421	0.235	0.501

Table B.7: Document-level BERT-based and SBERT distributions by mode (full sample $n = 113$). Source: `doc_bert_summary.csv`.

Metric	Mode	n	Mean	SD	Q1	Median	Q3	Min	Max
BERTScore Precision	Aggregated	113	0.890	0.014	0.880	0.891	0.900	0.861	0.922
	Form	37	0.889	0.014	0.880	0.889	0.899	0.862	0.912
	Ask	37	0.886	0.016	0.872	0.889	0.899	0.861	0.918
	Chat	39	0.895	0.012	0.890	0.896	0.902	0.867	0.922
BERTScore Recall	Aggregated	113	0.915	0.014	0.914	0.918	0.922	0.861	0.936
	Form	37	0.923	0.005	0.919	0.923	0.926	0.914	0.936
	Ask	37	0.908	0.017	0.908	0.915	0.918	0.861	0.925
	Chat	39	0.913	0.013	0.908	0.917	0.922	0.874	0.933
BERTScore F1	Aggregated	113	0.902	0.011	0.895	0.904	0.910	0.863	0.922
	Form	37	0.905	0.009	0.900	0.906	0.912	0.887	0.920
	Ask	37	0.897	0.013	0.892	0.896	0.906	0.863	0.918
	Chat	39	0.904	0.010	0.899	0.906	0.911	0.872	0.922
SBERT (ModernBERT)	Aggregated	113	0.987	0.004	0.985	0.988	0.990	0.971	0.994
	Form	37	0.986	0.004	0.984	0.988	0.989	0.974	0.992
	Ask	37	0.986	0.005	0.983	0.987	0.990	0.971	0.993
	Chat	39	0.990	0.002	0.988	0.990	0.991	0.985	0.994

Table B.8: Chapter-level BLEU mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.181 (0.080)	0.179 (0.077)	0.200 (0.085)	0.166 (0.076)
Purposes & Legal Bases	0.121 (0.060)	0.145 (0.057)	0.098 (0.038)	0.121 (0.070)
Personal Data Categories	0.081 (0.050)	0.110 (0.041)	0.077 (0.052)	0.058 (0.043)
Data Subjects & Recipients	0.097 (0.075)	0.123 (0.076)	0.081 (0.082)	0.088 (0.063)
Retention & Deletion	0.044 (0.076)	0.038 (0.044)	0.023 (0.073)	0.068 (0.094)
Technical & Org. Measures	0.080 (0.066)	0.078 (0.068)	0.070 (0.080)	0.091 (0.047)

Table B.9: Chapter-level ROUGE-1 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.571 (0.114)	0.550 (0.097)	0.605 (0.129)	0.559 (0.109)
Purposes & Legal Bases	0.414 (0.109)	0.457 (0.094)	0.373 (0.093)	0.410 (0.122)
Personal Data Categories	0.467 (0.164)	0.600 (0.084)	0.366 (0.120)	0.437 (0.174)
Data Subjects & Recipients	0.437 (0.116)	0.491 (0.111)	0.393 (0.128)	0.426 (0.086)
Retention & Deletion	0.260 (0.131)	0.252 (0.092)	0.212 (0.132)	0.314 (0.143)
Technical & Org. Measures	0.331 (0.130)	0.313 (0.137)	0.330 (0.157)	0.350 (0.090)

Table B.10: Chapter-level ROUGE-2 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.386 (0.145)	0.363 (0.131)	0.430 (0.158)	0.365 (0.138)
Purposes & Legal Bases	0.269 (0.109)	0.332 (0.093)	0.216 (0.077)	0.261 (0.119)
Personal Data Categories	0.298 (0.148)	0.425 (0.097)	0.198 (0.089)	0.272 (0.149)
Data Subjects & Recipients	0.250 (0.109)	0.317 (0.105)	0.215 (0.113)	0.220 (0.077)
Retention & Deletion	0.131 (0.113)	0.134 (0.071)	0.087 (0.107)	0.168 (0.137)
Technical & Org. Measures	0.176 (0.118)	0.166 (0.120)	0.163 (0.141)	0.197 (0.089)

Table B.11: Chapter-level ROUGE-L mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.534 (0.129)	0.518 (0.110)	0.576 (0.140)	0.511 (0.128)
Purposes & Legal Bases	0.351 (0.111)	0.404 (0.096)	0.299 (0.083)	0.351 (0.125)
Personal Data Categories	0.426 (0.168)	0.567 (0.085)	0.309 (0.107)	0.403 (0.179)
Data Subjects & Recipients	0.372 (0.121)	0.433 (0.114)	0.330 (0.131)	0.355 (0.096)
Retention & Deletion	0.229 (0.136)	0.218 (0.087)	0.183 (0.141)	0.283 (0.152)
Technical & Org. Measures	0.257 (0.135)	0.237 (0.140)	0.246 (0.158)	0.285 (0.100)

Table B.12: Chapter-level METEOR mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.471 (0.128)	0.456 (0.120)	0.506 (0.136)	0.453 (0.124)
Purposes & Legal Bases	0.400 (0.113)	0.452 (0.075)	0.348 (0.101)	0.401 (0.131)
Personal Data Categories	0.315 (0.120)	0.415 (0.074)	0.275 (0.111)	0.258 (0.104)
Data Subjects & Recipients	0.355 (0.109)	0.440 (0.098)	0.312 (0.101)	0.314 (0.077)
Retention & Deletion	0.314 (0.160)	0.332 (0.114)	0.245 (0.160)	0.362 (0.177)
Technical & Org. Measures	0.293 (0.113)	0.295 (0.108)	0.280 (0.145)	0.305 (0.079)

Table B.13: Chapter-level BERTScore Precision mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.857 (0.017)	0.855 (0.015)	0.862 (0.020)	0.855 (0.016)
Purposes & Legal Bases	0.873 (0.021)	0.872 (0.020)	0.869 (0.020)	0.877 (0.022)
Personal Data Categories	0.883 (0.022)	0.897 (0.015)	0.868 (0.019)	0.884 (0.022)
Data Subjects & Recipients	0.857 (0.028)	0.856 (0.025)	0.853 (0.035)	0.864 (0.023)
Retention & Deletion	0.824 (0.042)	0.814 (0.033)	0.817 (0.050)	0.840 (0.037)
Technical & Org. Measures	0.854 (0.033)	0.841 (0.038)	0.858 (0.031)	0.863 (0.027)

Table B.14: Chapter-level BERTScore Recall mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.882 (0.022)	0.879 (0.019)	0.887 (0.025)	0.879 (0.022)
Purposes & Legal Bases	0.902 (0.024)	0.911 (0.012)	0.892 (0.028)	0.904 (0.024)
Personal Data Categories	0.878 (0.041)	0.910 (0.016)	0.862 (0.042)	0.861 (0.039)
Data Subjects & Recipients	0.880 (0.026)	0.904 (0.014)	0.873 (0.024)	0.864 (0.022)
Retention & Deletion	0.889 (0.024)	0.890 (0.018)	0.879 (0.023)	0.897 (0.027)
Technical & Org. Measures	0.888 (0.025)	0.889 (0.018)	0.882 (0.032)	0.893 (0.022)

Table B.15: Chapter-level BERTScore F1 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.869 (0.019)	0.867 (0.016)	0.875 (0.022)	0.867 (0.017)
Purposes & Legal Bases	0.887 (0.018)	0.891 (0.015)	0.880 (0.018)	0.890 (0.020)
Personal Data Categories	0.880 (0.029)	0.903 (0.012)	0.865 (0.025)	0.872 (0.029)
Data Subjects & Recipients	0.868 (0.020)	0.879 (0.017)	0.862 (0.022)	0.863 (0.017)
Retention & Deletion	0.855 (0.031)	0.850 (0.024)	0.847 (0.035)	0.867 (0.031)
Technical & Org. Measures	0.871 (0.025)	0.864 (0.027)	0.870 (0.027)	0.878 (0.019)

Table B.16: Chapter-level SBERT (ModernBERT) cosine similarity mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.976 (0.005)	0.975 (0.004)	0.977 (0.006)	0.975 (0.005)
Purposes & Legal Bases	0.979 (0.008)	0.979 (0.006)	0.977 (0.009)	0.981 (0.008)
Personal Data Categories	0.973 (0.013)	0.982 (0.003)	0.969 (0.015)	0.970 (0.014)
Data Subjects & Recipients	0.971 (0.008)	0.973 (0.008)	0.970 (0.008)	0.971 (0.009)
Retention & Deletion	0.949 (0.016)	0.945 (0.013)	0.946 (0.018)	0.955 (0.014)
Technical & Org. Measures	0.971 (0.010)	0.967 (0.012)	0.972 (0.010)	0.975 (0.007)

Table B.17: Chapter-level SBERT (`all-MiniLM-L6-v2`) cosine similarity mean (SD) by mode (full sample $n = 113$). Reported as a robustness check on the ModernBERT pass.

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.687 (0.122)	0.663 (0.119)	0.736 (0.133)	0.662 (0.101)
Purposes & Legal Bases	0.741 (0.074)	0.763 (0.053)	0.740 (0.081)	0.721 (0.080)
Personal Data Categories	0.713 (0.122)	0.768 (0.063)	0.705 (0.149)	0.669 (0.118)
Data Subjects & Recipients	0.706 (0.094)	0.691 (0.087)	0.692 (0.115)	0.733 (0.072)
Retention & Deletion	0.519 (0.141)	0.516 (0.129)	0.474 (0.150)	0.566 (0.131)
Technical & Org. Measures	0.710 (0.095)	0.711 (0.073)	0.701 (0.110)	0.717 (0.099)

C Inferential Statistics in Full

This appendix reproduces the complete inferential test output referenced in Chapter 4. The within-subjects user-study tests use $n = 69$ (matched within-subjects, Friedman + Wilcoxon); the document-quality, timing, and confidence–quality analyses use $n = 113$ (between-groups, Kruskal–Wallis + Mann–Whitney U). The full sample-size derivation is reproduced first, followed by the user-study inferential tests, the NLP inferential tests, the confidence–quality correlations, and the timing inferential tests.

C.1 Sample-Size Derivation

All sample sizes used in the thesis are derived from the raw data files committed to the repository: `python/survey/docs/data.csv` for the user study (one row per participant) and `python/metrics/output/all_scores_combined.csv` for the NLP and judge scores (one row per LLM-generated document).

User-study subset	**n**
Recruited participants (raw)	73
Participants with at least one valid session ($p\_0001 \geq 1$)	69
Form responses (mode-level)	69
Ask responses (mode-level)	69
Chat responses (mode-level)	69
Matched-triple participants (valid response in all three modes)	69

Table C.1: User-study sample sizes after consent and quality filtering.

NLP / judge subset	**n**
Total documents with parseable NLP scores and judge ratings	113
Form documents	37
Ask documents	37
Chat documents	39
Unique participants represented	47
Participants with at least one document in each mode	28
Strict matched triple (exactly one document per mode)	26

Table C.2: NLP and judge sample sizes. The full $n = 113$ sample is used for the document-quality inferential tests (Kruskal–Wallis, Mann–Whitney U).

User-study $\times$ NLP join	**n**
Joined rows (document with matching user-study response)	106
Form joined rows	34
Ask joined rows	34
Chat joined rows	38
Unique participants in the joined sample	44
Participants with user-study $\times$ NLP coverage in all three modes	27

Table C.3: Triangulated samples used for confidence–quality analyses. The joined $n = 106$ is the canonical sample for the rank-inversion table; the within-subjects $n = 27$ is used for paired confidence $\times$ quality correlations.

C.2 User-Study Inferential Tests

C.2.1 Normality (Shapiro–Wilk, $n = 69$)

Table C.4: Shapiro–Wilk normality tests for user-study measures by mode.

Measure	Mode	W	p	Normal?
SUS	Form	0.944	0.004	No
SUS	Ask	0.976	0.210	Yes
SUS	Chat	0.927	0.001	No
R-TLX composite	Form	0.940	0.003	No
R-TLX composite	Ask	0.926	0.001	No
R-TLX composite	Chat	0.945	0.005	No
AI Support (6-item)	Form	0.923	< 0.001	No
AI Support (6-item)	Ask	0.965	0.048	No
AI Support (6-item)	Chat	0.952	0.010	No
AI Confidence (5_{ai})	Form	0.846	< 0.001	No
AI Confidence (5_{ai})	Ask	0.856	< 0.001	No
AI Confidence (5_{ai})	Chat	0.864	< 0.001	No
AI Reuse (6_{ai})	Form	0.799	< 0.001	No
AI Reuse (6_{ai})	Ask	0.872	< 0.001	No
AI Reuse (6_{ai})	Chat	0.861	< 0.001	No

Non-normal distributions are detected for nearly every measure. Non-parametric tests (Friedman omnibus and Wilcoxon signed-rank post-hoc) are used throughout.

C.2.2 Friedman Omnibus ($n = 69$ matched triple)

C.2.3 Pairwise Wilcoxon Signed-Rank Post-hoc (Bonferroni $\alpha = 0.0167$)

Post-hoc tests are reported for measures with a significant Friedman result and, for completeness, for SUS and the R-TLX composite. Sign convention: positive r means the first-named mode ranks higher. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

Table C.5: Friedman omnibus tests across the three modes for user-study measures. Kendall's W: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

Measure	χ^2	df	p	W	Significant
SUS (0–100)	5.51	2	0.064	0.040	No
R-TLX composite (1–21)	5.24	2	0.073	0.038	No
R-TLX Mental Demand	3.13	2	0.209	0.023	No
R-TLX Physical Demand	2.08	2	0.354	0.015	No
R-TLX Temporal Demand	12.77	2	0.002	0.093	**Yes**
R-TLX Effort	3.78	2	0.151	0.027	No
R-TLX Frustration	8.93	2	0.012	0.065	**Yes**
R-TLX Performance (pos.)	2.33	2	0.311	0.017	No
AI Confidence (5_{ai})	3.76	2	0.152	0.027	No
AI Support (6-item composite)	1.05	2	0.592	0.008	No
AI Reuse intention (6_{ai})	11.12	2	0.004	0.081	**Yes**

Table C.6: Pairwise Wilcoxon signed-rank post-hoc tests for user-study measures.

Measure	Pair	n	W	p	r	Sig (Bonf)
SUS	Form vs Ask	60	578.5	0.013	$+0.368$ medium	**Yes**
SUS	Form vs Chat	58	549.5	0.018	$+0.358$ medium	No
SUS	Ask vs Chat	61	885.5	0.666	-0.063 negligible	No
R-TLX composite	Form vs Ask	65	684.0	0.011	-0.362 medium	**Yes**
R-TLX composite	Form vs Chat	65	636.0	0.004	-0.407 medium	**Yes**
R-TLX composite	Ask vs Chat	63	995.5	0.932	$+0.012$ negligible	No
Temporal Demand	Form vs Ask	43	237.0	0.004	-0.499 medium	**Yes**
Temporal Demand	Form vs Chat	46	233.0	< 0.001	-0.569 large	**Yes**
Temporal Demand	Ask vs Chat	46	427.0	0.211	-0.210 small	No
Frustration	Form vs Ask	52	403.5	0.009	-0.414 medium	**Yes**
Frustration	Form vs Chat	57	519.5	0.014	-0.371 medium	**Yes**
Frustration	Ask vs Chat	52	629.0	0.584	$+0.087$ negligible	No
AI Reuse (6_{ai})	Form vs Ask	49	402.5	0.034	$+0.343$ medium	No
AI Reuse (6_{ai})	Form vs Chat	43	216.0	0.002	$+0.543$ large	**Yes**
AI Reuse (6_{ai})	Ask vs Chat	45	454.5	0.468	$+0.122$ small	No

C.2.4 Mode Preference Ranking

A chi-square goodness-of-fit test on the rank-1 vote distribution against a uniform null gives $\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$ (medium effect). The Friedman test on the full rank scores (1–3) is non-significant: $\chi^2(2) = 5.07$, $p = 0.079$, $W = 0.037$.

C.3 NLP Metric Inferential Tests

C.3.1 Kruskal–Wallis Omnibus ($n = 113$)

Table C.7: Kruskal–Wallis H tests on the unbalanced full document set ($n = 113$). Primary omnibus for document-quality inferential testing.

Metric	H	p	Sig ($\alpha = 0.05$)	ε^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (moderate)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (moderate)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	< 0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (moderate)
SBERT (ModernBERT)	17.43	< 0.001	Yes	0.140 (large)

C.3.2 Pairwise Mann–Whitney U ($n = 113$)

Pairwise Mann–Whitney U tests on the unbalanced full sample, serving as the post-hoc family where the Kruskal–Wallis omnibus is significant. Bonferroni-corrected $\alpha = 0.0167$. Only significant comparisons are listed; non-significant comparisons are omitted for brevity.

Ask is the significantly lowest-scoring mode (Ask < Form, Ask < Chat) on the metrics that reach significance, most clearly on BERTScore Recall, F1, ROUGE-2, and

Table C.8: Pairwise Mann–Whitney U post-hoc tests on the unbalanced full sample of $n = 113$ documents.

Metric	Comparison	p	r	Sig
BERTScore Recall	Form > Ask	< 0.001	0.756 (large)	Yes
BERTScore Recall	Form > Chat	< 0.001	0.512 (large)	Yes
BERTScore F1	Form > Ask	0.004	0.386 (medium)	Yes
BERTScore F1	Chat > Ask	0.011	0.340 (medium)	Yes
SBERT (ModernBERT)	Chat > Form	< 0.001	0.505 (large)	Yes
SBERT (ModernBERT)	Chat > Ask	< 0.001	0.453 (large)	Yes
ROUGE-2	Form > Ask	0.006	0.369 (medium)	Yes
ROUGE-2	Chat > Ask	0.011	0.342 (medium)	Yes
METEOR	Form > Ask	0.003	0.407 (medium)	Yes

METEOR; Form and Chat trade leadership depending on whether the metric emphasizes lexical recall or document-level semantic alignment.

C.4 Confidence–Quality Correlations

Spearman rank correlation ρ between the primary confidence measure (item 5_{ai}, "I am confident the AI identified the correct legal basis") and document-quality measures, computed on the user-study $\times$ NLP joined sample. Per-mode joined samples: Form $n = 34$, Ask $n = 34$, Chat $n = 38$; pooled $n = 106$.

Table C.9: Spearman rank correlations: primary confidence (5_{ai}) versus document-quality measures, by mode.

	5_{ai} vs BERTScore F1			5_{ai} vs Judge Overall		
Mode	n	ρ	p	n	ρ	p
Form	34	-0.22	0.22	34	$+0.07$	0.70
Ask	34	-0.23	0.19	34	-0.21	0.24
Chat	38	$+0.28$	0.09	38	$+0.41$	**0.012**
Pooled	106	-0.02	0.88	106	$+0.14$	0.14

For comparison, the secondary 6-item Perceived AI Support composite produces weaker mode-specific correlations with BERTScore F1 (Form $\rho = -0.22$, Ask $\rho =$

-0.04, Chat $\rho = +0.20$) and a small but significant pooled correlation with Judge Overall ($\rho = +0.21$, $p = 0.033$, $n = 106$).

C.5 Timing Inferential Tests

Inferential tests on the per-mode completion time recorded in the `Time elapsed` line emitted by ROPAgen at document submission (one value per participant–mode pairing). Sample sizes match the NLP document-level analysis: full $n = 113$ (Form 37, Ask 37, Chat 39); matched-triple within-subjects subset $n = 26$ participants. Non-parametric tests are used throughout, consistent with the NLP and user-study pipelines, as Shapiro–Wilk rejects normality on every mode (Form $W = 0.780$, $p < 0.001$; Ask $W = 0.912$, $p = 0.030$; Chat $W = 0.601$, $p < 0.001$). Source: `python/metrics/output/statistical_tests.md` (*Per-Mode Completion Time* section).

C.5.1 Kruskal–Wallis Omnibus ($n = 113$)

Table C.10: Kruskal–Wallis H test on per-mode completion time across the unbalanced full sample of $n = 113$ documents.

Measure	H	p	Sig ($\alpha = 0.05$)	ε^2
Completion time (s)	16.455	< 0.001	Yes	0.147 (large)

C.5.2 Pairwise Mann–Whitney U ($n = 113$)

Pairwise Mann–Whitney U tests on the unbalanced full sample, serving as the post-hoc family where the Kruskal–Wallis omnibus is significant. Bonferroni-corrected $\alpha = 0.0167$.

Form is significantly faster than both Ask and Chat (large and medium effects); Ask and Chat are statistically indistinguishable from one another.

Table C.11: Pairwise Mann–Whitney U post-hoc tests on per-mode completion time on the unbalanced full sample of $n = 113$ documents. Effect-size r is positive when the first-named mode is *slower*.

Pair	U	p	r	Median A (s)	Median B (s)	Sig (Bonf)
Form vs Ask	334.0	< 0.001	$+0.512$ large	420.0	858.0	**Yes**
Form vs Chat	420.0	0.002	$+0.418$ medium	420.0	717.0	**Yes**
Ask vs Chat	793.5	0.457	-0.100 negligible	858.0	717.0	No

D Acronyms

This appendix lists the acronyms used throughout the thesis, together with their expansions, for ease of reference.

Acronym	Expansion
API	Application Programming Interface
BDSG	Bundesdatenschutzgesetz (German Federal Data Protection Act)
BERTScore	Bidirectional Encoder Representations from Transformers Score
BLEU	Bilingual Evaluation Understudy
DBIS	Institute of Databases and Information Systems (University of Ulm)
DPO	Data Protection Officer
DSGVO	Datenschutz-Grundverordnung (German term for GDPR)
EU	European Union
GDPR	General Data Protection Regulation
HGB	Handelsgesetzbuch (German Commercial Code)
IfSG	Infektionsschutzgesetz (German Infection Protection Act)
LCS	Longest Common Subsequence
LLM	Large Language Model
METEOR	Metric for Evaluation of Translation with Explicit ORdering
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ROPA	Record of Processing Activities
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
R-TLX	Raw NASA Task Load Index
RQ	Research Question
SBERT	Sentence-BERT
SD	Standard Deviation
SUS	System Usability Scale
UX	User Experience

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Declaration

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.

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